SLAM 的过去、现在与未来：迈向鲁棒感知时代

Past, Present, and Future of Simultaneous Localization And Mapping: Towards the Robust-Perception Age

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核心摘要校订

根据原文摘要单独校订。正文仍为逐页机器翻译初稿；原始图表、公式和英文条目请查看紧邻原文页。

同时定位与建图（SLAM）是在构建环境模型，即地图的同时，估计其中运动机器人的状态。领域长期进展已经支持大规模真实应用，并逐步进入产业。本文综述当时的研究现状与未来方向，先介绍 SLAM 的标准问题形式，再考察长期建图的鲁棒性与规模、几何和语义地图、理论性能保证、主动 SLAM 与探索，以及其他前沿。本文同时具有立场文章与教程的作用，批判性分析尚未解决的问题，并讨论“机器人是否需要 SLAM”和“SLAM 是否已经解决”两个核心问题。

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}

Past, Present, and Future of Simultaneous Localization And Mapping: Towards the Robust-Perception Age

Cesar Cadena, Luca Carlone, Henry Carrillo, Yasir Latif,
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Abstract*—Simultaneous Localization And Mapping (SLAM) consists in the concurrent construction of a model of the environment (the *map*), and the estimation of the state of the robot moving within it. The SLAM community has made astonishing progress over the last 30 years, enabling large-scale real-world applications, and witnessing a steady transition of this technology to industry. We survey the current state of SLAM and consider future directions. We start by presenting what is now the *de-facto* standard formulation for SLAM. We then review related work, covering a broad set of topics including robustness and scalability in long-term mapping, metric and semantic representations for mapping, theoretical performance guarantees, active SLAM and exploration, and other new frontiers. This paper simultaneously serves as a position paper and tutorial to those who are users of SLAM. By looking at the published research with a critical eye, we delineate open challenges and new research issues, that still deserve careful scientific investigation. The paper also contains the authors' take on two questions that often animate discussions during robotics conferences: *Do robots need SLAM?* and *Is SLAM solved?

***Index Terms*—Robots, SLAM, Localization, Mapping, Factor graphs, Maximum a posteriori estimation, sensing, perception.**

Multimedia Material

Additional material for this paper, including an extended list of references (bibtex) and a table of pointers to online datasets for SLAM, can be found at `https://slam-future.github.io/`.

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This paper summarizes and extends the outcome of the workshop "The Problem of Mobile Sensors: Setting future goals and indicators of progress for SLAM" [25], held during the *Robotics: Science and System* (RSS) conference (Rome, July 2015).

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I. Introduction

SLAM comprises the simultaneous estimation of the state of a robot equipped with on-board sensors, and the construction of a model (the *map*) of the environment that the sensors are perceiving. In simple instances, the robot state is described by its pose (position and orientation), although other quantities may be included in the state, such as robot velocity, sensor biases, and calibration parameters. The map, on the other hand, is a representation of aspects of interest (e.g., position of landmarks, obstacles) describing the environment in which the robot operates.

The need to use a map of the environment is twofold. First, the map is often required to support other tasks; for instance, a map can inform path planning or provide an intuitive visualization for a human operator. Second, the map allows limiting the error committed in estimating the state of the robot. In the absence of a map, dead-reckoning would quickly drift over time; on the other hand, using a map, e.g., a set of distinguishable landmarks, the robot can "reset" its localization error by re-visiting known areas (so-called *loop closure*). Therefore, SLAM finds applications in all scenarios in which a prior map is not available and needs to be built.

In some robotics applications the location of a set of landmarks is known *a priori*. For instance, a robot operating on a factory floor can be provided with a manually-built map of artificial beacons in the environment. Another example is the case in which the robot has access to GPS (the GPS satellites can be considered as moving beacons at known locations). In such scenarios, SLAM may not be required if localization can be done reliably with respect to the known landmarks.

The popularity of the SLAM problem is connected with the emergence of indoor applications of mobile robotics. Indoor operation rules out the use of GPS to bound the localization error; furthermore, SLAM provides an appealing alternative to user-built maps, showing that robot operation is possible in the absence of an ad hoc localization infrastructure.

A thorough historical review of the first 20 years of the SLAM problem is given by Durrant-Whyte and Bailey in two surveys [7, 69]. These mainly cover what we call the *classical age* (1986-2004); the classical age saw the introduction of the main probabilistic formulations for SLAM, including approaches based on Extended Kalman Filters, Rao-Blackwellised Particle Filters, and maximum likelihood estimation; moreover, it delineated the basic challenges connected

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同步定位与测绘的过去、现在和未来：迈向鲁棒感知时代

塞萨尔·卡德纳、卢卡·卡龙、亨利·卡里略、亚西尔·拉蒂夫、达维德·斯卡拉穆扎、何塞·内拉、伊恩·里德、约翰·J·伦纳德

Abstract -同步定位与建图 (SLAM) 包括同时构建环境模型（map），并估计机器人在其中移动的状态。SLAM 社区在过去 30 年中取得了惊人的进步，实现了大规模的现实应用，并见证了该技术向工业的稳步过渡。我们调查了 SLAM 的现状并考虑了未来的方向。我们首先介绍目前 SLAM 的_事实上_ 标准公式。然后，我们回顾相关工作，涵盖广泛的主题，包括长期映射的鲁棒性和可扩展性、映射的度量和语义表示、理论性能保证、主动 SLAM 和探索以及其他新领域。本文同时作为 SLAM 用户的立场文件和教程。通过以批判的眼光看待已发表的研究，我们描绘了仍然值得仔细科学研究的开放挑战和新研究问题。该论文还包含作者对机器人会议期间经常引起讨论的两个问题的看法：_机器人需要 SLAM 吗？_和_SLAM 是否已解决？_

索引术语 -机器人、SLAM、定位、建图、因子图、最大后验估计、传感、感知。
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本文的其他材料，包括扩展的参考文献列表 (bibtex) 和 SLAM 在线数据集的指针表，可以在 https://slam-future.github.io/ 上找到。

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本文总结并扩展了在_机器人：科学与系统_ (RSS) 会议（罗马，2015 年 7 月）期间举行的研讨会“移动传感器问题：为 SLAM 设定未来目标和进展指标”[25] 的成果。

这项工作得到了以下赠款的部分支持：MINECOFEDER DPI2015-68905-P、Grupo DGA T04-FSE；ARC授予DP130104413、CE140100016和FL130100102； NCCR 机器人； PUJ 6601； EU-FP7-ICT项目 TRADR 609763、EU-H2020-688652 和 SERI-15.0284。
我

。简介

LAM 包括状态的同时估计

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配备机载传感器的机器人，以及传感器感知的环境模型（地图）的构建。在简单的情况下，机器人状态由其位姿（位置和方向）描述，尽管状态中可能包含其他量，例如机器人速度、传感器偏差和校准参数。另一方面，地图是描述机器人操作环境的感兴趣方面（例如地标、障碍物的位置）的表示。

使用环境地图的需要是双重的。首先，地图常常需要支持其他任务；例如，地图可以为路径规划提供信息或为操作员提供直观的可视化。其次，该地图可以限制估计机器人状态时所犯的错误。如果没有地图，航位推算法很快就会随着时间的推移而漂移。另一方面，使用地图，例如一组可区分的地标，机器人可以通过重新访问已知区域来“重置”其定位误差（所谓的“循环闭合”）。因此，SLAM在所有无法获得先验地图且需要构建先验地图的场景中都有应用。

在一些机器人应用中，一组地标的位置是先验已知的。例如，可以为在工厂车间运行的机器人提供手动构建的环境中人工信标地图。另一个例子是机器人可以访问 GPS 的情况（GPS 卫星可以被视为已知位置的移动信标）。在这种情况下，如果可以相对于已知地标可靠地完成定位，则可能不需要 SLAM。

SLAM 问题的流行与移动机器人室内应用的出现有关。室内操作排除使用GPS限制定位误差；此外，SLAM 为用户构建的地图提供了一种有吸引力的替代方案，表明在没有临时定位基础设施的情况下机器人操作也是可能的。

Durrant-Whyte 和 Bailey 在两项调查中对 SLAM 问题的前 20 年进行了彻底的历史回顾 [7, 69]。这些主要涵盖我们所说的_古典时代_（1986-2004）；经典时代引入了 SLAM 的主要概率公式，包括基于扩展卡尔曼滤波器、RaoBlackwellized 粒子滤波器和最大似然估计的方法；此外，它还描述了相关的基本挑战

to efficiency and robust data association. Two other excellent references describing the three main SLAM formulations of the classical age are the book of Thrun, Burgard, and Fox [240] and the chapter of Stachniss *et al.* [234, Ch. 46]. The subsequent period is what we call the *algorithmic-analysis age* (2004-2015), and is partially covered by Dissanayake *et al.* in [64]. The algorithmic analysis period saw the study of fundamental properties of SLAM, including observability, convergence, and consistency. In this period, the key role of sparsity towards efficient SLAM solvers was also understood, and the main open-source SLAM libraries were developed.

We review the main SLAM surveys to date in Table I, observing that most recent surveys only cover specific aspects or sub-fields of SLAM. The popularity of SLAM in the last 30 years is not surprising if one thinks about the manifold aspects that SLAM involves. At the lower level (called the *front-end* in Section II) SLAM naturally intersects other research fields such as computer vision and signal processing; at the higher level (that we later call the *back-end*), SLAM is an appealing mix of geometry, graph theory, optimization, and probabilistic estimation. Finally, a SLAM expert has to deal with practical aspects ranging from sensor calibration to system integration.

The present paper gives a broad overview of the current state of SLAM, and offers the perspective of part of the community on the open problems and future directions for the SLAM research. Our main focus is on metric and semantic SLAM, and we refer the reader to the recent survey by Lowry *et al.* [160], which provides a comprehensive review of vision-based place recognition and topological SLAM.

Before delving into the paper, we first discuss two questions that often animate discussions during robotics conferences: (1) do autonomous robots need SLAM? and (2) is SLAM solved as an academic research endeavor? We will revisit these questions at the end of the manuscript.

Answering the question "Do autonomous robots really need SLAM?" requires understanding what makes SLAM unique. SLAM aims at building a globally consistent representation of the environment, leveraging both ego-motion measurements and loop closures. The keyword here is "loop closure": if we sacrifice loop closures, SLAM reduces to odometry. In early applications, odometry was obtained by integrating wheel encoders. The pose estimate obtained from wheel odometry quickly drifts, making the estimate unusable after few meters [128, Ch. 6]; this was one of the main thrusts behind the development of SLAM: the observation of external landmarks is useful to reduce the trajectory drift and possibly correct it [185]. However, more recent odometry algorithms are based on visual and inertial information, and have very small drift ($< 0.5\%$ of the trajectory length [82]). Hence the question becomes legitimate: do we really need SLAM? Our answer is three-fold.

First of all, we observe that the SLAM research done over the last decade has itself produced the visual-inertial odometry algorithms that currently represent the state of the art, e.g., [163, 175]; in this sense Visual-Inertial Navigation (VIN) *is* SLAM: VIN can be considered a *reduced* SLAM system, in which the loop closure (or place recognition) module is disabled. More generally, SLAM has directly led to the study of sensor fusion under more challenging setups (i.e., no GPS, low quality sensors) than previously considered in other literature (e.g., inertial navigation in aerospace engineering).

TABLE I: Surveying the surveys and tutorials

Year	Topic	Reference
2006	Probabilistic approaches and data association	Durrant-Whyte and Bailey [7, 69]
2008	Filtering approaches	Aulinas *et al.* [6]
2011	SLAM back-end	Grisetti *et al.* [97]
2011	Observability, consistency and convergence	Dissanayake *et al.* [64]
2012	Visual odometry	Scaramuzza and Fraundofer [85, 218]
2016	Multi robot SLAM	Saeedi *et al.* [216]
2016	Visual place recognition	Lowry *et al.* [160]
2016	SLAM in the Handbook of Robotics	Stachniss *et al.* [234, Ch. 46]
2016	Theoretical aspects	Huang and Dissanayake [109]

The second answer regards the true topology of the environment. A robot performing odometry and neglecting loop closures interprets the world as an "infinite corridor" (Fig. 1-left) in which the robot keeps exploring new areas indefinitely. A loop closure event informs the robot that this "corridor" keeps intersecting itself (Fig. 1-right). The advantage of loop closure now becomes clear: by finding loop closures, the robot understands the real topology of the environment, and is able to find shortcuts between locations (e.g., point B and C in the map). Therefore, if getting the right topology of the environment is one of the merits of SLAM, why not simply drop the metric information and just do place recognition? The answer is simple: the metric information makes place recognition much simpler and more robust; the metric reconstruction informs the robot about loop closure opportunities and allows discarding spurious loop closures [150]. Therefore, while SLAM might be redundant in principle (an oracle place recognition module would suffice for topological mapping), SLAM offers a natural defense against wrong data association and perceptual aliasing, where similarly looking scenes, corresponding to distinct locations in the environment, would deceive place recognition. In this sense, the SLAM map provides a way to predict and validate future measurements: we believe that this mechanism is key to robust operation.

The third answer is that SLAM is needed for many applications that, either implicitly or explicitly, *do* require a globally consistent map. For instance, in many military and civilian applications, the goal of the robot is to explore an environment and report a map to the human operator, ensuring that full coverage of the environment has been obtained. Another example is the case in which the robot has to perform structural inspection (of a building, bridge, etc.); also in this case a globally consistent 3D reconstruction is a requirement for successful operation.

This question of "is SLAM solved?" is often asked within the robotics community, c.f. [87]. This question is difficult to answer because SLAM has become such a broad topic that the question is well posed only for a given robot/environment/performance combination. In particular, one can evaluate the maturity of the SLAM problem once the following aspects are specified:

- *robot*: type of motion (e.g., dynamics, maximum speed), available sensors (e.g., resolution, sampling rate), avail-

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高效和强大的数据关联。描述古典时代三种主要 SLAM 公式的另外两本优秀参考文献是 Thrun、Burgard 和 Fox 的书 [240] 以及 Stachniss et al. 的章节 [234，Ch.1]。 46]。随后的时期是我们所说的“算法分析时代”（2004-2015），Dissanayake 等人在[64]中部分涵盖了这一时期。算法分析阶段研究了 SLAM 的基本特性，包括可观测性、收敛性和一致性。在此期间，人们还了解了稀疏性对于高效 SLAM 求解器的关键作用，并开发了主要的开源 SLAM 库。

我们在表一中回顾了迄今为止主要的 SLAM 调查，发现最近的调查仅涵盖 SLAM 的特定方面或子领域。如果考虑一下 SLAM 涉及的多个方面，SLAM 在过去 30 年的流行就不足为奇了。在较低层次（第二节中称为“前端”）SLAM 自然地与计算机视觉和信号处理等其他研究领域相交叉；在更高的层面上（我们后来称之为_后端_），SLAM 是几何学、图论、优化和概率估计的有吸引力的组合。最后，SLAM 专家必须处理从传感器校准到系统集成等实际问题。

本文对 SLAM 的现状进行了广泛的概述，并提供了部分社区对 SLAM 研究的开放问题和未来方向的看法。我们的主要关注点是度量和语义 SLAM，我们建议读者参考 Lowry 等人最近的调查[160]，其中对基于视觉的位置识别和拓扑 SLAM 进行了全面的回顾。

在深入研究本文之前，我们首先讨论机器人会议期间经常引起讨论的两个问题：（1）自主机器人需要 SLAM 吗？ (2) SLAM 是作为一项学术研究工作来解决的吗？我们将在手稿的末尾重新讨论这些问题。

回答“自主机器人真的需要 SLAM 吗？”这个问题需要了解 SLAM 的独特之处。 SLAM 旨在利用自我运动测量和闭环来构建全球一致的环境表示。这里的关键词是“循环闭合”：如果我们牺牲循环闭合，SLAM 就会简化为里程计。在早期应用中，里程计是通过集成车轮编码器来获得的。从车轮里程计获得的姿态估计很快就会漂移，使得估计在几米后无法使用[128，Ch.1]。 6];这是 SLAM 发展背后的主要推动力之一：外部地标的观察有助于减少轨迹漂移并可能纠正它[185]。然而，最近的里程计算法基于视觉和惯性信息，并且具有非常小的漂移（< 0 . 轨迹长度的 5% [82]）。因此，这个问题就变得合理了：我们真的需要 SLAM 吗？我们的答案是三重的。

首先，我们观察到过去十年进行的 SLAM 研究本身已经产生了代表当前最先进技术的视觉惯性里程计算法，例如 [163, 175]

;从这个意义上说，视觉惯性导航 (VIN) is SLAM：VIN 可以被认为是一个_reduced_ SLAM 系统，其中闭环（或位置识别）模块被禁用。更一般地说，SLAM 直接导致了

表 I：调查问卷和教程

年份	主题	参考
2006	概率 方法 和数据关联	Durrant-Whyte 和 Bailey [7, 69]
2008	过滤方法	Aulinas et al. [6]
2011	SLAM 后端	Grisetti et al. [97]
2011	可观测性、一致性 和收敛性	Dissanayake et al. [64]

2012	视觉里程计	Scaramuzza 和 Fraundofer [85, 218]
2016	多机器人 SLAM	Saeedi et al. [216]
2016	视觉地点识别	Lowry et al. [160]
2016	机器人手册中的 SLAM
Stachniss et al. [234，Ch. 46]
2016	理论方面	Huang和Dissanayake [109]

在比其他文献（例如航空航天工程中的惯性导航）中考虑的更具挑战性的设置（即没有 GPS、低质量传感器）下研究传感器融合。

第二个答案涉及环境的真实拓扑。执行里程计并忽略闭环的机器人将世界解释为“无限走廊”（图 1-左），机器人在其中无限期地不断探索新区域。环路闭合事件通知机器人这条“走廊”不断与自身相交（图 1-右）。回环的优势现在变得显而易见：通过寻找回环，机器人了解环境的真实拓扑，并能够找到位置之间的捷径（例如地图中的 B 点和 C 点）。因此，如果获得正确的环境拓扑是 SLAM 的优点之一，为什么不简单地放弃度量信息而只进行位置识别呢？答案很简单：度量信息使地点识别变得更简单、更稳健；度量重建告知机器人有关环路闭合的机会，并允许丢弃虚假的环路闭合[150]。因此，虽然 SLAM 原则上可能是多余的（预言机地点识别模块足以进行拓扑映射），但 SLAM 提供了针对错误数据关联和感知混叠的天然防御，其中相似的场景（对应于环境中的不同位置）会欺骗地点识别。从这个意义上说，SLAM地图提供了一种预测和验证未来测量的方法：我们相信这种机制是稳健运行的关键。

第三个答案是许多应用程序都需要 SLAM，这些应用程序无论是隐式还是显式地都需要全局一致的地图。例如，在许多军事和民用应用中，机器人的目标是探索环境并向人类操作员报告地图，确保获得环境的完全覆盖。另一个例子是机器人必须执行结构检查（建筑物、桥梁等）；在这种情况下，全局一致的 3D 重建也是成功操作的必要条件。

“SLAM解决了吗？”这个问题在机器人社区中经常被问到，参见： [87]

。这个问题很难回答，因为 SLAM 已经成为一个如此广泛的话题，以至于这个问题仅适用于给定的机器人/环境/性能组合。特别是，一旦指定了以下几个方面，就可以评估 SLAM 问题的成熟度：

- robot：运动类型（例如，动态、最大速度）、可用传感器（例如，分辨率、采样率）、可用

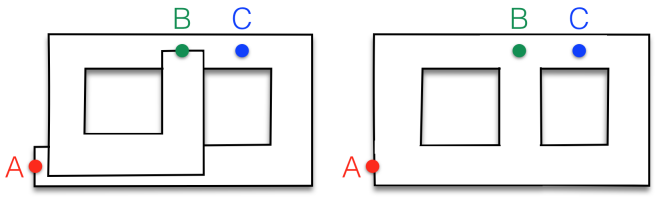


Fig. 1: Left: map built from odometry. The map is homotopic to a long corridor that goes from the starting position A to the final position B. Points that are close in reality (e.g., B and C) may be arbitrarily far in the odometric map. Right: map build from SLAM. By leveraging loop closures, SLAM estimates the actual topology of the environment, and "discovers" shortcuts in the map.

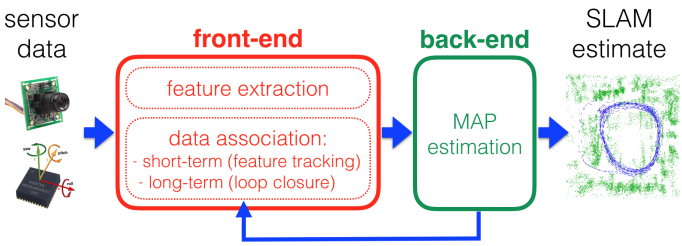


Fig. 2: Front-end and back-end in a typical SLAM system. The back-end can provide feedback to the front-end for loop closure detection and verification.

able computational resources;

- *environment*: planar or three-dimensional, presence of natural or artificial landmarks, amount of dynamic elements, amount of symmetry and risk of perceptual aliasing. Note that many of these aspects actually depend on the sensor-environment pair: for instance, two rooms may look identical for a 2D laser scanner (perceptual aliasing), while a camera may discern them from appearance cues;
- *performance requirements*: desired accuracy in the estimation of the state of the robot, accuracy and type of representation of the environment (e.g., landmark-based or dense), success rate (percentage of tests in which the accuracy bounds are met), estimation latency, maximum operation time, maximum size of the mapped area.

For instance, mapping a 2D indoor environment with a robot equipped with wheel encoders and a laser scanner, with sufficient accuracy ($<$ 10cm) and sufficient robustness (say, low failure rate), can be considered largely solved (an example of industrial system performing SLAM is the *Kuka Navigation Solution* [145]). Similarly, vision-based SLAM with slowly-moving robots (e.g., Mars rovers [166], domestic robots [114]), and visual-inertial odometry [94] can be considered mature research fields.

On the other hand, other robot/environment/performance combinations still deserve a large amount of fundamental research. Current SLAM algorithms can be easily induced to fail when either the motion of the robot or the environment are too challenging (e.g., fast robot dynamics, highly dynamic environments); similarly, SLAM algorithms are often unable to face strict performance requirements, e.g., high rate estimation for fast closed-loop control. This survey will provide a comprehensive overview of these open problems, among others.

In this paper, we argue that we are entering in a third era for SLAM, the *robust-perception age*, which is characterized by the following key requirements:

1) *robust performance*: the SLAM system operates with low failure rate for an extended period of time in a broad set of environments; the system includes fail-safe mechanisms and has self-tuning capabilities[1] in that it can adapt the selection of the system parameters to the scenario.
2) *high-level understanding*: the SLAM system goes beyond basic geometry reconstruction to obtain a high-level understanding of the environment (e.g., high-level geometry, semantics, physics, affordances);
3) *resource awareness*: the SLAM system is tailored to the available sensing and computational resources, and provides means to adjust the computation load depending on the available resources;
4) *task-driven perception*: the SLAM system is able to select relevant perceptual information and filter out irrelevant sensor data, in order to support the task the robot has to perform; moreover, the SLAM system produces adaptive map representations, whose complexity may vary depending on the task at hand.

[1]The SLAM community has been largely affected by the "curse of manual tuning", in that satisfactory operation is enabled by expert tuning of the system parameters (e.g., stopping conditions, thresholds for outlier rejection).

Paper organization. The paper starts by presenting a standard formulation and architecture for SLAM (Section II). Section III tackles robustness in life-long SLAM. Section IV deals with scalability. Section V discusses how to represent the geometry of the environment. Section VI extends the question of the environment representation to the modeling of semantic information. Section VII provides an overview of the current accomplishments on the theoretical aspects of SLAM. Section VIII broadens the discussion and reviews the active SLAM problem in which decision making is used to improve the quality of the SLAM results. Section IX provides an overview of recent trends in SLAM, including the use of unconventional sensors and deep learning. Section X provides final remarks. Throughout the paper, we provide many pointers to related work outside the robotics community. Despite its unique traits, SLAM is related to problems addressed in computer vision, computer graphics, and control theory, and cross-fertilization among these fields is a necessary condition to enable fast progress.

For the non-expert reader, we recommend to read Durrant-Whyte and Bailey's SLAM tutorials [7, 69] before delving in this position paper. The more experienced researchers can jump directly to the section of interest, where they will find a self-contained overview of the state of the art and open problems.

II. Anatomy of a Modern SLAM System

The architecture of a SLAM system includes two main components: the *front-end* and the *back-end*. The front-end abstracts sensor data into models that are amenable for estimation, while the back-end performs inference on the abstracted data produced by the front-end. This architecture is summarized in Fig. 2. We review both components, starting from the back-end.

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图 1：左：根据里程计构建的地图。该地图与从起始位置 A 到最终位置 B 的长走廊同伦。现实中接近的点（例如 B 和 C）在里程地图中可能任意远。右：从 SLAM 构建地图。通过利用闭环，SLAM 估计环境的实际拓扑，并“发现”地图中的捷径。
强大的计算资源；

- 环境：平面或三维、自然或人工地标的存在、动态元素的数量、对称性的数量以及感知混叠的风险。请注意，其中许多方面实际上取决于传感器-环境对：例如，对于 2D 激光扫描仪来说，两个房间可能看起来相同（感知混叠），而相机可以通过外观线索来辨别它们；

- 性能要求：机器人状态估计所需的精度、环境表示的精度和类型（例如，基于地标的或密集的）、成功率（满足精度界限的测试的百分比）、估计延迟、最大操作时间、映射区域的最大尺寸。

例如，使用配备轮式编码器和激光扫描仪的机器人绘制二维室内环境，具有足够的精度（< 10cm）和足够的鲁棒性（例如，低故障率），可以认为已经在很大程度上解决了（执行SLAM的工业系统的一个例子是_Kuka导航解决方案_[145]）。同样，基于视觉的慢速移动机器人（例如火星探测器[166]、家用机器人[114]）和视觉惯性里程计[94]可以被认为是成熟的研究领域。

另一方面，其他机器人/环境/性能组合仍然值得进行大量的基础研究。当机器人的运动或环境过于具有挑战性时（例如快速的机器人动力学、高度动态的环境），当前的 SLAM 算法很容易失败；同样，SLAM算法通常无法满足严格的性能要求，例如快速闭环控制的高速率估计。这项调查将对这些未决问题等进行全面概述。

在本文中，我们认为我们正在进入 SLAM 的第三个时代，即“鲁棒感知时代”，其特点是以下关键要求：

- 1) 鲁棒性能：SLAM系统在广泛的环境中长时间运行，故障率低；该系统包括故障安全机制并具有自调整功能1，可以根据场景调整系统参数的选择。

- 2) 高层理解：SLAM 系统超越基本几何重建，以获得对环境的高层理解（例如，高层几何、

> 1SLAM 社区在很大程度上受到“手动调整的诅咒”的影响，因为通过专家调整系统参数（例如停止条件、异常值拒绝阈值）才能实现令人满意的操作。

图 2：典型 SLAM 系统中的前端和后端。后端可以向前端提供反馈，用于闭环检测和验证。

语义、物理、可供性（Affordances））；

- 3)资源感知：SLAM系统针对可用的感知和计算资源进行定制，并提供根据可用资源调整计算负载的方法；

- 4）任务驱动的感知：SLAM系统能够选择相关的感知信息并过滤掉不相关的传感器数据，以支持机器人必须执行的任务；此外，SLAM 系统会生成自适应地图表示，其复杂性可能会根据手头的任务而有所不同。

论文组织。本文首先介绍 SLAM 的标准制定和架构（第二部分）。第三节讨论了终身 SLAM 的鲁棒性。第四节讨论可扩展性。第五节讨论如何表示环境的几何形状。第六节将环境表示问题扩展到语义信息的建模。第七节概述了 SLAM 理论方面的当前成就。第八节扩大了讨论并回顾了主动 SLAM 问题，其中决策用于提高 SLAM 结果的质量。第九节概述了 SLAM 的最新趋势，包括非常规传感器和深度学习的使用。第 X 节提供了最后的评论。在整篇论文中，我们提供了许多机器人社区之外相关工作的指导。尽管SLAM

具有独特的特点，但它与计算机视觉、计算机图形学和控制理论中解决的问题有关，这些领域之间的交叉融合是实现快速进步的必要条件。

对于非专家读者，我们建议在深入研究本立场文件之前阅读 DurrantWhyte 和 Bailey 的 SLAM 教程 [7, 69]。经验丰富的研究人员可以直接跳到感兴趣的部分，在那里他们会找到对最新技术和开放问题的独立概述。

二.现代 SLAM 系统剖析

SLAM 系统的架构包括两个主要组件：前端和后端。前端将传感器数据抽象为适合估计的模型，而后端则对前端产生的抽象数据进行推理。图 2 总结了该架构。我们从后端开始回顾这两个组件。

Maximum a posteriori (MAP) estimation and the SLAM back-end. The current *de-facto* standard formulation of SLAM has its origins in the seminal paper of Lu and Milios [161], followed by the work of Gutmann and Konolige [101]. Since then, numerous approaches have improved the efficiency and robustness of the optimization underlying the problem [63, 81, 100, 125, 192, 241]. All these approaches formulate SLAM as a maximum a posteriori estimation problem, and often use the formalism of *factor graphs* [143] to reason about the interdependence among variables.

Assume that we want to estimate an unknown variable $\mathcal{X}$; as mentioned before, in SLAM the variable $\mathcal{X}$ typically includes the trajectory of the robot (as a discrete set of poses) and the position of landmarks in the environment. We are given a set of measurements $Z = \{z_k : k = 1, \dots, m\}$ such that each measurement can be expressed as a function of $\mathcal{X}$, i.e., $z_k = h_k(\mathcal{X}_k) + \epsilon_k$, where $\mathcal{X}_k \subseteq \mathcal{X}$ is a subset of the variables, $h_k(\cdot)$ is a known function (the *measurement* or *observation* model), and ϵ_k is random measurement noise.

In MAP estimation, we estimate $\mathcal{X}$ by computing the assignment of variables $\mathcal{X}^\star$ that attains the maximum of the posterior $\mathrm{p}(\mathcal{X}|Z)$ (the *belief* over $\mathcal{X}$ given the measurements):

$$\mathcal{X}^\star \doteq \operatorname*{argmax}_{\mathcal{X}} \mathrm{p}(\mathcal{X}|Z) = \operatorname*{argmax}_{\mathcal{X}} \mathrm{p}(Z|\mathcal{X})\mathrm{p}(\mathcal{X}) \quad (1)$$

where the equality follows from the Bayes theorem. In (1), $\mathrm{p}(Z|\mathcal{X})$ is the likelihood of the measurements Z given the assignment $\mathcal{X}$, and $\mathrm{p}(\mathcal{X})$ is a prior probability over $\mathcal{X}$. The prior probability includes any prior knowledge about $\mathcal{X}$; in case no prior knowledge is available, $\mathrm{p}(\mathcal{X})$ becomes a constant (uniform distribution) which is inconsequential and can be dropped from the optimization. In that case MAP estimation reduces to *maximum likelihood estimation*. Note that, unlike Kalman filtering, MAP estimation does not require an explicit distinction between motion and observation model: both models are treated as factors and are seamlessly incorporated in the estimation process. Moreover, it is worth noting that Kalman filtering and MAP estimation return the same estimate in the linear Gaussian case, while this is not the case in general.

Assuming that the measurements Z are independent (i.e., the corresponding noises are uncorrelated), problem (1) factorizes into:

$$\mathcal{X}^\star = \operatorname*{argmax}_{\mathcal{X}} \mathrm{p}(\mathcal{X}) \prod_{k=1}^{m} \mathrm{p}(z_k|\mathcal{X}) = \operatorname*{argmax}_{\mathcal{X}} \mathrm{p}(\mathcal{X}) \prod_{k=1}^{m} \mathrm{p}(z_k|\mathcal{X}_k) \quad (2)$$

where, on the right-hand-side, we noticed that z_k only depends on the subset of variables in $\mathcal{X}_k$.

Problem (2) can be interpreted in terms of inference over a factors graph [143]. The variables correspond to nodes in the factor graph. The terms $\mathrm{p}(z_k|\mathcal{X}_k)$ and the prior $\mathrm{p}(\mathcal{X})$ are called *factors*, and they encode probabilistic constraints over a subset of nodes. A factor graph is a graphical model that encodes the dependence between the k-th factor (and its measurement z_k) and the corresponding variables $\mathcal{X}_k$. A first advantage of the factor graph interpretation is that it enables an insightful visualization of the problem. Fig. 3 shows an example of a factor graph underlying a simple SLAM problem. The figure shows the variables, namely, the robot poses, the landmark positions, and the camera calibration parameters, and the factors imposing constraints among these variables. A second advantage is generality: a factor graph can model complex inference problems with heterogeneous variables and factors, and arbitrary interconnections. Furthermore, the connectivity of the factor graph in turn influences the sparsity of the resulting SLAM problem as discussed below.

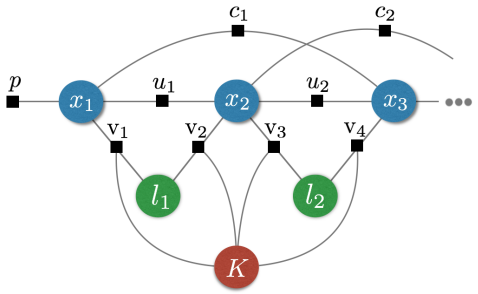


Fig. 3: **SLAM as a factor graph**: Blue circles denote robot poses at consecutive time steps $(x_1, x_2, \dots)$, green circles denote landmark positions $(l_1, l_2, \dots)$, red circle denotes the variable associated with the intrinsic calibration parameters (K). Factors are shown as black squares: the label "u" marks factors corresponding to odometry constraints, "v" marks factors corresponding to camera observations, "c" denotes loop closures, and "p" denotes prior factors.

In order to write (2) in a more explicit form, assume that the measurement noise ϵ_k is a zero-mean Gaussian noise with information matrix Ω_k (inverse of the covariance matrix). Then, the measurement likelihood in (2) becomes:

$$\mathrm{p}(z_k|\mathcal{X}_k) \propto \exp(-\frac{1}{2}||h_k(\mathcal{X}_k) - z_k||^2_{\Omega_k}) \quad (3)$$

where we use the notation $||e||^2_\Omega = e^\mathsf{T}\Omega e$. Similarly, assume that the prior can be written as: $\mathrm{p}(\mathcal{X}) \propto \exp(-\frac{1}{2}||h_0(\mathcal{X}) - z_0||^2_{\Omega_0})$, for some given function $h_0(\cdot)$, prior mean z_0, and information matrix Ω_0. Since maximizing the posterior is the same as minimizing the *negative log-posterior*, the MAP estimate in (2) becomes:

$$\mathcal{X}^\star = \operatorname*{argmin}_{\mathcal{X}} -\log\left(\mathrm{p}(\mathcal{X}) \prod_{k=1}^{m} \mathrm{p}(z_k|\mathcal{X}_k)\right) = \operatorname*{argmin}_{\mathcal{X}} \sum_{k=0}^{m} ||h_k(\mathcal{X}_k) - z_k||^2_{\Omega_k} \quad (4)$$

which is a nonlinear least squares problem, as in most problems of interest in robotics, $h_k(\cdot)$ is a nonlinear function. Note that the formulation (4) follows from the assumption of Normally distributed noise. Other assumptions for the noise distribution lead to different cost functions; for instance, if the noise follows a Laplace distribution, the squared ℓ_2-norm in (4) is replaced by the ℓ_1-norm. To increase resilience to outliers, it is also common to substitute the squared ℓ_2-norm in (4) with robust loss functions (e.g., Huber or Tukey loss) [112].

The computer vision expert may notice a resemblance between problem (4) and *bundle adjustment* (BA) in Structure from Motion [244]; both (4) and BA indeed stem from a maximum a posteriori formulation. However, two key features

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最大后验 (MAP) 估计和 SLAM 后端。当前 SLAM 的事实上的标准制定起源于 Lu 和 Milios [161] 的开创性论文，以及 Gutmann 和 Konolige [101] 的工作。从那时起，许多方法提高了问题基础优化的效率和鲁棒性[63,81,100,125,192,241]。所有这些方法都将 SLAM 表述为最大后验估计问题，并且经常使用_因子图_[143]的形式来推理变量之间的相互依赖关系。

假设我们想要估计一个未知变量 X ；如前所述，在 SLAM 中，变量 X 通常包括机器人的轨迹（作为一组离散的姿势）和环境中地标的位置。我们得到一组测量值 Z = {zk : k = 1 , 。 。 。 , m} 使得每个测量值可以表示为 X 的函数，即 zk = hk (Xk)+ □k ，其中 Xk ⊆X 是变量的子集， hk (·) 是已知函数（测量 或 观察 模型）， □k 是随机测量噪声。

在 MAP 估计中，我们通过计算变量 _X__⋆_ 的分配来估计 X，该变量获得后验 p(X|Z) 的最大值（给定测量值的_X_上的_信念_）：

X ⋆ = . argmax p(X|Z) = argmax p(Z|X)p(X) (1)
X X

其中等式由贝叶斯定理得出。在 (1) 中，p(Z|X) 是给定分配 X 的情况下测量 Z 的可能性，p(X) 是 X 的先验概率。先验概率包括关于 X 的任何先验知识；如果没有先验知识，p(X) 就会变成一个常数（均匀分布），这是无关紧要的，可以从优化中删除。在这种情况下，MAP 估计减少为_最大似然估计_。请注意，与卡尔曼滤波不同，MAP 估计不需要运动模型和观测模型之间的明确区别：这两个模型都被视为因素，并且无缝地合并到估计过程中。此外，值得注意的是，卡尔曼滤波和 MAP 估计在线性高斯情况下返回相同的估计，但一般情况并非如此。

假设测量_Z_是独立的（即相应的噪声不相关），问题（1）分解为：

m
X ⋆ = argmax p(X) ∏ p(zk|X) =
X
k =1
m
argmax p(X) ∏ p(zk|Xk) (2)
X
k =1

在右侧，我们注意到 zk 仅依赖于 Xk 中的变量子集。

问题（2）可以用因子图的推理来解释[143]。变量对应于因子图中的节点。项 p(zk|Xk) 和先验 p(X) 称为 factors ，它们对节点子集上的概率约束进行编码。因子图是一种图形模型，对第_k_ 个因子（及其测量_zk_ ）与相应变量_Xk_ 之间的依赖性进行编码。因子图解释的第一个优点是它可以提供富有洞察力的分析

无花果

。图 3：SLAM 作为因子图：蓝色圆圈表示连续时间步长的机器人姿势 (x 1 , x 2 , . . .)，绿色圆圈表示地标位置 (l 1 , l 2 , . . .)，红色圆圈表示与内在校准参数相关的变量 (K)。因子显示为黑色方块：标签“u”标记与里程计约束相对应的因子，“v”标记与摄像机观察相对应的因子，“c”表示闭环，“p”表示先验因子。

问题的可视化。图 3 显示了简单 SLAM 问题背后的因子图示例。图中显示了变量，即机器人位姿、地标位置和相机标定参数，以及这些变量之间施加约束的因素。第二个优点是通用性：因子图可以对具有异构变量和因子以及任意互连的复杂推理问题进行建模。此外，因子图的连通性反过来又会影响所得 SLAM 问题的稀疏性，如下所述。

为了以更明确的形式写出(2)，假设测量噪声_□k_是具有信息矩阵Ω_k_（协方差矩阵的逆矩阵）的零均值高斯噪声。那么，(2)中的测量似然变为：

p(zk|Xk) ∝ exp(− 1 2 ||hk (Xk) − zk|| Ω 2 k) (3)

其中我们使用符号 _||e||_2 Ω=_e_TΩ_e_。类似地，假设先验可以写为： p(X) __∝_ exp(__−<u>1</u> 2||h_0(X)− z 0 _||_2 Ω0)，对于某个给定函数_h_0(·)，先验平均值_z_0，以及信息矩阵Ω0。由于最大化后验与最小化_负对数后验_相同，因此（2）中的 MAP 估计变为：

m
=
X ⋆ = argmin − log p(X) − p(zk|Xk)
X
k =1
m
argmin ||hk (Xk) − zk|| 2 Ω k (4)
X
k =0

这是一个非线性最小二乘问题，就像在机器人学中大多数感兴趣的问题中一样，hk (·) 是一个非线性函数。请注意，公式 (4) 是根据正态分布噪声的假设得出的。噪声分布的其他假设会导致不同的成本函数；例如，如果噪声服从拉普拉斯分布，则 (4) 中的平方 ℓ 2-范数将替换为 ℓ 1-范数。为了提高对异常值的恢复能力，通常用稳健的损失函数（例如 Huber 或 Tukey 损失）替代 (4) 中的平方 ℓ 2-范数 [112]。

计算机视觉专家可能会注意到问题（4）和Structure from Motion [244]中的_捆绑调整_（BA）之间的相似之处； (4) 和 BA 确实都源于最大后验公式。然而，有两个关键特征

make SLAM unique. First, the factors in (4) are not constrained to model projective geometry as in BA, but include a broad variety of sensor models, e.g., inertial sensors, wheel encoders, GPS, to mention a few. For instance, in laser-based mapping, the factors usually constrain relative poses corresponding to different viewpoints, while in direct methods for visual SLAM, the factors penalize differences in pixel intensities across different views of the same portion of the scene. The second difference with respect to BA is that, in a SLAM scenario, problem (4) typically needs to be solved *incrementally*: new measurements are made available at each time step as the robot moves.

The minimization problem (4) is commonly solved via successive linearizations, e.g., the Gauss-Newton or the Levenberg-Marquardt methods (alternative approaches, based on convex relaxations and Lagrangian duality are reviewed in Section VII). Successive linearization methods proceed iteratively, starting from a given initial guess $\hat{\mathcal{X}}$, and approximate the cost function at $\hat{\mathcal{X}}$ with a quadratic cost, which can be optimized in closed form by solving a set of linear equations (the so called *normal equations*). These approaches can be seamlessly generalized to variables belonging to smooth manifolds (e.g., rotations), which are of interest in robotics [1, 82].

The key insight behind modern SLAM solvers is that the matrix appearing in the normal equations is sparse and its sparsity structure is dictated by the topology of the underlying factor graph. This enables the use of fast linear solvers [125, 126, 146, 204]. Moreover, it allows designing *incremental* (or *online*) solvers, which update the estimate of $\mathcal{X}$ as new observations are acquired [125, 126, 204]. Current SLAM libraries (e.g., GTSAM [61], g2o [146], Ceres [214], iSAM [126], and SLAM++ [204]) are able to solve problems with tens of thousands of variables in few seconds. The hands-on tutorials [61, 97] provide excellent introductions to two of the most popular SLAM libraries; each library also includes a set of examples showcasing real SLAM problems.

The SLAM formulation described so far is commonly referred to as *maximum a posteriori* estimation, *factor graph optimization*, *graph-SLAM*, *full smoothing*, or *smoothing and mapping* (SAM). A popular variation of this framework is *pose graph optimization*, in which the variables to be estimated are poses sampled along the trajectory of the robot, and each factor imposes a constraint on a pair of poses.

MAP estimation has been proven to be more accurate and efficient than original approaches for SLAM based on nonlinear filtering. We refer the reader to the surveys [7, 69] for an overview on filtering approaches, and to [236] for a comparison between filtering and smoothing. We remark that some SLAM systems based on EKF have also been demonstrated to attain state-of-the-art performance. Excellent examples of EKF-based SLAM systems include the Multi-State Constraint Kalman Filter of Mourikis and Roumeliotis [175], and the VIN systems of Kottas *et al.* [139] and Hesch *et al.* [105]. Not surprisingly, the performance mismatch between filtering and MAP estimation gets smaller when the linearization point for the EKF is accurate (as it happens in visual-inertial navigation problems), when using sliding-window filters, and when potential sources of inconsistency in the EKF are taken care of [105, 108, 139].

As discussed in the next section, MAP estimation is usually performed on a pre-processed version of the sensor data. In this regard, it is often referred to as the SLAM *back-end*.

Sensor-dependent SLAM front-end. In practical robotics applications, it might be hard to write directly the sensor measurements as an analytic function of the state, as required in MAP estimation. For instance, if the raw sensor data is an image, it might be hard to express the intensity of each pixel as a function of the SLAM state; the same difficulty arises with simpler sensors (e.g., a laser with a single beam). In both cases the issue is connected with the fact that we are not able to design a sufficiently general, yet tractable representation of the environment; even in the presence of such a general representation, it would be hard to write an analytic function that connects the measurements to the parameters of such a representation.

For this reason, before the SLAM back-end, it is common to have a module, the *front-end*, that extracts relevant features from the sensor data. For instance, in vision-based SLAM, the front-end extracts the pixel location of few distinguishable points in the environment; pixel observations of these points are now easy to model within the back-end. The front-end is also in charge of associating each measurement to a specific landmark (say, 3D point) in the environment: this is the so called *data association*. More abstractly, the data association module associates each measurement z_k with a subset of unknown variables $\mathcal{X}_k$ such that $z_k = h_k(\mathcal{X}_k) + \epsilon_k$. Finally, the front-end might also provide an initial guess for the variables in the nonlinear optimization (4). For instance, in feature-based monocular SLAM the front-end usually takes care of the landmark initialization, by triangulating the position of the landmark from multiple views.

A pictorial representation of a typical SLAM system is given in Fig. 2. The data association module in the front-end includes a short-term data association block and a long-term one. Short-term data association is responsible for associating corresponding features in consecutive sensor measurements; for instance, short-term data association would track the fact that 2 pixel measurements in consecutive frames are picturing the same 3D point. On the other hand, long-term data association (or loop closure) is in charge of associating new measurements to older landmarks. We remark that the back-end usually feeds back information to the front-end, e.g., to support loop closure detection and validation.

The pre-processing that happens in the front-end is sensor dependent, since the notion of *feature* changes depending on the input data stream we consider.

III. Long-term autonomy I: Robustness

A SLAM system might be fragile in many aspects: failure can be algorithmic[2] or hardware-related. The former class includes failure modes induced by limitation of the existing SLAM algorithms (e.g., difficulty to handle extremely dynamic or harsh environments). The latter includes failures due

[2]We omit the (large) class of software-related failures. The non-expert reader must be aware that integration and testing are key aspects of SLAM and robotics in general.

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让SLAM独一无二。首先，(4) 中的因素并不局限于 BA 中的射影几何模型，而是包括各种传感器模型，例如惯性传感器、车轮编码器、GPS 等等。例如，在基于激光的测绘中，这些因素通常会限制与不同视点相对应的相对姿势，而在视觉 SLAM 的直接方法中，这些因素会惩罚场景同一部分的不同视图之间的像素强度差异。与 BA 相关的第二个区别是，在 SLAM 场景中，问题 (4) 通常需要增量解决：随着机器人移动，每个时间步都会提供新的测量结果。

最小化问题（4）通常通过连续线性化来解决，例如高斯-牛顿或莱文伯格-马夸特方法（第七节回顾了基于凸松弛和拉格朗日对偶性的替代方法）。连续的线性化方法从给定的初始猜测 _X_ˆ 开始迭代进行，并以二次成本逼近 _X_ˆ 处的成本函数，可以通过求解一组线性方程（所谓的_正规方程_）以封闭形式进行优化。这些方法可以无缝地推广到属于平滑流形的变量（例如旋转），这在机器人学中很有趣[1, 82]。

现代 SLAM 求解器背后的关键见解是，正规方程中出现的矩阵是稀疏的，其稀疏结构由底层因子图的拓扑决定。这使得可以使用快速线性求解器[125、126、146、204]。此外，它允许设计_增量_（或_在线_）求解器，当获取新的观测值时更新_X_的估计[125,126,204]。当前的 SLAM 库（例如 GTSAM [61]、g2o [146]、Ceres [214]、iSAM [126] 和 SLAM++ [204]）能够在几秒钟内解决具有数万个变量的问题。实践教程 [61, 97] 提供了对两个最流行的 SLAM 库的精彩介绍；每个库还包含一组展示真实 SLAM 问题的示例。

迄今为止描述的 SLAM 公式通常被称为“最大后验估计”、“因子图优化”、“图 SLAM”、“完全平滑”或“平滑和映射”（SAM）。该框架的一个流行变体是_位姿图优化_，其中要估计的变量是沿机器人轨迹采样的位姿，每个因素对一对位姿施加约束。

MAP 估计已被证明比基于非线性滤波的 SLAM 原始方法更加准确和高效。我们建议读者参考调查 [7, 69] 以了解过滤方法的概述，并参考 [236] 以了解过滤和平滑之间的比较。我们注意到，一些基于 EKF 的 SLAM 系统也已被证明能够达到最先进的性能。基于 EKF 的 SLAM 系统的优秀示例包括 Mourikis 和 Roumeliotis 的多状态约束卡尔曼滤波器 [175]，以及 Kottas et al. [139] 和 Hesch et al. [105] 的 VIN 系统

。毫不奇怪，当 EKF 的线性化点准确时（如在视觉惯性导航问题中发生的情况）、使用滑动窗口滤波器时以及当潜在的不一致来源时，滤波和 MAP 估计之间的性能失配会变得更小。

EKF 得到处理[105,108,139]。

如下一节所述，MAP 估计通常是在传感器数据的预处理版本上执行的。在这方面，它通常被称为 SLAM_back-end_。

依赖于传感器的 SLAM 前端。在实际的机器人应用中，可能很难按照 MAP 估计的要求直接将传感器测量结果写为状态的解析函数。例如，如果原始传感器数据是图像，则可能很难将每个像素的强度表示为 SLAM 状态的函数；更简单的传感器（例如单光束激光器）也会出现同样的困难。在这两种情况下，问题都与我们无法设计一个足够普遍但易于处理的环境表示有关。即使存在这样的一般表示，也很难编写将测量结果与这种表示的参数连接起来的分析函数。

因此，在 SLAM 后端之前，通常有一个模块，即 front-end ，用于从传感器数据中提取相关特征。例如，在基于视觉的SLAM中，前端提取环境中少数可区分点的像素位置；现在可以轻松在后端对这些点的像素观察进行建模。前端还负责将每个测量值与环境中的特定地标（例如 3D 点）相关联：这就是所谓的_数据关联_。更抽象地，数据关联模块将每个测量_zk_与未知变量_Xk_的子集相关联，使得_zk_=hk(Xk)+□k。

最后，前端还可以为非线性优化中的变量提供初始猜测 (4)。例如，在基于特征的单目 SLAM 中，前端通常通过从多个视图对地标的位置进行三角测量来处理地标初始化。

图2给出了典型SLAM系统的图示。前端的数据关联模块包括短期数据关联块和长期数据关联块。短期数据关联负责关联连续传感器测量中的相应特征；例如，短期数据关联将跟踪连续帧中的 2 个像素测量描绘相同 3D 点的事实。另一方面，长期数据关联（或闭环）负责将新测量值与旧地标相关联。我们注意到后端通常会向前端反馈信息，例如支持回环检测和验证。

前端发生的预处理取决于传感器，因为 feature 的概念会根据我们考虑的输入数据流而变化。
三．长期自主性 I：稳健性

SLAM 系统在许多方面可能都很脆弱：故障可能是算法2或与硬件相关

。前一类包括由现有 SLAM 算法的限制引起的故障模式（例如，难以处理极端动态或恶劣的环境）。后者包括由于

> 2我们忽略了（大量）与软件相关的故障。非专业读者必须意识到，集成和测试通常是 SLAM 和机器人技术的关键方面。

to sensor or actuator degradation. Explicitly addressing these failure modes is crucial for long-term operation, where one can no longer make simplifying assumptions about the structure of the environment (e.g., mostly static) or fully rely on on-board sensors. In this section we review the main challenges to algorithmic robustness. We then discuss open problems, including robustness against hardware-related failures.

One of the main sources of algorithmic failures is data association. As mentioned in Section II data association matches each measurement to the portion of the state the measurement refers to. For instance, in feature-based visual SLAM, it associates each visual feature to a specific landmark. Perceptual aliasing, the phenomenon in which different sensory inputs lead to the same sensor signature, makes this problem particularly hard. In the presence of perceptual aliasing, data association establishes erroneous measurement-state matches (outliers, or false positives), which in turn result in wrong estimates from the back-end. On the other hand, when data association decides to incorrectly reject a sensor measurement as spurious (false negatives), fewer measurements are used for estimation, at the expense of estimation accuracy.

The situation is made worse by the presence of unmodeled dynamics in the environment including both short-term and seasonal changes, which might deceive short-term and long-term data association. A fairly common assumption in current SLAM approaches is that the world remains unchanged as the robot moves through it (in other words, landmarks are static). This *static world assumption* holds true in a single mapping run in small scale scenarios, as long as there are no *short term dynamics* (e.g., people and objects moving around). When mapping over longer time scales and in large environments, change is inevitable.

Another aspect of robustness is that of doing SLAM in harsh environments such as underwater [73, 131]. The challenges in this case are the limited visibility, the constantly changing conditions, and the impossibility of using conventional sensors (e.g., laser range finder).

Brief Survey. Robustness issues connected to incorrect data association can be addressed in the front-end and/or in the back-end of a SLAM system. Traditionally, the front-end has been entrusted with establishing correct data association. Short-term data association is the easier one to tackle: if the sampling rate of the sensor is relatively fast, compared to the dynamics of the robot, tracking features that correspond to the same 3D landmark is easy. For instance, if we want to track a 3D point across consecutive images and assuming that the framerate is sufficiently high, standard approaches based on descriptor matching or optical flow [218] ensure reliable tracking. Intuitively, at high framerate, the viewpoint of the sensor (camera, laser) does not change significantly, hence the features at time $t+1$ (and its appearance) remain close to the ones observed at time t.[3] Long-term data association in the front-end is more challenging and involves loop closure *detection* and *validation*. For loop closure detection at the front-end, a brute-force approach which detects features in the current measurement (e.g., image) and tries to match them against all previously detected features quickly becomes impractical. Bag-of-words models [226] avoid this intractability by quantizing the feature space and allowing more efficient search. Bag-of-words can be arranged into hierarchical vocabulary trees [189] that enable efficient lookup in large-scale datasets. Bag-of-words-based techniques such as [53] have shown reliable performance on the task of single session loop closure detection. However, these approaches are not capable of handling severe illumination variations as visual words can no longer be matched. This has led to develop new methods that explicitly account for such variations by matching sequences [173], gathering different visual appearances into a unified representation [48], or using spatial as well as appearance information [106]. A detailed survey on visual place recognition can be found in Lowry *et al.* [160]. Feature-based methods have also been used to detect loop closures in laser-based SLAM front-ends; for instance, Tipaldi *et al.* [242] propose FLIRT features for 2D laser scans.

Loop closure validation, instead, consists of additional geometric verification steps to ascertain the quality of the loop closure. In vision-based applications, RANSAC is commonly used for geometric verification and outlier rejection, see [218] and the references therein. In laser-based approaches, one can validate a loop closure by checking how well the current laser scan matches the existing map (i.e., how small is the residual error resulting from scan matching).

Despite the progress made to robustify loop closure detection at the front-end, in presence of perceptual aliasing, it is unavoidable that wrong loop closures are fed to the back-end. Wrong loop closures can severely corrupt the quality of the MAP estimate [238]. In order to deal with this issue, a recent line of research [33, 150, 191, 238] proposes techniques to make the SLAM back-end resilient against spurious measurements. These methods reason on the validity of loop closure constraints by looking at the residual error induced by the constraints during optimization. Other methods, instead, attempt to detect outliers *a priori*, that is, before any optimization takes place, by identifying incorrect loop closures that are not supported by the odometry [215].

In dynamic environments, the challenge is twofold. First, the SLAM system has to detect, discard, or track changes. While mainstream approaches attempt to discard the dynamic portion of the scene [180], some works include dynamic elements as part of the model [11, 253]. The second challenge regards the fact that the SLAM system has to model permanent or semi-permanent changes, and understand how and when to update the map accordingly. Current SLAM systems that deal with dynamics either maintain multiple (time-dependent) maps of the same location [60], or have a single representation parameterized by some time-varying parameter [140].

Open Problems. In this section we review open problems and novel research questions arising in long-term SLAM.

Failsafe SLAM and recovery: Despite the progress made on the SLAM back-end, current SLAM solvers are still vulnerable in the presence of outliers. This is mainly due to the fact that virtually all robust SLAM techniques are based on iterative optimization of nonconvex costs. This has two consequences:

[3] In hindsight, the fact that short-term data association is much easier and more reliable than the long-term one is what makes (visual, inertial) odometry simpler than SLAM.

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导致传感器或执行器退化。明确解决这些故障模式对于长期运行至关重要，因为人们不能再对环境结构（例如，大部分是静态的）做出简化的假设或完全依赖机载传感器。在本节中，我们回顾算法鲁棒性的主要挑战。然后我们讨论开放问题，包括针对硬件相关故障的鲁棒性。

算法失败的主要来源之一是数据关联。正如第二节中提到的，数据关联将每个测量值与测量值所指的状态部分相匹配。例如，在基于特征的视觉 SLAM 中，它将每个视觉特征与特定的地标相关联。感知混叠是一种不同的感官输入导致相同的传感器特征的现象，使得这个问题变得特别困难。在存在感知混叠的情况下，数据关联会建立错误的测量状态匹配（异常值或误报），进而导致后端的错误估计。另一方面，当数据关联决定错误地拒绝传感器测量作为虚假（假阴性）时，用于估计的测量会减少，但代价是估计准确性。

由于环境中存在未建模的动态（包括短期和季节性变化），情况变得更糟，这可能会欺骗短期和长期数据关联。当前 SLAM 方法中一个相当常见的假设是，当机器人在其中移动时，世界保持不变（换句话说，地标是静态的）。只要不存在“短期动态”（例如，人员和物体四处移动），这种“静态世界假设”在小规模场景中运行的单个映射中就成立。当在更长的时间尺度和更大的环境中绘制地图时，变化是不可避免的。

鲁棒性的另一个方面是在水下等恶劣环境中进行 SLAM [73, 131]。这种情况下的挑战是可见度有限、条件不断变化以及无法使用传统传感器（例如激光测距仪）。

简短调查。与不正确的数据关联相关的鲁棒性问题可以在 SLAM 系统的前端和/或后端得到解决。传统上，前端被委托建立正确的数据关联。短期数据关联是更容易解决的问题：如果传感器的采样率相对较快，与机器人的动力学相比，跟踪与相同 3D 地标相对应的特征就很容易。例如，如果我们想要跟踪连续图像上的 3D 点并假设帧速率足够高，基于描述符匹配或光流 [218] 的标准方法可以确保可靠的跟踪。直观上，在高帧速率下，传感器（相机、激光）的视点不会发生显着变化，因此 t + 1 时刻的特征（及其外观）仍然接近于 t 时刻观察到的特征。3前端的长期数据关联更具挑战性，涉及闭环_检测_和_验证_。对于前端的闭环检测，采用暴力方法检测特征

> 3事后看来，短期数据关联比长期数据关联更容易、更可靠，这一事实使得（视觉、惯性）里程计比 SLAM 更简单。

当前测量（例如图像）并尝试将它们与所有先前检测到的特征进行匹配很快就变得不切实际。词袋模型 [226]通过量化特征空间并允许更有效的搜索来避免这种棘手的问题。词袋可以排列成分层词汇树[189]，从而能够在大规模数据集中进行高效查找。基于词袋的技术（例如[53]）已经在单会话循环闭合检测任务上表现出了可靠的性能。然而，这些方法无法处理严重的照明变化，因为视觉单词不再匹配。这导致开发出新的方法，通过匹配序列[173]、将不同的视觉外观收集到统一的表示中[48]或使用空间和外观信息[106]来明确地解释这种变化。关于视觉位置识别的详细调查可以在 Lowry et al. [160] 中找到。基于特征的方法也被用来检测基于激光的 SLAM 前端的闭环；例如，Tipaldi et al. [242] 提出了用于 2D 激光扫描的 FLIRT 特征。

相反，环路闭合验证包括额外的几何验证步骤，以确定环路闭合的质量。在基于视觉的应用中，RANSAC 通常用于几何验证和异常值拒绝，请参阅[218]及其中的参考文献。在基于激光的方法中，可以通过检查当前激光扫描与现有地图的匹配程度（即扫描匹配产生的残余误差有多小）来验证环路闭合。

尽管在增强前端的回环检测方面取得了进展，但在存在感知混叠的情况下，错误的回环被馈送到后端是不可避免的。错误的闭环可能会严重影响 MAP 估计的质量 [238]。为了解决这个问题，最近的一系列研究

[33,150,191,238] 提出了使 SLAM 后端能够抵抗虚假测量的技术。这些方法通过查看优化过程中约束引起的残余误差来推理闭环约束的有效性。相反，其他方法尝试先验地检测异常值，即在进行任何优化之前，通过识别里程计不支持的不正确的闭环来检测异常值[215]。

在动态环境中，挑战是双重的。首先，SLAM 系统必须检测、丢弃或跟踪变化。虽然主流方法试图放弃场景的动态部分 [180]，但一些作品将动态元素作为模型的一部分 [11, 253]。第二个挑战是 SLAM 系统必须对永久或半永久变化进行建模，并了解如何以及何时相应地更新地图。当前处理动态的 SLAM 系统要么维护同一位置的多个（与时间相关的）地图 [60]，要么具有由某些时变参数参数化的单个表示形式 [140]。

未决问题

。在本节中，我们回顾长期 SLAM 中出现的开放性问题和新颖的研究问题。

_故障安全 SLAM 和恢复：_尽管 SLAM 后端取得了进展，但当前的 SLAM 求解器在存在异常值的情况下仍然容易受到攻击。这主要是因为几乎所有鲁棒 SLAM 技术都是基于非凸成本的迭代优化。这有两个后果：

first, the outlier rejection outcome depends on the quality of the initial guess fed to the optimization; second, the system is inherently fragile: the inclusion of a single outlier degrades the quality of the estimate, which in turn degrades the capability of discerning outliers later on. These types of failures lead to an incorrect linearization point from which recovery is not trivial, especially in an incremental setup. An ideal SLAM solution should be *fail-safe* and *failure-aware*, i.e., the system needs to be aware of imminent failure (e.g., due to outliers or degeneracies) and provide recovery mechanisms that can re-establish proper operation. None of the existing SLAM approaches provides these capabilities. A possible way to achieve this is a tighter integration between the front-end and the back-end, but how to achieve that is still an open question.

Robustness to HW failure: While addressing hardware failures might appear outside the scope of SLAM, these failures impact the SLAM system, and the latter can play a key role in detecting and mitigating sensor and locomotion failures. If the accuracy of a sensor degrades due to malfunctioning, off-nominal conditions, or aging, the quality of the sensor measurements (e.g., noise, bias) does not match the noise model used in the back-end (c.f. eq. (3)), leading to poor estimates. This naturally poses different research questions: how can we detect degraded sensor operation? how can we adjust sensor noise statistics (covariances, biases) accordingly? more generally, how do we resolve conflicting information from different sensors? This seems crucial in safety-critical applications (e.g., self-driving cars) in which misinterpretation of sensor data may put human life at risk.

Metric Relocalization: While appearance-based, as opposed to feature-based, methods are able to close loops between day and night sequences or between different seasons, the resulting loop closure is topological in nature. For metric relocalization (i.e., estimating the relative pose with respect to the previously built map), feature-based approaches are still the norm; however, current feature descriptors lack sufficient invariance to work reliably under such circumstances. Spatial information, inherent to the SLAM problem, such as trajectory matching, might be exploited to overcome these limitations. Additionally, mapping with one sensor modality (e.g., 3D lidar) and localizing in the same map with a different sensor modality (e.g., camera) can be a useful addition. The work of Wolcott *et al.* [260] is an initial step in this direction.

Time varying and deformable maps: Mainstream SLAM methods have been developed with the rigid and static world assumption in mind; however, the real world is non-rigid both due to dynamics as well as the inherent deformability of objects. An ideal SLAM solution should be able to reason about dynamics in the environment including non-rigidity, work over long time periods generating "all terrain" maps and be able to do so in real time. In the computer vision community, there have been several attempts since the 80s to recover shape from non-rigid objects but with restrictive applicability. Recent results in non-rigid SfM such as [91, 96] are less restrictive but only work in small scenarios. In the SLAM community, Newcombe *et al.* [182] have address the non-rigid case for small-scale reconstruction. However, addressing the problem of non-rigid maps at a large scale is still largely unexplored.

Automatic parameter tuning: SLAM systems (in particular, the data association modules) require extensive parameter tuning in order to work correctly for a given scenario. These parameters include thresholds that control feature matching, RANSAC parameters, and criteria to decide when to add new factors to the graph or when to trigger a loop closing algorithm to search for matches. If SLAM has to work "out of the box" in arbitrary scenarios, methods for automatic tuning of the involved parameters need to be considered.

IV. Long-term autonomy II: Scalability

While modern SLAM algorithms have been successfully demonstrated mostly in indoor building-scale environments, in many application endeavors, robots must operate for an extended period of time over larger areas. These applications include ocean exploration for environmental monitoring, non-stop cleaning robots in our ever changing cities, or large-scale precision agriculture. For such applications the size of the factor graph underlying SLAM can grow unbounded, due to the continuous exploration of new places and the increasing time of operation. In practice, the computational time and memory footprint are bounded by the resources of the robot. Therefore, it is important to design SLAM methods whose computational and memory complexity remains bounded.

In the worst-case, successive linearization methods based on *direct* linear solvers imply a memory consumption which grows quadratically in the number of variables. When using iterative linear solvers (e.g., the conjugate gradient [62]) the memory consumption grows linearly in the number of variables. The situation is further complicated by the fact that, when re-visiting a place multiple times, factor graph optimization becomes less efficient as nodes and edges are continuously added to the same spatial region, compromising the sparsity structure of the graph.

In this section we review some of the current approaches to control, or at least reduce, the growth of the size of the problem and discuss open challenges.

Brief Survey. We focus on two ways to reduce the complexity of factor graph optimization: (i) *sparsification methods*, which trade off information loss for memory and computational efficiency, and (ii) *out-of-core and multi-robot methods*, which split the computation among many robots/processors.

Node and edge sparsification: This family of methods addresses scalability by reducing the number of nodes *added* to the graph, or by *pruning* less "informative" nodes and factors. Ila *et al.* [115] use an information-theoretic approach to add only non-redundant nodes and highly-informative measurements to the graph. Johannsson *et al.* [120], when possible, avoid adding new nodes to the graph by inducing new constraints between existing nodes, such that the number of variables grows only with size of the explored space and not with the mapping duration. Kretzschmar *et al.* [141] propose an information-based criterion for determining which nodes to marginalize in pose graph optimization. Carlevaris-Bianco and Eustice [28], and Mazuran *et al.* [170] introduce the *Generic Linear Constraint* (GLC) factors and the *Nonlinear Graph Sparsification* (NGS) method, respectively. These methods

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首先，异常值拒绝结果取决于优化过程中初始猜测的质量；其次，该系统本质上是脆弱的：包含单个异常值会降低估计的质量，进而降低以后识别异常值的能力。这些类型的故障会导致不正确的线性化点，从该点进行恢复并非易事，尤其是在增量设置中。理想的 SLAM 解决方案应该是“故障安全”和“故障感知”，即系统需要意识到即将发生的故障（例如，由于异常值或退化）并提供可以重新建立正确操作的恢复机制。现有的 SLAM 方法均不提供这些功能。实现这一目标的一种可能方法是前端和后端之间更紧密的集成，但如何实现这一点仍然是一个悬而未决的问题。

_硬件故障的鲁棒性：_虽然解决硬件故障可能出现在 SLAM 范围之外，但这些故障会影响 SLAM 系统，而后者可以在检测和减轻传感器和运动故障方面发挥关键作用。如果传感器的精度由于故障、非标称条件或老化而降低，则传感器测量的质量（例如噪声、偏差）与后端使用的噪声模型不匹配（参见等式（3）），从而导致估计结果不佳。这自然会提出不同的研究问题：我们如何检测退化的传感器操作？我们如何相应地调整传感器噪声统计数据（协方差、偏差）？更一般地说，我们如何解决来自不同传感器的冲突信息？这在安全关键型应用（例如自动驾驶汽车）中似乎至关重要，在这些应用中，传感器数据的误解可能会危及人类生命。

_度量重定位：_与基于特征相反，基于外观的方法能够在白天和夜晚序列之间或不同季节之间闭合循环，但所得的循环闭合本质上是拓扑结构的。对于度量重定位（即估计相对于先前构建的地图的相对姿态），基于特征的方法仍然是常态；然而，当前的特征描述符缺乏足够的不变性来在这种情况下可靠地工作。SLAM 问题固有的空间信息（例如轨迹匹配）可以用来克服这些限制。此外，使用一种传感器模式（例如 3D 激光雷达）进行绘图并使用不同的传感器模式（例如相机）在同一地图中进行定位可能是一个有用的补充。 Wolcott et al. [260] 的工作是朝这个方向迈出的第一步。

_时变和可变形地图：_主流 SLAM 方法是在考虑到刚性和静态世界假设的情况下开发的；然而，由于动力学以及物体固有的可变形性，现实世界是非刚性的。理想的 SLAM 解决方案应该能够推理环境中的动态（包括非刚性）、长时间工作生成“全地形”地图，并且能够实时执行此操作。在计算机视觉领域，自 80 年代以来已经进行了多次尝试从非刚性物体恢复形状，但适用性有限

。非刚性 SfM 的最新结果（例如 [91, 96]）限制较少，但仅适用于小场景。在 SLAM 社区中，Newcombe et al. [182] 已经解决了小规模重建的非刚性情况。然而，解决大比例尺非刚性地图问题在很大程度上仍未得到探索。

自动参数调整： SLAM 系统（特别是数据关联模块）需要大量参数调整才能在给定场景下正常工作。这些参数包括控制特征匹配的阈值、RANSAC 参数以及决定何时向图中添加新因素或何时触发闭环算法来搜索匹配的标准。如果 SLAM 必须在任意场景下“开箱即用”，则需要考虑自动调整相关参数的方法。
四．长期自治 II：可扩展性

虽然现代 SLAM 算法主要在室内建筑规模环境中得到成功演示，但在许多应用努力中，机器人必须在更大的区域内长时间运行。这些应用包括用于环境监测的海洋探索、不断变化的城市中的不间断清洁机器人或大规模精准农业。对于此类应用程序，由于对新地点的不断探索和操作时间的增加，SLAM 底层的因子图的大小可以无限增长。实际上，计算时间和内存占用受到机器人资源的限制。因此，设计计算和内存复杂度保持有限的 SLAM 方法非常重要。

在最坏的情况下，基于_direct_线性求解器的连续线性化方法意味着内存消耗随着变量数量呈二次方增长。当使用迭代线性求解器（例如，共轭梯度[62]）时，内存消耗随着变量的数量线性增长。事实上，当多次重新访问一个地方时，由于节点和边不断添加到同一空间区域，因子图优化变得效率较低，从而损害了图的稀疏结构，因此情况变得更加复杂。

在本节中，我们回顾一些当前控制或至少减少问题规模增长的方法，并讨论开放的挑战。

简短调查。我们专注于降低因子图优化复杂性的两种方法：（i）稀疏方法，它以信息损失换取内存和计算效率，以及（ii）核外和多机器人方法，它将计算分配给许多机器人/处理器。

_节点和边稀疏化：_这一系列方法通过减少_添加_到图中的节点数量，或通过_修剪_更少的“信息丰富”节点和因子来解决可扩展性问题。 Ila et al. [115] 使用信息论方法仅向图中添加非冗余节点和高信息量测量。Johannsson et al. [120]，在可能的情况下，通过在现有节点之间引入新的约束来避免向图中添加新节点，这样变量的数量仅随着探索空间的大小而增长，而不是随着映射持续时间而增长

。 Kretzschmar_et al._[141]提出了一种基于信息的标准，用于确定在位姿图优化中边缘化哪些节点。Carlevaris-Bianco 和 Eustice [28] 以及 Mazuran et al. [170] 分别介绍了_通用线性约束_ (GLC) 因子和_非线性图稀疏化_ (NGS) 方法。这些方法

operate on the Markov blanket of a marginalized node and compute a sparse approximation of the blanket. Huang *et al.* [107] sparsify the Hessian matrix (arising in the normal equations) by solving an ℓ_1-regularized minimization problem.

Another line of work that allows reducing the number of parameters to be estimated over time is the *continuous-time trajectory estimation*. The first SLAM approach of this class was proposed by Bibby and Reid using cubic-splines to represent the continuous trajectory of the robot [12]. In their approach the nodes in the factor graph represented the control-points (knots) of the spline which were optimized in a sliding window fashion. Later, Furgale *et al.* [88] proposed the use of basis functions, particularly B-splines, to approximate the robot trajectory, within a batch-optimization formulation. Sliding-window B-spline formulations were also used in SLAM with rolling shutter cameras, with a landmark-based representation by Patron-Perez *et al.* [196] and with a semi-dense direct representation by Kim *et al.* [133]. More recently, Mueggler *et al.* [177] applied the continuous-time SLAM formulation to event-based cameras. Bosse *et al.* [21] extended the continuous 3D scan-matching formulation from [19] to a large-scale SLAM application. Later, Anderson *et al.* [4] and Dubé *et al.* [67] proposed more efficient implementations by using wavelets or sampling non-uniform knots over the trajectory, respectively. Tong *et al.* [243] changed the parametrization of the trajectory from basis curves to a Gaussian process representation, where nodes in the factor graph are actual robot poses and any other pose can be interpolated by computing the posterior mean at the given time. An expensive batch Gauss-Newton optimization is needed to solve for the states in this first proposal. Barfoot *et al.* [3] then proposed a Gaussian process with an exactly sparse inverse kernel that drastically reduces the computational time of the batch solution.

Out-of-core (parallel) SLAM: Parallel *out-of-core* algorithms for SLAM split the computation (and memory) load of factor graph optimization among multiple processors. The key idea is to divide the factor graph into different subgraphs and optimize the overall graph by alternating local optimization of each subgraph, with a global refinement. The corresponding approaches are often referred to as *submapping algorithms*, an idea that dates back to the initial attempts to tackle large-scale maps [18]. Ni *et al.* [187] and Zhao *et al.* [267] present submapping approaches for factor graph optimization, organizing the submaps in a binary tree structure. Grisetti *et al.* [98] propose a hierarchy of submaps: whenever an observation is acquired, the highest level of the hierarchy is modified and only the areas which are substantially affected are changed at lower levels. Some methods approximately decouple localization and mapping in two threads that run in parallel like Klein and Murray [135]. Other methods resort to solving different stages in parallel: inspired by [223], Strasdat *et al.* [235] take a two-stage approach and optimize first a local pose-features graph and then a pose-pose graph; Williams *et al.* [259] split factor graph optimization in a high-frequency filter and low-frequency smoother, which are periodically synchronized.

Distributed multi robot SLAM: One way of mapping a large-scale environment is to deploy multiple robots doing SLAM, and divide the scenario in smaller areas, each one mapped by a different robot. This approach has two main variants: the *centralized* one, where robots build submaps and transfer the local information to a central station that performs inference [66, 210], and the *decentralized* one, where there is no central data fusion and the agents leverage local communication to reach consensus on a common map. Nerurkar *et al.* [181] propose an algorithm for cooperative localization based on distributed conjugate gradient. Araguez *et al.* [5] investigate consensus-based approaches for map merging. Knuth and Barooah [137] estimate 3D poses using distributed gradient descent. In Lazaro *et al.* [151], robots exchange portions of their factor graphs, which are approximated in the form of condensed measurements to minimize communication. Cunnigham *et al.* [54] use Gaussian elimination, and develop an approach, called DDF-SAM, in which each robot exchanges a Gaussian marginal over the *separators* (i.e., the variables shared by multiple robots). A recent survey on multi-robot SLAM approaches can be found in [216].

While Gaussian elimination has become a popular approach it has two major shortcomings. First, the marginals to be exchanged among the robots are dense, and the communication cost is quadratic in the number of separators. This motivated the use of sparsification techniques to reduce the communication cost [197]. The second reason is that Gaussian elimination is performed on a linearized version of the problem, hence approaches such as DDF-SAM [54] require good linearization points and complex bookkeeping to ensure consistency of the linearization points across the robots. An alternative approach to Gaussian elimination is the Gauss-Seidel approach of Choudhary *et al.* [47], which implies a communication burden which is linear in the number of separators.

Open Problems. Despite the amount of work to reduce complexity of factor graph optimization, the literature has large gaps on other aspects related to long-term operation.

Map representation: A fairly unexplored question is how to store the map during long-term operation. Even when memory is not a tight constraint, e.g. data is stored on the cloud, raw representations as point clouds or volumetric maps (see also Section V) are wasteful in terms of memory; similarly, storing feature descriptors for vision-based SLAM quickly becomes cumbersome. Some initial solutions have been recently proposed for localization against a compressed known map [163], and for memory-efficient dense reconstruction [136].

Learning, forgetting, remembering: A related open question for long-term mapping is how often to update the information contained in the map and how to decide when this information becomes outdated and can be discarded. When is it fine, if ever, to forget? In which case, what can be forgotten and what is essential to maintain? Can parts of the map be "off-loaded" and recalled when needed? While this is clearly task-dependent, no grounded answer to these questions has been proposed in the literature.

Robust distributed mapping: While approaches for outlier rejection have been proposed in the single robot case, the literature on multi robot SLAM barely deals with the problem of outliers. Dealing with spurious measurements is particularly challenging for two reasons. First, the robots might not share a common reference frame, making it harder to detect and

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对边缘化节点的马尔可夫毯进行操作并计算毯的稀疏近似。 Huang et al. [107] 通过求解 ℓ 1-正则化最小化问题来稀疏 Hessian 矩阵（出现在正规方程中）。

允许随着时间的推移减少要估计的参数数量的另一项工作是_连续时间轨迹估计_。此类的第一个 SLAM 方法是由 Bibby 和 Reid 提出的，使用三次样条来表示机器人的连续轨迹[12]。在他们的方法中，因子图中的节点代表样条线的控制点（结），这些控制点以滑动窗口的方式进行了优化。后来，Furgale 等人[88]提出在批量优化公式中使用基函数，特别是 B 样条曲线来近似机器人轨迹。滑动窗口 B 样条公式也用于带有卷帘快门相机的 SLAM，Patron-Perez 等人 [196] 采用基于地标的表示，Kim 等人 [133] 采用半密集直接表示。最近，Mueggler 等人[177]将连续时间 SLAM 公式应用于基于事件的相机。 Bosse et al. [21] 将连续 3D 扫描匹配公式从 [19] 扩展到大规模 SLAM 应用。后来，Anderson et al. [4] 和 Dub´e et al. [67] 分别通过使用小波或在轨迹上采样非均匀结提出了更有效的实现。 Tong et al. [243] 将轨迹的参数化从基本曲线更改为高斯过程表示，其中因子图中的节点是实际的机器人姿势，任何其他姿势都可以通过计算给定时间的后验平均值来插值。需要昂贵的批量高斯牛顿优化来解决第一个提案中的状态。 Barfoot et al. [3] 然后提出了一种具有精确稀疏逆内核的高斯过程，大大减少了批量解决方案的计算时间。

Out-of-core（并行）SLAM： SLAM 的并行_out-of-core_算法将因子图优化的计算（和内存）负载分散到多个处理器之间。关键思想是将因子图划分为不同的子图，并通过交替对每个子图进行局部优化和全局细化来优化整体图。相应的方法通常被称为“子映射算法”，这个想法可以追溯到处理大规模地图的最初尝试 [18]。 Ni et al. [187] 和Zhao et al. [267] 提出了因子图优化的子映射方法，以二叉树结构组织子图。 Grisetti_et al._[98]提出了子图的层次结构：每当获取观测值时，层次结构的最高级别就会被修改，并且只有受到严重影响的区域才会在较低级别发生变化。一些方法近似解耦两个并行运行的线程中的定位和映射，如 Klein 和 Murray [135]。其他方法采用并行解决不同阶段的方法：受[223]的启发，Strasdat_et al._[235]采用两阶段方法，首先优化局部姿态特征图，然后优化姿态-姿态图

; Williams et al. [259] 周期性同步的高频滤波器和低频平滑器中的分割因子图优化。

_分布式多机器人SLAM：_绘制大规模环境的一种方法是部署多个机器人进行SLAM，并将场景划分为较小的区域，每个区域

由不同的机器人绘制。这种方法有两种主要变体：一种是“集中式”，机器人构建子地图并将本地信息传输到执行推理的中央站 [66, 210]；另一种是“分散式”，其中没有中央数据融合，智能体利用本地通信在公共地图上达成共识。 Nerurkar 等人[181]提出了一种基于分布式共轭梯度的协作定位算法。 Araguez et al. [5] 研究基于共识的地图合并方法。 Knuth 和 Barooah [137] 使用分布式梯度下降估计 3D 姿势。在 Lazaro 等人[151]中，机器人交换其因子图的部分内容，这些部分以压缩测量的形式进行近似，以最大限度地减少通信。 Cunnigham_等人_[54]使用高斯消除，并开发了一种称为 DDF-SAM 的方法，其中每个机器人在_分隔符_（即多个机器人共享的变量）上交换高斯边际。最近关于多机器人 SLAM 方法的调查可以在 [216] 中找到。

虽然高斯消除法已成为一种流行的方法，但它有两个主要缺点。首先，机器人之间交换的边际是密集的，并且通信成本是分离器数量的二次方。这促使人们使用稀疏化技术来降低通信成本[197]。第二个原因是高斯消除是在问题的线性化版本上执行的，因此 DDF-SAM [54] 等方法需要良好的线性化点和复杂的簿记，以确保机器人之间线性化点的一致性。高斯消除的另一种方法是 Choudhary 等人的 Gauss-Seidel 方法 [47]，这意味着通信负担与分隔符的数量呈线性关系。

开放问题。尽管降低因子图优化复杂性的工作量很大，但文献在与长期操作相关的其他方面仍存在很大差距。

_地图表示：_一个相当未探索的问题是如何在长期操作期间存储地图。即使内存不是一个严格的限制，例如数据存储在云上，点云或体积图（另见第五节）的原始表示在内存方面是浪费的；同样，存储基于视觉的 SLAM 的特征描述符很快就会变得很麻烦。最近提出了一些初始解决方案，用于针对压缩的已知地图进行定位[163]，以及用于内存高效的密集重建[136]。

_学习、遗忘、记忆：_长期制图的一个相关的开放问题是多久更新一次地图中包含的信息以及如何决定这些信息何时过时并可以丢弃

。如果有的话，什么时候可以忘记？在这种情况下，什么是可以忘记的，什么是必须维持的？地图的某些部分可以“卸载”并在需要时调用吗？虽然这显然取决于任务，但文献中尚未对这些问题提出有根据的答案。

_鲁棒分布式建图：_虽然在单机器人情况下提出了异常值拒绝的方法，但多机器人 SLAM 的文献几乎没有涉及异常值问题。由于两个原因，处理杂散测量尤其具有挑战性。首先，机器人可能不共享共同的参考系，这使得检测和识别变得更加困难。

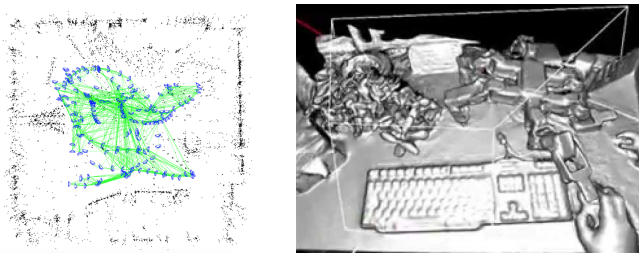


Fig. 4: Left: feature-based map of a room produced by ORB-SLAM [179]. Right: dense map of a desktop produced by DTAM [184].

reject wrong loop closures. Second, in the distributed setup, the robots have to detect outliers from very partial and local information. An early attempt to tackle this issue is [84], in which robots actively verify location hypotheses using a rendezvous strategy before fusing information. Indelman *et al.* [117] propose a probabilistic approach to establish a common reference frame in the face of spurious measurements.

Resource-constrained platforms: Another relatively unexplored issue is how to adapt existing SLAM algorithms to the case in which the robotic platforms have *severe* computational constraints. This problem is of great importance when the size of the platform is scaled down, e.g., mobile phones, micro aerial vehicles, or robotic insects [261]. Many SLAM algorithms are too expensive to run on these platforms, and it would be desirable to have algorithms in which one can tune a "knob" that allows to gently trade off accuracy for computational cost. Similar issues arise in the multi-robot setting: how can we guarantee reliable operation for multi robot teams when facing tight bandwidth constraints and communication dropout? The "version control" approach of Cieslewski *et al.* [49] is a first study in this direction.

V. Representation I: Metric Map Models

This section discusses how to model geometry in SLAM. More formally, a *metric representation* (or metric map) is a symbolic structure that encodes the geometry of the environment. We claim that understanding how to choose a suitable metric representation for SLAM (and extending the set or representations currently used in robotics) will impact many research areas, including long-term navigation, physical interaction with the environment, and human-robot interaction.

Geometric modeling appears much simpler in the 2D case, with only two predominant paradigms: *landmark-based maps* and *occupancy grid maps*. The former models the environment as a sparse set of landmarks, the latter discretizes the environment in cells and assigns a probability of occupation to each cell. The problem of standardization of these representations in the 2D case has been tackled by the *IEEE RAS Map Data Representation Working Group*, which recently released a standard for 2D maps in robotics [113]; the standard defines the two main metric representations for planar environments (plus topological maps) in order to facilitate data exchange, benchmarking, and technology transfer.

The question of 3D geometry modeling is more delicate, and the understanding of how to efficiently model 3D geometry during mapping is in its infancy. In this section we review metric representations, taking a broad perspective across robotics, computer vision, computer aided design (CAD), and computer graphics. Our taxonomy draws inspiration from [80, 209, 221], and includes pointers to more recent work.

Landmark-based sparse representations. Most SLAM methods represent the scene as a set of *sparse* 3D landmarks corresponding to discriminative features in the environment (e.g., lines, corners) [179]; one example is shown in Fig. 4(left). These are commonly referred to as *landmark-based* or *feature-based* representations, and have been widespread in mobile robotics since early work on localization and mapping, and in computer vision in the context of *Structure from Motion* [2, 244]. A common assumption underlying these representations is that the landmarks are distinguishable, i.e., sensor data measure some geometric aspect of the landmark, but also provide a *descriptor* which establishes a (possibly uncertain) data association between each measurement and the corresponding landmark. Previous work also investigates different 3D landmark parameterizations, including global and local Cartesian models, and inverse depth parametrization [174]. While a large body of work focuses on the estimation of point features, the robotics literature includes extensions to more complex geometric landmarks, including lines, segments, or arcs [162].

Low-level raw dense representations. Contrary to landmark-based representations, dense representations attempt to provide high-resolution models of the 3D geometry; these models are more suitable for obstacle avoidance, or for visualization and rendering, see Fig. 4(right). Among dense models, *raw representations* describe the 3D geometry by means of a large unstructured set of points (i.e., *point clouds*) or polygons (i.e., *polygon soup* [222]). Point clouds have been widely used in robotics, in conjunction with stereo and RGB-D cameras, as well as 3D laser scanners [190]. These representations have recently gained popularity in monocular SLAM, in conjunction with the use of *direct methods* [118, 184, 203], which estimate the trajectory of the robot and a 3D model directly from the intensity values of all the image pixels. Slightly more complex representations are *surfel maps*, which encode the geometry as a set of disks [104, 257]. While these representations are visually pleasant, they are usually cumbersome as they require storing a large amount of data. Moreover, they give a low-level description of the geometry, neglecting, for instance, the topology of the obstacles.

Boundary and spatial-partitioning dense representations. These representations go beyond unstructured sets of low-level primitives (e.g., points) and attempt to explicitly represent surfaces (or *boundaries*) and volumes. These representations lend themselves better to tasks such as motion or footstep planning, obstacle avoidance, manipulation, and other physics-based reasoning, such as contact reasoning. Boundary representations (b-reps) define 3D objects in terms of their surface boundary. Particularly simple boundary representations are plane-based models, which have been used for mapping by Castle *et al.* [44] and Kaess [124, 162]. More general b-reps include *curve-based representations* (e.g., tensor product of NURBS or B-splines), *surface mesh models* (connected sets

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图 4：左：ORB-SLAM [179] 生成的基于特征的房间地图。右图：DTAM [184] 生成的密集桌面地图。

拒绝错误的循环闭合。其次，在分布式设置中，机器人必须从非常局部和局部的信息中检测异常值。解决这个问题的早期尝试是[84]，其中机器人在融合信息之前使用会合策略主动验证位置假设。 Indelman_et al._[117]提出了一种概率方法来在面对虚假测量时建立一个共同的参考系。

_资源受限的平台：_另一个相对未探索的问题是如何使现有的 SLAM 算法适应机器人平台具有_严重_计算限制的情况。当平台尺寸缩小时，例如手机、微型飞行器或机器人昆虫，这个问题就变得非常重要[261]。许多 SLAM 算法的成本太高，无法在这些平台上运行，因此需要一种可以调整“旋钮”的算法，从而可以在计算成本和准确性之间进行温和的权衡。类似的问题也出现在多机器人环境中：当面临严格的带宽限制和通信中断时，如何保证多机器人团队的可靠运行？ Cieslewski et al. [49] 的“版本控制”方法是这个方向的第一个研究。

V. 表示 I：度量地图模型

本节讨论如何在 SLAM 中对几何进行建模。更正式地说，“度量表示”（或度量映射）是一种对环境几何形状进行编码的符号结构。我们声称，了解如何为 SLAM 选择合适的度量表示（并扩展当前机器人技术中使用的表示集或表示）将影响许多研究领域，包括长期导航、与环境的物理交互以及人机交互。

在 2D 情况下，几何建模显得简单得多，只有两种主要范例：_基于地标的地图_和_占用网格地图_。前者将环境建模为一组稀疏的地标，后者将环境离散化为单元格，并为每个单元格分配占用概率。 2D 情况下这些表示的标准化问题已由 IEEE RAS 地图数据表示工作组解决，该工作组最近发布了机器人技术中 2D 地图的标准 [113]；该标准定义了平面环境的两种主要度量表示（以及拓扑图），以促进数据交换、基准测试和技术转让。

3D几何建模的问题比较微妙，如何高效建模3D几何的理解

测绘还处于起步阶段。在本节中，我们从机器人技术、计算机视觉、计算机辅助设计 (CAD) 和计算机图形学的广阔角度回顾度量表示。我们的分类法从[80,209,221]中汲取灵感，并包括对最近工作的指导。

基于地标的稀疏表示。大多数 SLAM 方法将场景表示为一组稀疏的 3D 地标，对应于环境中的判别特征（例如线条、角点）[179]；一个例子如图4（左）所示

。这些通常被称为“基于地标”或“基于特征”表示，自定位和地图绘制的早期工作以来，在移动机器人技术以及“运动结构”背景下的计算机视觉中广泛使用[2, 244]。这些表示背后的一个常见假设是地标是可区分的，即传感器数据测量地标的某些几何方面，但还提供一个_描述符_，它在每个测量值和相应地标之间建立（可能不确定）数据关联。之前的工作还研究了不同的 3D 地标参数化，包括全局和局部笛卡尔模型以及逆深度参数化 [174]。虽然大量工作集中在点特征的估计上，但机器人文献包括对更复杂的几何地标的扩展，包括直线、线段或弧[162]。

低级原始密集表示。与基于地标的表示相反，密集表示试图提供 3D 几何的高分辨率模型；这些模型更适合避障，或者可视化和渲染，见图4（右）。在密集模型中，_原始表示_通过大量非结构化点集（即_点云_）或多边形（即_多边形汤_ [222]）来描述 3D 几何形状。点云已广泛应用于机器人领域，与立体相机和 RGB-D 相机以及 3D 激光扫描仪结合使用 [190]。这些表示最近在单目 SLAM 中得到了普及，结合使用“直接方法”[118,184,203]，直接根据所有图像像素的强度值估计机器人的轨迹和 3D 模型。稍微复杂的表示

是 surfel 映射 ，它将几何图形编码为一组磁盘 [104, 257]。虽然这些表示在视觉上令人愉悦，但它们通常很麻烦，因为它们需要存储大量数据。此外，它们给出了几何结构的低级描述，忽略了障碍物的拓扑结构等。

边界和空间分区密集表示。这些表示超越了低级基元（例如点）的非结构化集合，并尝试显式表示表面（或_边界_）和体积。这些表示更适合诸如运动或足迹规划、避障、操纵和其他基于物理的推理（例如接触推理）等任务。边界表示 (b-reps) 根据表面边界定义 3D 对象。特别简单的边界表示是基于平面的模型，Castle et al. [44] 和 Kaess [124, 162] 已将其用于映射。更一般的 b-rep 包括_基于曲线的表示_（例如，NURBS 或 B 样条的张量积）、表面网格模型（连通集

of polygons), and *implicit surface representations*. The latter specify the surface of a solid as the zero crossing of a function defined on $\mathbb{R}^3$ [16]; examples of functions include *radial-basis functions* [38], *signed-distance function* [55], and *truncated signed-distance function* (TSDF) [264]. TSDF are currently a popular representation for vision-based SLAM in robotics, attracting increasing attention after the seminal work [183]. Mesh models have been also used in [257, 258].

Spatial-partitioning representations define 3D objects as a collection of contiguous non-intersecting primitives. The most popular spatial-partitioning representation is the so called *spatial-occupancy enumeration*, which decomposes the 3D space into identical cubes (*voxels*), arranged in a regular 3D grid. More efficient partitioning schemes include *octree*, *Polygonal Map octree*, and *Binary Space-Partitioning tree* [80, §12.6]. In robotics, octree representations have been used for 3D mapping [75], while commonly used *occupancy grid maps* [71] can be considered as probabilistic variants of spatial-partitioning representations. In 3D environments without hanging obstacles, 2.5D elevation maps have been also used [23]. Before moving to higher-level representations, let us better understand how sparse (feature-based) representations (and algorithms) compare to dense ones in visual SLAM.

Which one is best: feature-based or direct methods? Feature-based approaches are quite mature, with a long history of success [59]. They allow to build accurate and robust SLAM systems with automatic relocation and loop closing [179]. However, such systems depend on the availability of features in the environment, the reliance on detection and matching thresholds, and on the fact that most feature detectors are optimized for speed rather than precision. On the other hand, direct methods work with the raw pixel information and dense-direct methods exploit all the information in the image, even from areas where gradients are small; thus, they can outperform feature-based methods in scenes with poor texture, defocus, and motion blur [184, 203]. However, they require high computing power (GPUs) for real-time performance. Furthermore, how to jointly estimate dense structure and motion is still an open problem (currently they can be only be estimated subsequently to one another). To avoid the caveats of feature-based methods there are two alternatives. *Semi-dense* methods overcome the high-computation requirement of dense method by exploiting only pixels with strong gradients (i.e., edges) [72, 83]; *semi-direct* methods instead leverage both sparse features (such as corners or edges) and direct methods [83] and are proven to be the most efficient [83]; additionally, because they rely on sparse features, they allow joint estimation of structure and motion.

High-level object-based representations. While point clouds and boundary representations are currently dominating the landscape of dense mapping, we envision that higher-level representations, including objects and solid shapes, will play a key role in the future of SLAM. Early techniques to include object-based reasoning in SLAM are "SLAM++" from Salas-Moreno *et al.* [217], the work from Civera *et al.* [50], and Dame *et al.* [56]. Solid representations explicitly encode the fact that real objects are three-dimensional rather than 1D (i.e., points), or 2D (surfaces). Modeling objects as solid shapes allows associating physical notions, such as volume and mass, to each object, which is definitely important for robots which have to interact the world. Luckily, existing literature from CAD and computer graphics paved the way towards these developments. In the following, we list few examples of solid representations that have not yet been used in a SLAM context:

- *Parameterized Primitive Instancing:* relies on the definition of *families* of objects (e.g., cylinder, sphere). For each family, one defines a set of parameters (e.g., radius, height), that uniquely identifies a member (or *instance*) of the family. This representation may be of interest for SLAM since it enables the use of extremely compact models, while still capturing many elements in man-made environments.
- *Sweep representations:* define a solid as the sweep of a 2D or 3D object along a trajectory through space. Typical sweeps representations include translation sweep (or *extrusion*) and rotation sweep. For instance, a cylinder can be represented as a translation sweep of a circle along an axis that is orthogonal to the plane of the circle. Sweeps of 2D cross-sections are known as *generalized cylinders* in computer vision [13], and they have been used in robotic grasping [200]. This representation seems particularly suitable to reason on the occluded portions of the scene, by leveraging symmetries.
- *Constructive solid geometry:* defines complex solids by means of boolean operations between primitives [209]. An object is stored as a tree in which the leaves are the primitives and the edges represent operations. This representation can model fairly complicated geometry and is extensively used in computer graphics.

We conclude this review by mentioning that other types of representations exist, including feature-based models in CAD [220], dictionary-based representations [266], affordance-based models [134], generative and procedural models [172], and scene graphs [121]. In particular, dictionary-based representations, which define a solid as a combination of atoms in a dictionary, have been considered in robotics and computer vision, with dictionary learned from data [266] or based on existing repositories of object models [149, 157].

Open Problems. The following problems regarding metric representation for SLAM deserve a large amount of fundamental research, and are still vastly unexplored.

High-level, expressive representations in SLAM: While most of the robotics community is currently focusing on point clouds or TSDF to model 3D geometry, these representations have two main drawbacks. First, they are wasteful of memory. For instance, both representations use many parameters (i.e., points, voxels) to encode even a simple environment, such as an empty room (this issue can be partially mitigated by the so-called voxel hashing [188]). Second, these representations do not provide any high-level understanding of the 3D geometry. For instance, consider the case in which the robot has to figure out if it is moving in a room or in a corridor. A point cloud does not provide readily usable information about the type of environment (i.e., room vs. corridor). On the other hand, more sophisticated models (e.g., parameterized primitive instancing) would provide easy ways to discern the two scenes (e.g., by looking at the parameters

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多边形）和_隐式表面表示_。后者将实体的表面指定为 R3 [16] 上定义的函数的零交叉点；函数的示例包括_径向基函数_ [38]、有符号距离函数 [55] 和_截断有符号距离函数_ (TSDF) [264]。 TSDF 目前是机器人技术中基于视觉的 SLAM 的流行代表，在开创性的工作之后吸引了越来越多的关注 [183]。网格模型也已在[257, 258]中使用。

空间分区表示将 3D 对象定义为连续的不相交图元的集合。最流行的空间分区表示是所谓的 空间占用枚举，它将 3D 空间分解为相同的立方体（voxels），并排列在规则的 3D 网格中。更有效的分区方案包括 octree 、Polygonal Map octree 和 Binary Space-Partitioning tree [80, § 12.6]。在机器人技术中，八叉树表示已用于 3D 映射 [75]，而常用的_占用网格图_[71] 可以被视为空间分区表示的概率变体。在没有悬挂障碍物的 3D 环境中，也使用了 2.5D 高程图 [23]。在转向更高级别的表示之前，让我们更好地了解视觉 SLAM 中稀疏（基于特征）表示（和算法）与密集表示（和算法）的比较。

_哪一种最好：基于特征的方法或直接方法？_基于特征的方法相当成熟，具有悠久的成功历史[59]。它们允许构建具有自动重定位和闭环功能的精确且强大的 SLAM 系统 [179]。然而，此类系统取决于环境中特征的可用性、对检测和匹配阈值的依赖，以及大多数特征检测器针对速度而不是精度进行优化的事实。另一方面，直接方法使用原始像素信息，而密集直接方法利用图像中的所有信息，即使是梯度很小的区域；因此，在纹理较差、散焦和运动模糊的场景中，它们的性能优于基于特征的方法 [184, 203]。然而，它们需要高计算能力（GPU）才能实现实时性能。此外，如何联合估计密集结构和运动仍然是一个悬而未决的问题（目前它们只能依次估计）。为了避免基于特征的方法的警告，有两种选择。半密集方法通过仅利用具有强梯度的像素（即边缘）来克服密集方法的高计算要求[72, 83]；相反，半直接方法同时利用稀疏特征（例如角或边缘）和直接方法[83]，并且被证明是最有效的[83]；此外，由于它们依赖于稀疏特征，因此它们允许对结构和运动进行联合估计。

基于对象的高级表示。虽然点云和边界表示目前在密集地图领域占据主导地位，但我们预计更高级别的表示（包括对象和实体形状）将在 SLAM 的未来中发挥关键作用

。在 SLAM 中包含基于对象的推理的早期技术是 SalasMoreno et al. [217] 的“SLAM++”、Civera et al. [50] 和 Dame et al. [56] 的工作。实体表示明确地编码了这样一个事实：真实对象是三维的，而不是一维（即点）或二维（表面）。将对象建模为实体形状

允许将物理概念（例如体积和质量）与每个对象相关联，这对于必须与世界互动的机器人来说绝对是重要的。幸运的是，现有的 CAD 和计算机图形学文献为这些发展铺平了道路。下面，我们列出了一些尚未在 SLAM 环境中使用的实体表示示例：

-_参数化原始实例化：_依赖于对象_族_的定义（例如，圆柱体、球体）。对于每个族，定义一组参数（例如，半径、高度），唯一标识该族的成员（或_实例_）。这种表示可能会引起 SLAM 的兴趣，因为它可以使用极其紧凑的模型，同时仍然捕获人造环境中的许多元素。

- 扫描表示： 将实体定义为 2D 或 3D 对象沿着空间轨迹的扫描。典型的扫描表示包括平移扫描（或_挤压_）和旋转扫描。例如，圆柱体可以表示为圆沿着与圆平面正交的轴的平移扫描。二维横截面的扫描在计算机视觉中被称为“广义圆柱体”[13]，并且它们已被用于机器人抓取[200]。这种表示似乎特别适合通过利用对称性对场景的被遮挡部分进行推理。

-_构造实体几何：_通过基元之间的布尔运算定义复杂实体[209]。对象存储为树，其中叶子是基元，边代表操作。这种表示可以模拟相当复杂的几何形状，并广泛应用于计算机图形学中。

我们在总结这篇评论时提到存在其他类型的表示，包括 CAD 中基于特征的模型 [220]、基于字典的表示 [266]、基于 可供性（Affordance） 的模型 [134]、生成和过程模型 [172] 以及场景图 [121]。特别是，基于字典的表示，将实体定义为字典中原子的组合，已在机器人技术和计算机视觉中得到考虑，字典是从数据中学习的[266]或基于现有的对象模型存储库[149, 157]。

开放问题。以下关于 SLAM 度量表示的问题值得进行大量的基础研究，但仍未得到充分探索。

SLAM 中的高级、富有表现力的表示： 虽然大多数机器人社区目前都专注于点云或 TSDF 来建模 3D 几何，但这些表示有两个主要缺点。首先，它们浪费内存。例如，两种表示都使用许多参数（即点、体素）来编码甚至是简单的环境，例如空房间（这个问题可以通过所谓的体素散列[188]部分缓解）

。其次，这些表示不能提供对 3D 几何的任何高级理解。例如，考虑机器人必须确定它是在房间还是走廊中移动的情况。点云不提供有关环境类型（即房间与走廊）的易于使用的信息。另一方面，更复杂的模型（例如，参数化的原始实例）将提供简单的方法来辨别两个场景（例如，通过查看参数

defining the primitive). Therefore, the use of higher-level representations in SLAM carries three promises. First, using more compact representations would provide a natural tool for map compression in large-scale mapping. Second, high-level representations would provide a higher-level description of objects geometry which is a desirable feature to facilitate data association, place recognition, semantic understanding, and human-robot interaction; these representations would also provide a powerful support for SLAM, enabling to reason about occlusions, leverage shape priors, and inform the inference/mapping process of the physical properties of the objects (e.g., weight, dynamics). Finally, using rich 3D representations would enable interactions with existing standards for construction and management of modern buildings, including CityGML [193] and IndoorGML [194]. No SLAM techniques can currently build higher-level representations, beyond point clouds, mesh models, surfels models, and TSDFs. Recent efforts in this direction include [17, 51, 231].

Optimal Representations: While there is a relatively large body of literature on different representations for 3D geometry, few works have focused on understanding which criteria should guide the choice of a specific representation. Intuitively, in simple indoor environments one should prefer parametrized primitives since few parameters can sufficiently describe the 3D geometry; on the other hand, in complex outdoor environments, one might prefer mesh models. Therefore, how should we compare different representations and how should we choose the "optimal" representation? Requicha [209] identifies few basic properties of solid representations that allow comparing different representation. Among these properties we find: *domain* (the set of real objects that can be represented), *conciseness* (the "size" of a representation for storage and transmission), *ease of creation* (in robotics this is the "inference" time required for the construction of the representation), and *efficacy in the context of the application* (this depends on the tasks for which the representation is used). Therefore, the "optimal" representation is the one that enables preforming a given task, while being concise and easy to create. Soatto and Chiuso [229] define the optimal representation as a minimal sufficient statistics to perform a given task, and its maximal invariance to nuisance factors. Finding a general yet tractable framework to choose the best representation for a task remains an open problem.

Automatic, Adaptive Representations: Traditionally, the choice of a representation has been entrusted to the roboticist designing the system, but this has two main drawbacks. First, the design of a suitable representation is a time-consuming task that requires an expert. Second, it does not allow any flexibility: once the system is designed, the representation of choice cannot be changed; ideally, we would like a robot to use more or less complex representations depending on the task and the complexity of the environment. The automatic design of optimal representations will have a large impact on long-term navigation.

VI. Representation II: Semantic Map Models

Semantic mapping consists in associating semantic concepts to geometric entities in a robot's surroundings. Recently, the limitations of purely geometric maps have been recognized and this has spawned a significant and ongoing body of work in semantic mapping of environments, in order to enhance robot's autonomy and robustness, facilitate more complex tasks (e.g. avoid muddy-road while driving), move from path-planning to task-planning, and enable advanced human-robot interaction [9, 26, 217]. These observations have led to different approaches for semantic mapping which vary in the numbers and types of semantic concepts and means of associating them with different parts of the environments. As an example, Pronobis and Jensfelt [206] label different rooms, while Pillai and Leonard [201] segment several known objects in the map. With the exception of few approaches, semantic parsing at the basic level was formulated as a classification problem, where simple mapping between the sensory data and semantic concepts has been considered.

Semantic vs. topological SLAM. As mentioned in Section I, topological mapping drops the metric information and only leverages place recognition to build a graph in which the nodes represent distinguishable "places", while edges denote reachability among places. We note that *topological* mapping is radically different from *semantic* mapping. While the former requires recognizing a previously seen place (disregarding whether that place is a kitchen, a corridor, etc.), the latter is interested in classifying the place according to semantic labels. A comprehensive survey on vision-based topological SLAM is presented in Lowry *et al.* [160], and some of its challenges are discussed in Section III. In the rest of this section we focus on semantic mapping.

Semantic SLAM: Structure and detail of concepts. The unlimited number of, and relationships among, concepts for humans opens a more philosophical and task-driven decision about the level and organization of the semantic concepts. The detail and organization depend on the context of what, and where, the robot is supposed to perform a task, and they impact the complexity of the problem at different stages. A semantic representation is built by defining the following aspects:

- *Level/Detail of semantic concepts:* For a given robotic task, e.g. "going from room A to room B", coarse categories (rooms, corridor, doors) would suffice for a successful performance, while for other tasks, e.g. "pick up a tea cup", finer categories (table, tea cup, glass) are needed.
- *Organization of semantic concepts:* The semantic concepts are not exclusive. Even more, a single entity can have an unlimited number of properties or concepts. A chair can be "movable" and "sittable"; a dinner table can be "movable" and "unsittable". While the chair and the table are pieces of furniture, they share the movable property but with different usability. Flat or hierarchical organizations, sharing or not some properties, have to be designed to handle this multiplicity of concepts.

Brief Survey. There are three main ways to attack semantic mapping, and assign semantic concepts to data.

SLAM helps Semantics: The first robotic researchers working on semantic mapping started by the straightforward approach of segmenting the metric map built by a classical SLAM system into semantic concepts. An early work was that of Mozos *et al.* [176], which builds a geometric map using a

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定义原语）。因此，在 SLAM 中使用更高级别的表示具有三个承诺。首先，使用更紧凑的表示将为大规模测绘中的地图压缩提供自然的工具。其次，高级表示将提供对象几何的高级描述，这是促进数据关联、位置识别、语义理解和人机交互的理想功能；这些表示还将为 SLAM 提供强大的支持，能够推理遮挡、利用形状先验，并为物体物理属性（例如重量、动力学）的推理/映射过程提供信息。最后，使用丰富的 3D 表示将能够与现代建筑的施工和管理的现有标准进行交互，包括 CityGML [193] 和 IndoorGML [194]。目前，没有任何 SLAM 技术可以构建超出点云、网格模型、面元模型和 TSDF 之外的更高级别的表示。最近在这个方向上的努力包括[17,51,231]。

最佳表示： 虽然有相对大量的关于 3D 几何的不同表示的文献，但很少有作品关注于理解哪些标准应指导特定表示的选择。直观上，在简单的室内环境中，人们应该更喜欢参数化基元，因为很少有参数可以充分描述 3D 几何形状；另一方面，在复杂的室外环境中，人们可能更喜欢网格模型。那么，我们应该如何比较不同的表示以及如何选择“最优”的表示呢？ Requicha [209] 确定了实体表示的一些基本属性，可以比较不同的表示。在这些属性中，我们发现：域（可以表示的真实对象的集合）、简洁性（用于存储和传输的表示的“大小”）、易于创建（在机器人技术中，这是构建表示所需的“推理”时间）以及_应用程序上下文中的功效_（这取决于使用表示的任务）。因此，“最佳”表示是能够执行给定任务，同时简洁且易于创建的表示。 Soatto 和 Chiuso [229] 将最佳表示定义为执行给定任务的最小足够统计量，及其对干扰因素的最大不变性。寻找一个通用但易于处理的框架来选择任务的最佳表示仍然是一个悬而未决的问题。

_自动、自适应表示：_传统上，表示的选择被委托给设计系统的机器人专家，但这有两个主要缺点。首先，设计合适的表示是一项耗时的任务，需要专家的帮助。其次，它不允许任何灵活性：一旦系统被设计出来，选择的表示就不能改变；理想情况下，我们希望机器人根据任务和环境的复杂性使用或多或少复杂的表示。最佳表示的自动设计将对长期导航产生很大影响。
六．表示二：语义地图模型

语义映射包括将语义概念与机器人周围的几何实体相关联。最近，

人们已经认识到纯几何地图的局限性，这在环境语义映射方面催生了大量持续的工作，以增强机器人的自主性和鲁棒性，促进更复杂的任务（例如驾驶时避免泥泞的道路），从路径规划转向任务规划，并实现先进的人机交互[9,26,217]。这些观察结果导致了不同的语义映射方法，这些方法在语义概念的数量和类型以及将它们与环境的不同部分相关联的方式方面有所不同。例如，Pronobis 和 Jensfelt [206] 标记了不同的房间，而 Pillai 和 Leonard [201] 在地图中分割了几个已知的对象。除了少数方法之外，基本级别的语义解析被表述为分类问题，其中考虑了感知数据和语义概念之间的简单映射。

语义 SLAM 与拓扑 SLAM。正如第一节中提到的，拓扑映射丢弃了度量信息，仅利用地点识别来构建一个图，其中节点代表可区分的“地点”，而边表示地点之间的可达性。我们注意到，拓扑映射与语义映射完全不同。前者需要识别以前见过的地方（不管该地方是厨房、走廊等），后者感兴趣的是根据语义标签对地方进行分类。 Lowry 等人[160]对基于视觉的拓扑 SLAM 进行了全面的调查，第三节讨论了它的一些挑战。在本节的其余部分中，我们将重点关注语义映射。

语义 SLAM：概念的结构和细节。人类概念的无限数量和概念之间的关系开启了关于语义概念的层次和组织的更加哲学和任务驱动的决策。细节和组织取决于机器人执行任务的内容和地点的背景，并且它们会影响不同阶段问题的复杂性。语义表示是通过定义以下方面来构建的：

- _语义概念的级别/细节：_对于给定的机器人任务，例如“从房间 A 到房间 B”，粗类别（房间、走廊、门）足以成功执行，而对于其他任务，例如“拿起茶杯”，需要更精细的类别（桌子、茶杯、玻璃杯）。

- _语义概念的组织：_语义概念不是排他性的。更重要的是，单个实体可以拥有无限数量的属性或概念。椅子可以是“可动的”和“可坐的”；餐桌可以是“可移动的”和“不可坐的”。椅子和桌子虽然都是家具，但它们具有可移动的属性，但用途不同。扁平或分层组织，无论是否共享某些属性，都必须设计为处理这种概念的多样性。

简短调查。攻击语义映射以及将语义概念分配给数据有三种主要方法。

SLAM 有助于语义学： 第一批研究语义映射的机器人研究人员首先采用简单的方法，将经典 SLAM 系统构建的度量映射分割为语义概念。 Mozos et al. [176] 的早期工作是使用

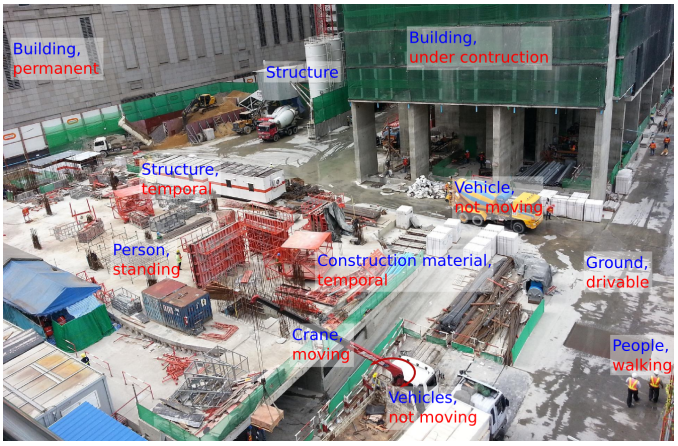


Fig. 5: Semantic understanding allows humans to predict changes in the environment at different time scales. For instance, in the construction site shown in the figure, humans account for the motion of the crane and expect the crane-truck not to move in the immediate future, while at the same time we can predict the semblance of the site which will allow us to localize even after the construction finishes. This is possible because we reason on the functional properties and interrelationships of the entities in the environment. Enhancing our robots with similar capabilities is an open problem for semantic SLAM.

2D laser scan and then fuses the classified semantic places from each robot pose through an associative Markov network in an offline manner. Similarly, Lai *et al.* [148] build a 3D map from RGB-D sequences to then carry out an offline object classification. An online semantic mapping system was later proposed by Pronobis *et al.* [206], who combine three layers of reasoning (sensory, categorical, and place) to build a semantic map of the environment using laser and camera sensors. More recently, Cadena *et al.* [26] use motion estimation, and interconnect a coarse semantic segmentation with different object detectors to outperform the individual systems. Pillai and Leonard [201] use a monocular SLAM system to boost the performance in the task of object recognition in videos.

Semantics helps SLAM: Soon after the first semantic maps came out, another trend started by taking advantage of known semantic classes or objects. The idea is that if we can recognize objects or other elements in a map then we can use our prior knowledge about their geometry to improve the estimation of that map. First attempts were done in small scale by Castle *et al.* [44] and by Civera *et al.* [50] with a monocular SLAM with sparse features, and by Dame *et al.* [56] with a dense map representation. Taking advantage of RGB-D sensors, Salas-Moreno *et al.* [217] propose a SLAM system based on the detection of known objects in the environment.

Joint SLAM and Semantics inference: Researchers with expertise in both computer vision and robotics realized that they could perform monocular SLAM and map segmentation within a joint formulation. The online system of Flint *et al.* [79] presents a model that leverages the Manhattan world assumption to segment the map in the main planes in indoor scenes. Bao *et al.* [9] propose one of the first approaches to jointly estimate camera parameters, scene points and object labels using both geometric and semantic attributes in the scene. In their work, the authors demonstrate the improved object recognition performance and robustness, at the cost of a run-time of 20 minutes per image-pair, and the limited number of object categories makes the approach impractical for online robot operation. In the same line, Häne *et al.* [102] solve a more specialized class-dependent optimization problem in outdoors scenarios. Although still offline, Kundu *et al.* [147] reduce the complexity of the problem by a late fusion of the semantic segmentation and the metric map, a similar idea was proposed earlier by Sengupta *et al.* [219] using stereo cameras. It should be noted that [147] and [219] focus only on the mapping part and they do not refine the early computed poses in this late stage. Recently, a promising online system was proposed by Vineet *et al.* [251] using stereo cameras and a dense map representation.

Open Problems. The problem of including semantic information in SLAM is in its infancy, and, contrary to metric SLAM, it still lacks a cohesive formulation. Fig. 5 shows a construction site as a simple example where we can find the challenges discussed below.

Consistent semantic-metric fusion: Although some progress has been done in terms of temporal fusion of, for instance, per frame semantic evidence [219, 251], the problem of consistently fusing several sources of semantic information with metric information coming at different points in time is still open. Incorporating the confidence or uncertainty of the semantic categorization in the already well known factor graph formulation for the metric representation is a possible way to go for a joint semantic-metric inference framework.

Semantic mapping is much more than a categorization problem: The semantic concepts are evolving to more specialized information such as affordances and actionability[4] of the entities in the map and the possible interactions among different active agents in the environment. How to represent these properties, and interrelationships, are questions to answer for high level human-robot interaction.

Ignorance, awareness, and adaptation: Given some prior knowledge, the robot should be able to reason about new concepts and their semantic representations, that is, it should be able to discover new objects or classes in the environment, learning new properties as result of active interaction with other robots and humans, and adapting the representations to slow and abrupt changes in the environment over time. For example, suppose that a wheeled-robot needs to classify whether a terrain is drivable or not, to inform its navigation system. If the robot finds some mud on a road, that was previously classified as drivable, the robot should learn a new class depending on the grade of difficulty of crossing the muddy region, or adjust its classifier if another vehicle stuck in the mud is perceived.

Semantic-based reasoning[5]*:* As humans, the semantic representations allow us to compress and speed-up reasoning about the environment, while assessing accurate metric representations takes us some effort. Currently, this is not the case for robots. Robots can handle (colored) metric representation but they do not truly exploit the semantic concepts. Our robots

[4]The term *affordances* refers to the set of possible actions on a given object/environment by a given agent [92], while the term *actionability* includes the expected utility of these actions.

[5]Reasoning in the sense of localization and mapping. This is only a sub-area of the vast area of *Knowledge Representation and Reasoning* in the field of Artificial Intelligence that deals with solving complex problems, like having a dialogue in natural language or inferring a person's mood.

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图 5：语义理解使人类能够预测不同时间尺度的环境变化。例如，在图中所示的建筑工地中，人类考虑了起重机的运动，并期望起重机车在不久的将来不会移动，同时我们可以预测现场的外观，这使得我们即使在施工结束后也能进行定位。这是可能的，因为我们对环境中实体的功能属性和相互关系进行推理。增强我们的机器人具有类似的功能是语义 SLAM 的一个悬而未决的问题。

2D 激光扫描，然后通过关联马尔可夫网络以离线方式融合每个机器人姿势的分类语义位置。类似地，Lai 等人[148]从 RGB-D 序列构建 3D 地图，然后进行离线对象分类。 Pronobis等人后来提出了在线语义地图系统[206]，他们结合了三层推理（感觉、分类和位置），使用激光和摄像头传感器构建环境的语义地图。最近，Cadena 等人 [26] 使用运动估计，并将粗略语义分割与不同的对象检测器互连，以超越各个系统。Pillai 和 Leonard [201] 使用单目 SLAM 系统来提高视频中对象识别任务的性能。

语义学有助于 SLAM： 第一个语义地图问世后不久，另一个趋势是利用已知的语义类或对象。这个想法是，如果我们可以识别地图中的对象或其他元素，那么我们就可以使用有关其几何形状的先验知识来改进该地图的估计。 Castle et al. [44] 和 Civera et al. [50] 进行了小规模的首次尝试，使用具有稀疏特征的单目 SLAM，以及 Dame et al. [56] 使用密集地图表示进行了小规模尝试。 Salas-Moreno 等人[217]利用 RGB-D 传感器提出了一种基于环境中已知物体检测的 SLAM 系统。

联合 SLAM 和语义推理： 具有计算机视觉和机器人技术专业知识的研究人员意识到，他们可以在联合公式中执行单目 SLAM 和地图分割。 Flint 等人的在线系统[79]提出了一个模型，该模型利用曼哈顿世界假设来分割室内场景中主平面的地图。 Bao et al. [9] 提出了使用场景中的几何和语义属性联合估计相机参数、场景点和对象标签的首批方法之一。在他们的工作中，作者展示了改进的对象识别性能和鲁棒性，但代价是每个图像对运行 20 分钟，并且有限的对象类别数量使得该方法对于实时运行来说不切实际。

线机器人操作。同样，Héane 等人[102]解决了户外场景中更专业的类相关优化问题。尽管仍处于离线状态，Kundu_et al._ [147] 通过语义分割和度量图的后期融合来降低问题的复杂性，Sengupta_et al._ [219] 早些时候使用立体相机提出了类似的想法

。应该注意的是，[147]和[219]只关注映射部分，他们没有细化后期阶段的早期计算姿势。最近，Vineet et al. [251] 使用立体相机和密集地图表示提出了一种有前景的在线系统。

开放问题。在 SLAM 中包含语义信息的问题还处于起步阶段，并且与度量 SLAM 相反，它仍然缺乏一个有凝聚力的表述。图 5 显示了一个建筑工地作为一个简单示例，我们可以在其中找到下面讨论的挑战。

_一致的语义-度量融合：_尽管在时间融合方面已经取得了一些进展，例如，每帧语义证据 [219, 251]，但将多个语义信息源与不同时间点的度量信息一致融合的问题仍然悬而未决。将语义分类的置信度或不确定性纳入已知的度量表示的因子图公式中是联合语义度量推理框架的一种可能方法。

_语义映射不仅仅是一个分类问题：_语义概念正在演变为更专业的信息，例如 可供性（Affordances） 和映射中实体的可操作性4以及环境中不同活动智能体之间可能的交互。如何表示这些属性和相互关系是高级人机交互需要回答的问题。

_无知、意识和适应：_给定一些先验知识，机器人应该能够推理新概念及其语义表示，也就是说，它应该能够发现环境中的新对象或类别，通过与其他机器人和人类的主动交互来学习新属性，并使表示适应环境随时间的缓慢和突然变化。例如，假设轮式机器人需要对地形是否可行驶进行分类，以通知其导航系统。

如果机器人在道路上发现一些先前被分类为可驾驶的泥浆，则机器人应该根据穿越泥浆区域的难度等级来学习新的类别，或者如果感知到另一辆车陷入泥浆中，则调整其分类器。

_基于语义的推理_5：作为人类，语义表示允许我们压缩和加速对环境的推理，同时评估准确的度量表示需要我们付出一些努力。目前，机器人的情况并非如此。机器人可以处理（彩色）度量表示，但它们并没有真正利用语义概念。我们的机器人

> 4术语_可供性（Affordances）_指的是给定智能体对给定对象/环境的可能操作集合[92]，而术语_actionability_包括这些操作的预期效用。

5定位和映射意义上的推理。这只是人工智能领域“知识表示和推理”广阔领域的一个子领域，人工智能领域涉及解决复杂问题，例如用自然语言进行对话或推断人的情绪。

are currently unable to effectively, and efficiently localize and continuously map using the semantic concepts (categories, relationships and properties) in the environment. For instance, when detecting a car, a robot should infer the presence of a planar ground under the car (even if occluded) and when the car moves the map update should only refine the hallucinated ground with the new sensor readings. Even more, the same update should change the global pose of the car as a whole in a single and efficient operation as opposed to update, for instance, every single voxel.

VII. New theoretical tools for SLAM

This section discusses recent progress towards establishing performance guarantees for SLAM algorithms, and elucidates open problems. The theoretical analysis is important for three main reasons. First, SLAM algorithms and implementations are often tested in few problem instances and it is hard to understand how the corresponding results generalize to new instances. Second, theoretical results shed light on the intrinsic properties of the problem, revealing aspects that may be counter-intuitive during empirical evaluation. Third, a true understanding of the structure of the problem allows pushing the algorithmic boundaries, enabling to extend the set of real-world SLAM instances that can be solved.

Early theoretical analysis of SLAM algorithms were based on the use of EKF; we refer the reader to [64, 255] for a comprehensive discussion, on consistency and observability of EKF SLAM.[6] Here we focus on factor graph optimization approaches. Besides the practical advantages (accuracy, efficiency), factor graph optimization provides an elegant framework which is more amenable to analysis.

In the absence of priors, MAP estimation reduces to maximum likelihood estimation. Consequently, without priors, SLAM inherits all the properties of maximum likelihood estimators: the estimator in (4) is consistent, asymptotically Gaussian, asymptotically efficient, and invariant to transformations in the Euclidean space [171, Theorems 11-1,2]. Some of these properties are lost in presence of priors (e.g., the estimator is no longer invariant [171, page 193]).

In this context we are more interested in *algorithmic properties*: does a given algorithm converge to the MAP estimate? How can we improve or check convergence? What is the breakdown point in presence of spurious measurements?

Brief Survey. Most SLAM algorithms are based on iterative nonlinear optimization [63, 99, 125, 126, 192, 204]. SLAM is a nonconvex problem and iterative optimization can only guarantee local convergence. When an algorithm converges to a local minimum[7] it usually returns an estimate that is completely wrong and unsuitable for navigation (Fig. 6). State-of-the-art iterative solvers fail to converge to a global minimum of the cost for relatively small noise levels [32, 37].

[6]Interestingly, the lack of observability manifests itself very clearly in factor graph optimization, since the linear system to be solved in iterative methods becomes rank-deficient; this enables the design of techniques that can explicitly deal with problems that are not fully observable [265].

[7]We use the term "local minimum" to denote a minimum of the cost which does not attain the globally optimal objective.

Failure to converge in iterative methods has triggered efforts towards a deeper understanding of the SLAM problem. Huang and collaborators [110] pioneered this effort, with initial works discussing the nature of the nonconvexity in SLAM. Huang *et al.* [111] discuss the number of minima in small pose graph optimization problems. Knuth and Barooah [138] investigate the growth of the error in the absence of loop closures. Carlone [29] provides estimates of the basin of convergence for the Gauss-Newton method. Carlone and Censi [32] show that rotation estimation can be solved in closed form in 2D and show that the corresponding estimate is unique. The recent use of alternative maximum likelihood formulations (e.g., assuming Von Mises noise on rotations [34, 211]) has enabled even stronger results. Carlone and Dellaert [31, 36] show that under certain conditions (strong duality) that are often encountered in practice, the maximum likelihood estimate is unique and pose graph optimization can be solved globally, via (convex) semidefinite programming (SDP). A very recent overview on theoretical aspects of SLAM is given in [109].

As mentioned earlier, the theoretical analysis is sometimes the first step towards the design of better algorithms. Besides the dual SDP approach of [31, 36], other authors proposed convex relaxation to avoid convergence to local minima. These contributions include the work of Liu *et al.* [159] and Rosen *et al.* [211]. Another successful strategy to improve convergence consists in computing a suitable *initialization* for iterative nonlinear optimization. In this regard, the idea of solving for the rotations first and to use the resulting estimate to bootstrap nonlinear iteration has been demonstrated to be very effective in practice [20, 30, 32, 37]. Khosoussi *et al.* [130] leverage the (approximate) separability between translation and rotation to speed up optimization.

Recent theoretical results on the use of Lagrangian duality in SLAM also enabled the design of *verification techniques*: given a SLAM estimate these techniques are able to judge whether such estimate is optimal or not. Being able to ascertain the quality of a given SLAM solution is crucial to design failure detection and recovery strategies for safety-critical applications. The literature on verification techniques for SLAM is very recent: current approaches [31, 36] are able to perform verification by solving a sparse linear system and are guaranteed to provide a correct answer as long as strong duality holds (more on this point later).

We note that these results, proposed in a robotics context, provide a useful complement to related work in other communities, including localization in multi agent systems [46, 199, 202, 245, 254], structure from motion in computer vision [86, 95, 103, 168], and cryo-electron microscopy [224, 225].

Open Problems. Despite the unprecedented progress of the last years, several theoretical questions remain open.

Generality, guarantees, verification: The first question regards the generality of the available results. Most results on guaranteed global solutions and verification techniques have been proposed in the context of pose graph optimization. Can these results be generalized to arbitrary factor graphs? Moreover, most theoretical results assume the measurement noise to be isotropic or at least to be structured. Can we generalize existing results to arbitrary noise models?

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目前无法使用环境中的语义概念（类别、关系和属性）进行有效、高效的本地化和连续映射。例如，当检测汽车时，机器人应该推断汽车下方是否存在平坦地面（即使被遮挡），并且当汽车移动时，地图更新应该仅使用新的传感器读数来细化幻觉地面。更重要的是，相同的更新应该通过一次有效的操作来改变整个汽车的全局姿态，而不是更新，例如，每个单独的体素。

七． SLAM 的新理论工具

本节讨论为 SLAM 算法建立性能保证的最新进展，并阐明未解决的问题。理论分析之所以重要，主要有以下三个原因。首先，SLAM 算法和实现通常在少数问题实例中进行测试，很难理解相应的结果如何推广到新实例。其次，理论结果揭示了问题的内在属性，揭示了实证评估过程中可能违反直觉的方面。第三，对问题结构的真正理解可以突破算法边界，从而扩展可以解决的现实世界 SLAM 实例集。

早期SLAM算法的理论分析都是基于EKF的使用；我们建议读者参考 [64, 255] 来全面讨论 EKF SLAM 的一致性和可观察性。6这里我们重点关注因子图优化方法。除了实际优势（准确性、效率）之外，因子图优化还提供了一个更易于分析的优雅框架。

在没有先验的情况下，MAP 估计简化为最大似然估计。因此，在没有先验的情况下，SLAM 继承了最大似然估计器的所有属性：(4) 中的估计器是一致的、渐近高斯的、渐近有效的、并且对欧几里德空间中的变换不变[171，定理 11-1,2]。其中一些属性在存在先验的情况下会丢失（例如，估计器不再是不变的[171，第 193 页]）。

在这种情况下，我们更感兴趣的是_算法属性_：给定的算法是否收敛于 MAP 估计？我们如何改进或检查收敛性？存在杂散测量值时的击穿点是多少？

简短调查。大多数 SLAM 算法基于迭代非线性优化 [63,99,125,126,192,204]。 SLAM是一个非凸问题，迭代优化只能保证局部收敛。当算法收敛到局部最小值7时，它通常会返回完全错误且不适合导航的估计（图 6）。对于相对较小的噪声水平，最先进的迭代求解器无法收敛到全局最小成本 [32, 37]。

> 6有趣的是，可观测性的缺乏在因子图优化中表现得非常明显，因为用迭代方法求解的线性系统变得秩亏；这使得设计能够显式处理不完全可观察的问题的技术成为可能[265]。

> 7我们使用术语“局部最小值”来表示未达到全局最优目标的成本最小值。

迭代方法的收敛失败引发了对 SLAM 问题更深入理解的努力。 Huang 和合作者 [110] 开创了这项工作，最初的工作讨论了 SLAM 中非凸性的本质。 Huang_et al._[111]讨论了小姿态图优化问题中的最小值的数量。 Knuth 和 Barooah [138] 研究了在没有闭环的情况下误差的增长。 Carlone [29] 提供了高斯-牛顿方法收敛盆地的估计。 Carlone 和 Censi [32] 表明旋转估计可以以二维封闭形式求解，并表明相应的估计是唯一的。最近使用替代最大似然公式（例如，假设旋转上存在 Von Mises 噪声 [34, 211]）已经获得了更强的结果。 Carlone 和 Dellaert [31, 36] 表明，在实践中经常遇到的某些条件（强对偶性）下，最大似然估计是唯一的，并且位姿图优化可以通过（凸）半定规划（SDP）全局解决。 [109] 中给出了 SLAM 理论方面的最新概述。

如前所述，理论分析有时是设计更好算法的第一步。除了[31, 36]的对偶SDP方法之外，其他作者提出了凸松弛以避免收敛到局部最小值。这些贡献包括 Liu et al. [159] 和 Rosen et al. [211] 的工作。提高收敛性的另一个成功策略是计算适合迭代非线性优化的_初始化_。在这方面，首先求解旋转并使用所得估计来引导

非线性迭代的想法已被证明在实践中非常有效[20,30,32,37]。 Khosoussi et al. [130] 利用平移和旋转之间的（近似）可分离性来加速优化。

最近在 SLAM 中使用拉格朗日对偶性的理论结果也使得验证技术的设计成为可能：给定 SLAM 估计，这些技术能够判断这种估计是否是最优的。能够确定给定 SLAM 解决方案的质量对于为安全关键型应用设计故障检测和恢复策略至关重要。关于 SLAM 验证技术的文献是最近出现的：当前的方法 [31, 36] 能够通过求解稀疏线性系统来执行验证，并且只要强对偶性成立，就保证提供正确的答案（稍后会详细介绍这一点）。

我们注意到，在机器人技术背景下提出的这些结果为其他社区的相关工作提供了有用的补充，包括多智能体系统中的定位[46,199,202,245,254]，计算机视觉中的运动结构[86,95,103,168]和冷冻电子显微镜[224,225]。

开放问题。尽管过去几年取得了前所未有的进步，但一些理论问题仍然悬而未决。

一般性、保证、验证： 第一个问题涉及可用结果的一般性

。关于有保证的全局解决方案和验证技术的大多数结果都是在位姿图优化的背景下提出的。这些结果可以推广到任意因子图吗？此外，大多数理论结果假设测量噪声是各向同性的或至少是结构化的。我们可以将现有结果推广到任意噪声模型吗？

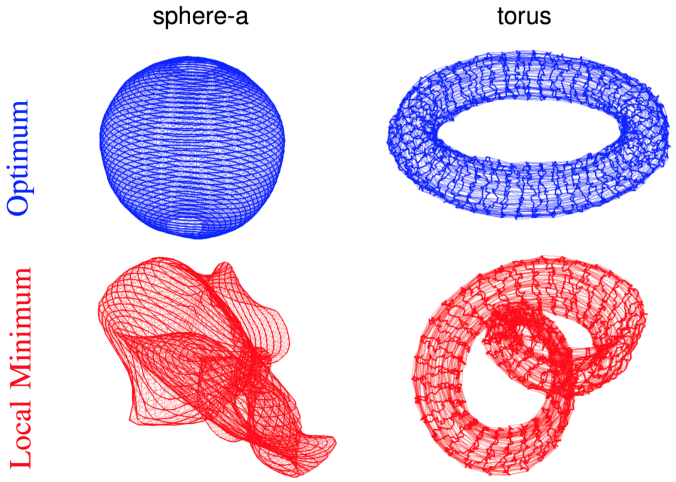


Fig. 6: The back-bone of most SLAM algorithms is the MAP estimation of the robot trajectory, which is computed via non-convex optimization. The figure shows trajectory estimates for two simulated benchmarking problems, namely `sphere-a` and `torus`, in which the robot travels on the surface of a sphere and a torus. The top row reports the correct trajectory estimate, corresponding to the global optimum of the optimization problem. The bottom row shows incorrect trajectory estimates resulting from convergence to local minima. Recent theoretical tools are enabling detection of wrong convergence episodes, and are opening avenues for failure detection and recovery techniques.

Weak or Strong duality? The works [31, 36] show that when strong duality holds SLAM can be solved globally; moreover, they provide empirical evidence that strong duality holds in most problem instances encountered in practical applications. The outstanding problem consists in establishing *a priori* conditions under which strong duality holds. We would like to answer the question "given a set of sensors (and the corresponding measurement noise statistics) and a factor graph structure, does strong duality hold?". The capability to answer this question would define the domain of applications in which we can compute (or verify) global solutions to SLAM. This theoretical investigation would also provide fundamental insights in sensor design and active SLAM (Section VIII).

Resilience to outliers: The third question regards estimation in the presence of spurious measurements. While recent results provide strong guarantees for pose graph optimization, no result of this kind applies in the presence of outliers. Despite the work on robust SLAM (Section III) and new modeling tools for the non-Gaussian noise case [212], the design of global techniques that are resilient to outliers and the design of verification techniques that can certify the correctness of a given estimate in presence of outliers remain open.

VIII. Active SLAM

So far we described SLAM as an estimation problem that is carried out passively by the robot, i.e. the robot performs SLAM *given* the sensor data, but without acting deliberately to collect it. In this section we discuss how to leverage a robot's motion to improve the mapping and localization results.

The problem of controlling robot's motion in order to minimize the uncertainty of its map representation and localization is usually named *active SLAM*. This definition stems from the well-known Bajcsy's active perception [8] and Thrun's robotic exploration [240, Ch. 17] paradigms.

Brief Survey. The first proposal and implementation of an active SLAM algorithm can be traced back to Feder [77] while the name was coined in [152]. However, active SLAM has its roots in ideas from artificial intelligence and robotic exploration that can be traced back to the early eighties (c.f. [10]). Thrun in [239] concluded that solving the *exploration-exploitation* dilemma, i.e., finding a balance between visiting new places (exploration) and reducing the uncertainty by re-visiting known areas (exploitation), provides a more efficient alternative with respect to random exploration or pure exploitation.

Active SLAM is a decision making problem and there are several general frameworks for decision making that can be used as backbone for exploration-exploitation decisions. One of these frameworks is the Theory of Optimal Experimental Design (TOED) [198] which, applied to active SLAM [41, 43], allows selecting future robot action based on the predicted map uncertainty. Information theoretic [164, 208] approaches have been also applied to active SLAM [40, 232]; in this case decision making is usually guided by the notion of *information gain*. Control theoretic approaches for active SLAM include the use of Model Predictive Control [152, 153]. A different body of works formulates active SLAM under the formalism of Partially Observably Markov Decision Process (POMDP) [123], which in general is known to be computationally intractable; approximate but tractable solutions for active SLAM include Bayesian Optimization [169] or efficient Gaussian beliefs propagation [195], among others.

A popular framework for active SLAM consists of selecting the best future action among a finite set of alternatives. This family of active SLAM algorithms proceeds in three main steps [15, 35]: 1) The robot identifies possible locations to explore or exploit, i.e. vantage locations, in its current estimate of the map; 2) The robot computes the utility of visiting each vantage point and selects the action with the highest utility; and 3) The robot carries out the selected action and decides if it is necessary to continue or to terminate the task. In the following, we discuss each point in details.

Selecting vantage points: Ideally, a robot executing an active SLAM algorithm should evaluate every possible action in the robot and map space, but the computational complexity of the evaluation grows exponentially with the search space which proves to be computationally intractable in real applications [24, 169]. In practice, a small subset of locations in the map is selected, using techniques such as frontier-based exploration [127, 262]. Recent works [250] and [116] have proposed approaches for continuous-space planning under uncertainty that can be used for active SLAM; currently these approaches can only guarantee convergence to locally optimal policies. Another recent continuous-domain avenue for active SLAM algorithms is the use of potential fields. Some examples are [249], which uses convolution techniques to compute entropy and select the robot's actions, and [122], which resorts to the solution of a boundary value problem.

Computing the utility of an action: Ideally, to compute the utility of a given action the robot should reason about the evolution of the posterior over the robot pose and the map, taking into account future (controllable) actions and future

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图 6：大多数 SLAM 算法的支柱是机器人轨迹的 MAP 估计，它是通过非凸优化计算的。该图显示了两个模拟基准问题（即球体-a 和环面）的轨迹估计，其中机器人在球体和环面的表面上行进。最上面一行报告了正确的轨迹估计，对应于优化问题的全局最优值。底行显示了由于收敛到局部最小值而导致的错误轨迹估计。最近的理论工具能够检测错误的收敛事件，并为故障检测和恢复技术开辟了途径。

弱对偶性还是强对偶性？ 著作[31, 36]表明，当强对偶性成立时，SLAM 可以全局解决；此外，它们提供的经验证据表明，在实际应用中遇到的大多数问题实例中都存在强对偶性。突出的问题在于建立强二元性成立的先验条件。我们想回答“给定一组传感器（以及相应的测量噪声统计数据）和因子图结构，强对偶性是否成立？”的问题。回答这个问题的能力将定义我们可以计算（或验证）SLAM 全局解决方案的应用领域。这项理论研究还将为传感器设计和主动 SLAM 提供基本见解（第八节）。

_对异常值的恢复能力：_第三个问题涉及存在虚假测量值时的估计。虽然最近的结果为位姿图优化提供了强有力的保证，但这种结果不适用于存在异常值的情况。尽管针对非高斯噪声情况 [212] 进行了鲁棒 SLAM（第 III 节）和新建模工具的研究，但对异常值具有弹性的全局技术的设计以及可以在存在异常值的情况下证明给定估计的正确性的验证技术的设计仍然处于开放状态。

八．主动SLAM

到目前为止，我们将 SLAM 描述为机器人被动执行的估计问题，即机器人在给定传感器数据的情况下执行 SLAM，但没有刻意采取行动来收集数据。在本节中，我们讨论如何利用机器人的运动来改进映射和定位结果。

控制机器人运动以最小化其地图表示和定位的不确定性的问题通常被称为_主动 SLAM_。这个定义源于著名的Bajcsy的主动感知[8]和Thrun的机器人探索[240，Ch.1]。 17]范式。

简短调查。主动 SLAM 算法的第一个提议和实现可以追溯到 Feder [77]，而这个名字是在 [152] 中创造的。然而，主动 SLAM 的根源可以追溯到 80 年代初的人工智能和机器人探索思想（参见 [10]）。 Thrun 在 [239]中得出的结论是，解决“探索-利用”困境，即在访问新地方（探索）和通过重新访问已知区域（开发）来减少不确定性之间找到平衡，为随机探索或纯粹开发提供了更有效的替代方案。

主动 SLAM 是一个决策问题，有几种通用的决策框架可以用作探索-利用决策的支柱。这些框架之一是最优实验设计理论（TOED）[198]，它应用于主动 SLAM [41, 43]，允许根据预测的地图不确定性选择未来的机器人动作。信息论 [164, 208] 方法也已应用于主动 SLAM [40, 232]；在这种情况下，决策通常以“信息增益”的概念为指导。主动 SLAM 的控制理论方法包括使用模型预测控制 [152, 153]。另一些工作在部分可观察马尔可夫决策过程（POMDP）[123]的形式下制定了主动 SLAM，众所周知，这在计算上是困难的；主动 SLAM 的近似但易于处理的解决方案包括贝叶斯优化 [169] 或有效的高斯置信传播 [195] 等。

主动 SLAM 的流行框架包括在一组有限的替代方案中选择最佳的未来行动。这一系列主动 SLAM 算法分三个主要步骤进行 [15, 35]： 1) 机器人在其当前估计的地图中识别可能要探索或利用的位置，即有利位置；2）机器人计算访问每个有利点的效用，并选择效用最高的动作； 3) 机器人执行选定的动作并决定是否有必要继续或终止任务。下面，我们详细讨论每一点。

_选择有利点：_理想情况下，执行主动 SLAM 算法的机器人应该评估机器人和地图空间中的每一个可能的动作，但评估的计算复杂性随着搜索空间呈指数增长，这在实际应用中被证明在计算上很困难 [24, 169]。在实践中，使用基于边界的探索等技术选择地图中的一小部分位置[127, 262]。最近的工作[250]和[116]提

出了不确定性下连续空间规划的方法，可用于主动SLAM；目前这些方法只能保证收敛到局部最优策略。主动 SLAM 算法的另一个最新连续域途径是使用势场。一些例子是[249]，它使用卷积技术来计算熵并选择机器人的动作，以及[122]，它诉诸边界值问题的解决方案。

_计算一个动作的效用：_理想情况下，为了计算给定动作的效用，机器人应该推理机器人姿势和地图上后验的演变，同时考虑到未来（可控）动作和未来

(unknown) measurements. If such posterior were known, an information-theoretic function, as the information gain, could be used to rank the different actions [22, 233]. However, computing this joint probability analytically is, in general, computationally intractable [35, 76, 233]. In practice, one resorts to approximations. Initial work considered the uncertainty of the map and the robot to be independent [246] or conditionally independent [233]. Most of these approaches define the utility as a linear combination of metrics that quantify robot and map uncertainties [22, 35]. One drawback of this approach is that the scale of the numerical values of the two uncertainties is not comparable, i.e. the map uncertainty is often orders of magnitude larger than the robot one, so manual tuning is required to correct it. Approaches to tackle this issue have been proposed for particle-filter-based SLAM [35], and for pose graph optimization [40].

The Theory of Optimal Experimental Design (TOED) [198] can also be used to account for the utility of performing an action. In the TOED, every action is considered as a stochastic design, and the comparison among designs is done using their associated covariance matrices via the so-called optimality criteria, e.g. *A-opt*, *D-opt* and *E-opt*. A study about the usage of optimality criteria in active SLAM can be found in [42, 43].

Executing actions or terminating exploration: While executing an action is usually an easy task, using well-established techniques from motion planning, the decision on whether or not the exploration task is complete, is currently an open challenge that we discuss in the following paragraph.

Open Problems. Several problems still need to be addressed, for active SLAM to have impact in real applications.

Fast and accurate predictions of future states: In active SLAM each action of the robot should contribute to reduce the uncertainty in the map and improve the localization accuracy; for this purpose, the robot must be able to forecast the effect of future actions on the map and robots localization. The forecast has to be fast to meet latency constraints and precise to effectively support the decision process. In the SLAM community it is well known that loop closings are important to reduce uncertainty and to improve localization and mapping accuracy. Nonetheless, efficient methods for forecasting the occurrence and the effect of a loop closing are yet to be devised. Moreover, predicting the effects of future actions is still a computational expensive task [116]. Recent approaches to forecasting future robot states can be found in the machine learning literature, and involve the use of spectral techniques [230] and deep learning [252].

Enough is enough: When do you stop doing active SLAM? Active SLAM is a computationally expensive task: therefore a natural question is when we can stop doing active SLAM and switch to classical (passive) SLAM in order to focus resources on other tasks. Balancing active SLAM decisions and exogenous tasks is critical, since in most real-world tasks, active SLAM is only a means to achieve an intended goal. Additionally, having a stopping criteria is a necessity because at some point it is provable that more information would lead not only to a diminishing return effect but also, in case of contradictory information, to an unrecoverable state (e.g. several wrong loop closures). Uncertainty metrics from TOED, which are task oriented, seem promising as stopping criteria, compared to information-theoretic metrics which are difficult to compare across systems [39].

Performance guarantees: Another important avenue is to look for mathematical guarantees for active SLAM and for near-optimal policies. Since solving the problem exactly is intractable, it is desirable to have approximation algorithms with clear performance bounds. Examples of this kind of effort is the use of submodularity [93] in the related field of active sensors placement.

IX. New Frontiers: Sensors and Learning

The development of new sensors and the use of new computational tools have often been key drivers for SLAM. Section IX-A reviews unconventional and new sensors, as well as the challenges and opportunities they pose in the context of SLAM. Section IX-B discusses the role of (deep) learning as an important frontier for SLAM, analyzing the possible ways in which this tool is going to improve, affect, or even restate, the SLAM problem.

A. New and Unconventional Sensors for SLAM

Besides the development of new algorithms, progress in SLAM (and mobile robotics in general) has often been triggered by the availability of novel sensors. For instance, the introduction of 2D laser range finders enabled the creation of very robust SLAM systems, while 3D lidars have been a main thrust behind recent applications, such as autonomous cars. In the last ten years, a large amount of research has been devoted to vision sensors, with successful applications in augmented reality and vision-based navigation.

Sensing in robotics has been mostly dominated by lidars and conventional vision sensors. However, there are many alternative sensors that can be leveraged for SLAM, such as depth, light-field, and event-based cameras, which are now becoming a commodity hardware, as well as magnetic, olfaction, and thermal sensors.

Brief Survey. We review the most relevant new sensors and their applications for SLAM, postponing a discussion on open problems to the end of this section.

Range cameras: Light-emitting depth cameras are not new sensors, but they became commodity hardware in 2010 with the advent the Microsoft Kinect game console. They operate according to different principles, such as structured light, time of flight, interferometry, or coded aperture. Structure-light cameras work by triangulation; thus, their accuracy is limited by the distance between the cameras and the pattern projector (structured light). By contrast, the accuracy of Time-of-Flight (ToF) cameras only depends on the time-of-flight measurement device; thus, they provide the highest range accuracy (sub millimeter at several meters). ToF cameras became commercially available for civil applications around the year 2000 but only began to be used in mobile robotics in 2004 [256]. While the first generation of ToF and structured-light cameras was characterized by low signal-to-noise ratio and high price, they soon became popular for video-game applications, which contributed to making them affordable

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（未知）测量。如果知道这样的后验，则可以使用信息论函数（作为信息增益）对不同的动作进行排序[22, 233]。然而，一般来说，通过分析方式计算这种联合概率在计算上是困难的[35,76,233]。在实践中，人们会求助于近似值。最初的工作认为地图和机器人的不确定性是独立的[246]或有条件独立的[233]。这些方法中的大多数将效用定义为量化机器人和映射不确定性的指标的线性组合[22, 35]。这种方法的一个缺点是两个不确定性的数值范围不具有可比性，即地图不确定性通常比机器人的不确定性大几个数量级，因此需要手动调整来纠正它。已经针对基于粒子滤波器的 SLAM [35] 和位姿图优化 [40] 提出了解决这个问题的方法。

最优实验设计理论（TOED）[198]也可以用来解释执行某个动作的效用。在 TOED 中，每个动作都被视为随机设计，并且设计之间的比较是通过所谓的最优性标准使用其相关的协方差矩阵来完成的。 A-opt 、D-opt 和 E-opt 。关于主动 SLAM 中最优性标准的使用的研究可以在 [42, 43] 中找到。

_执行动作或终止探索：_虽然执行动作通常是一项简单的任务，但使用运动规划中成熟的技术，决定探索任务是否完成，目前是一个公开的挑战，我们将在下一段中讨论。

开放问题。为了使主动 SLAM 在实际应用中产生影响，仍有几个问题需要解决。

快速准确地预测未来状态： 在主动 SLAM 中，机器人的每个动作都应有助于减少地图中的不确定性并提高定位精度；为此，机器人必须能够预测未来动作对地图和机器人定位的影响。预测必须快速以满足延迟限制，并且必须精确才能有效支持决策过程。在 SLAM 社区中，众所周知，闭环对于减少不确定性以及提高定位和建图精度非常重要。尽管如此，尚未设计出预测闭环发生和影响的有效方法。此外，预测未来行动的影响仍然是一项计算成本高昂的任务[116]。最近预测未来机器人状态的方法可以在机器学习文献中找到，并且涉及谱技术[230]和深度学习[252]的使用。

够了：什么时候停止执行主动 SLAM？ 主动 SLAM 是一项计算成本高昂的任务：因此，一个自然的问题是我们何时可以停止执行主动 SLAM 并切换到经典（被动）SLAM，以便将资源集中在其他任务上。平衡主动 SLAM 决策和外生任务至关重要，因为在大多数现实任务中，主动 SLAM 只是实现预期目标的一种手段

。此外，有一个停止标准是必要的，因为在某些时候，可以证明更多的信息不仅会导致回报效应递减，而且在信息矛盾的情况下，还会导致不可恢复的状态（例如，几个错误的闭环）。不确定性指标来自

与难以跨系统比较的信息论指标相比，面向任务的 TOED 似乎很有希望作为停止标准 [39]。

_性能保证：_另一个重要途径是寻找主动 SLAM 和近乎最优策略的数学保证。由于精确地解决该问题是棘手的，因此需要具有明确性能界限的近似算法。这种努力的例子是在有源传感器放置的相关领域中使用子模块化[93]。
九．新领域：传感器和学习

新传感器的开发和新计算工具的使用通常是 SLAM 的关键驱动力。第 IX-A 节回顾了非常规和新型传感器，以及它们在 SLAM 背景下带来的挑战和机遇。第 IX-B 节讨论了（深度）学习作为 SLAM 重要前沿领域的作用，分析了该工具改进、影响甚至重述 SLAM 问题的可能方式。
一个。用于 SLAM 的新型非常规传感器

除了新算法的开发之外，SLAM（以及一般的移动机器人技术）的进步往往是由新型传感器的可用性引发的。例如，2D 激光测距仪的引入使得能够创建非常强大的 SLAM 系统，而 3D 激光雷达一直是自动驾驶汽车等近期应用的主要推动力。在过去的十年中，大量的研究致力于视觉传感器，并在增强现实和基于视觉的导航方面取得了成功的应用。

机器人技术中的传感主要由激光雷达和传统视觉传感器主导。然而，有许多替代传感器可用于 SLAM，例如深度、光场和基于事件的相机，这些传感器现已成为商品硬件，以及磁、嗅觉和热传感器。

简短调查。我们回顾了最相关的新型传感器及其 SLAM 应用，并将对未决问题的讨论推迟到本节末尾。

_范围相机：_发光深度相机并不是新传感器，但随着 Microsoft Kinect 游戏机的出现，它们在 2010 年成为商品硬件。它们根据不同的原理工作，例如结构光、飞行时间、干涉测量或编码孔径。结构光相机通过三角测量工作；因此，它们的精度受到相机和图案投影仪（结构光）之间的距离的限制。相比之下，飞行时间（ToF）相机的精度仅取决于飞行时间测量设备；因此，它们提供最高的测距精度（几米处亚毫米）。 ToF 相机于 2000 年左右开始用于民用领域，但直到 2004 年才开始用于移动机器人 [256]

。虽然第一代 ToF 和结构光相机具有低信噪比和高价格的特点，但它们很快在视频游戏应用中流行起来，这使得它们变得价格实惠

and improving their accuracy. Since range cameras carry their own light source, they also work in dark and untextured scenes, which enabled the achievement of remarkable SLAM results [183].

Light-field cameras: Contrary to standard cameras, which only record the light intensity hitting each pixel, a light-field camera (also known as plenoptic camera), records both the intensity and the direction of light rays [186]. One popular type of light-field camera uses an array of micro lenses placed in front of a conventional image sensor to sense intensity, color, and directional information. Because of the manufacturing cost, commercially available light-field cameras still have relatively low resolution ($<$ 1MP), which is being overcome by current technological effort. Light-field cameras offer several advantages over standard cameras, such as depth estimation, noise reduction [57], video stabilization [227], isolation of distractors [58], and specularity removal [119]. Their optics also offers wide aperture and wide depth of field compared with conventional cameras [14].

Event-based cameras: Contrarily to standard frame-based cameras, which send entire images at fixed frame rates, event-based cameras, such as the Dynamic Vision Sensor (DVS) [156] or the Asynchronous Time-based Image Sensor (ATIS) [205], only send the local pixel-level changes caused by movement in a scene at the time they occur.

They have five key advantages compared to conventional frame-based cameras: a temporal latency of 1ms, an update rate of up to 1MHz, a dynamic range of up to 140dB (vs 60-70dB of standard cameras), a power consumption of 20mW (vs 1.5W of standard cameras), and very low bandwidth and storage requirements (because only intensity changes are transmitted). These properties enable the design of a new class of SLAM algorithms that can operate in scenes characterized by high-speed motion [89] and high-dynamic range [132, 207], where standard cameras fail. However, since the output is composed of a sequence of asynchronous events, traditional frame-based computer-vision algorithms are not applicable. This requires a *paradigm shift* from the traditional computer vision approaches developed over the last fifty years. Event-based real-time localization and mapping algorithms have recently been proposed [132, 207]. The design goal of such algorithms is that each incoming event can asynchronously change the estimated state of the system, thus, preserving the event-based nature of the sensor and allowing the design of microsecond-latency control algorithms [178].

Open Problems. The main bottleneck of active range cameras is the maximum range and interference with other external light sources (such as sun light); however, these weaknesses can be improved by emitting more light power.

Light-field cameras have been rarely used in SLAM because they are usually thought to increase the amount of data produced and require more computational power. However, recent studies have shown that they are particularly suitable for SLAM applications because they allow formulating the motion estimation problem as a linear optimization and can provide more accurate motion estimates if designed properly [65].

Event-based cameras are revolutionary image sensors that overcome the limitations of standard cameras in scenes characterized by high dynamic range and high speed motion. Open problems concern a full characterization of the sensor noise and sensor non idealities: event-based cameras have a complicated analog circuitry, with nonlinearities and biases that can change the sensitivity of the pixels, and other dynamic properties, which make the events susceptible to noise. Since a single event does not carry enough information for state estimation and because an event camera generate on average $100,000$ events a second, it can become intractable to do SLAM at the discrete times of the single events due to the rapidly growing size of the state space. Using a continuous-time framework [12], the estimated trajectory can be approximated by a smooth curve in the space of rigid-body motions using basis functions (e.g., cubic splines), and optimized according to the observed events [177]. While the temporal resolution is very high, the spatial resolution of event-based cameras is relatively low (QVGA), which is being overcome by current technological effort [155]. Newly developed event sensors overcome some of the original limitations: an ATIS sensor sends the magnitude of the pixel-level brightness; a DAVIS sensor [155] can output both frames and events (this is made possible by embedding a standard frame-based sensor and a DVS into the same pixel array). This will allow tracking features and motion in the blind time between frames [144].

We conclude this section with some general observations on the use of novel sensing modalities for SLAM.

Other sensors: Most SLAM research has been devoted to range and vision sensors. However, humans or animals are able to improve their sensing capabilities by using tactile, olfaction, sound, magnetic, and thermal stimuli. For instance, tactile cues are used by blind people or rodents for haptic exploration of objects, olfaction is used by bees to find their way home, magnetic fields are used by homing pigeons for navigation, sound is used by bats for obstacle detection and navigation, while some snakes can see infrared radiation emitted by hot objects. Unfortunately, these alternative sensors have not been considered in the same depth as range and vision sensors to perform SLAM. Haptic SLAM can be used for tactile exploration of an object or of a scene [237, 263]. Olfaction sensors can be used to localize gas or other odor sources [167]. Although ultrasound-based localization was predominant in early mobile robots, their use has rapidly declined with the advent of cheap optical range sensors. Nevertheless, animals, such as bats, can navigate at very high speeds using only echo localization. Thermal sensors offer important cues at night and in adverse weather conditions [165]. Local anomalies of the ambient magnetic field, present in many indoor environments, offer an excellent cue for localization [248]. Finally, pre-existing wireless networks, such as WiFi, can be used to improve robot navigation without any prior knowledge of the location of the antennas [78].

Which sensor is best for SLAM? A question that naturally arises is: what will be the next sensor technology to drive future long-term SLAM research? Clearly, the performance of a given algorithm-sensor pair for SLAM depends on the sensor and algorithm parameters, and on the environment [228]. A complete treatment of how to choose algorithms and sensors to achieve the best performance has not been found

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并提高其准确性。由于测距相机带有自己的光源，因此它们也可以在黑暗和无纹理的场景中工作，从而实现了显着的 SLAM 效果[183]。

光场相机：与仅记录照射到每个像素的光强度的标准相机相反，光场相机（也称为全光相机）记录光线的强度和方向[186]。一种流行的光场相机使用放置在传统图像传感器前面的微透镜阵列来感测强度、颜色和方向信息。由于制造成本的原因，商用光场相机仍然具有相对较低的分辨率（< 1MP），当前的技术努力正在克服这一问题。与标准相机相比，光场相机具有多种优势，例如深度估计、降噪[57]、视频稳定[227]、干扰物隔离[58]和镜面反射消除[119]。与传统相机相比，它们的光学器件还提供大光圈和大景深[14]。

_基于事件的相机：_与以固定帧速率发送整个图像的标准基于帧的相机相反，基于事件的相机，例如动态视觉传感器（DVS）[156]或基于异步时间的图像传感器（ATIS）[205]，仅发送由场景中的运动发生时引起的局部像素级变化。

与传统的基于帧的相机相比，它们具有五个关键优势：1ms的时间延迟、高达1MHz的更新速率、高达140dB的动态范围（标准相机为6070dB）、20mW的功耗（标准相机为1.5W）以及非常低的带宽和存储要求（因为仅传输强度变化）。这些特性使得新型 SLAM 算法的设计成为可能，该算法可以在以高速运动[89] 和高动态范围 [132, 207] 为特征的场景中运行，而标准相机却无法做到这一点。然而，由于输出由一系列异步事件组成，因此传统的基于帧的计算机视觉算法不适用。这需要对过去五十年开发的传统计算机视觉方法进行“范式转变”。最近提出了基于事件的实时定位和地图算法[132, 207]。此类算法的设计目标是每个传入事件都可以异步更改系统的估计状态，从而保留传感器基于事件的性质并允许设计微秒延迟控制算法[178]。

开放问题。主动测距相机的主要瓶颈是最大范围和对其他外部光源（例如太阳光）的干扰；然而，这些弱点可以通过发射更多的光功率来改善。

光场相机很少用于 SLAM，因为它们通常被认为会增加产生的数据量并且需要更多的计算能力。然而，最近的研究表明，它们特别适合 SLAM 应用，因为它们允许将运动估计问题表述为线性优化，并且如果设计得当，可以提供更准确的运动估计[65]。

基于事件的相机是革命性的图像传感器，克服了标准相机在场景特征方面的局限性

具有高动态范围和高速运动的特点。未解决的问题涉及传感器噪声和传感器非理想性的完整表征：基于事件的相机具有复杂的模拟电路，具有可以改变像素灵敏度的非线性和偏差以及其他动态属性，这使得事件容易受到噪声的影响。由于单个事件没有携带足够的信息用于状态估计，并且事件相机平均每秒生成 100，000 个事件，因此由于状态空间大小的快速增长，在单个事件的离散时间执行 SLAM 可能变得很困难。使用连续时间框架[12]，估计轨迹可以使用基函数（例如三次样条）通过刚体运动空间中的平滑曲线来近似，并根据观察到的事件进行优化[177]。虽然时间分辨率非常高，但基于事件的相机的空间分辨率相对较低（QVGA），当前的技术努力正在克服这一问题[155]。新开发的事件传感器克服了一些原始限制：ATIS 传感器发送像素级亮度的大小； DAVIS 传感器 [155] 可以输出帧和事件（这是通过将标准的基于帧的传感器和 DVS 嵌入到同一像素阵列中来实现的）。这将允许在帧之间的盲区内跟踪特征和运动[144]。

我们通过一些关于 SLAM 新型传感模式使用的一般观察来结束本节。

其他传感器： 大多数 SLAM 研究都致力于范围和视觉传感器。然而，人类或动物能够通过使用触觉、嗅觉、声音、磁和热刺激来提高其感知能力。例如，盲人或啮齿动物利用触觉线索对物体进行触觉探索，蜜蜂利用嗅觉来寻找回家的路，信鸽利用磁场进行导航，蝙蝠利用声音进行障碍物检测和导航，而一些蛇可以看到热物体发出的红外辐射。不幸的是，这些替代传感器尚未被考虑与距离和视觉传感器相同的深度来执行 SLAM。触觉 SLAM 可用于物体或场景的触觉探索 [237, 263]。嗅觉传感器可用于定位气体或其他气味源[167]。尽管基于超声波的定位在早期移动机器人中占主导地位，但随着廉价光学距离传感器的出现，它们的使用迅速下降。然而，蝙蝠等动物仅使用回声定位就可以以非常高的速度导航。热传感器在夜间和恶劣天气条件下提供重要线索[165]。许多室内环境中存在的环境磁场的局部异常为定位提供了极好的线索[248]。最后，现有的无线网络（例如 WiFi）可用于改进机器人导航，而无需事先了解天线的位置 [78]。

哪种传感器最适合 SLAM？ 自然出现的一个问题是：推动未来长期 SLAM 研究的下一个传感器技术是什么？显然，SLAM 的给定算法-传感器对的性能取决于传感器和算法参数以及环境 [228]。尚未找到关于如何选择算法和传感器以实现最佳性能的完整处理

yet. A preliminary study by Censi et al. [45], has shown that the performance for a given task also depends on the power available for sensing. It also suggests that the optimal sensing architecture may have multiple sensors that might be instantaneously switched on and off according to the required performance level or measure the same phenomenon through different physical principles for robustness [68].

B. Deep Learning

It would be remiss of a paper that purports to consider future directions in SLAM not to make mention of deep learning. Its impact in computer vision has been transformational, and, at the time of writing this article, it is already making significant inroads into traditional robotics, including SLAM.

Researchers have already shown that it is possible to learn a deep neural network to regress the inter-frame pose between two images acquired from a moving robot directly from the original image pair [52], effectively replacing the standard geometry of visual odometry. Likewise it is possible to localize the 6DoF of a camera with regression forest [247] and with deep convolutional neural network [129], and to estimate the depth of a scene (in effect, the map) from a single view solely as a function of the input image [27, 70, 158].

This does not, in our view, mean that traditional SLAM is dead, and it is too soon to say whether these methods are simply curiosities that show what can be done in principle, but which will not replace traditional, well-understood methods, or if they will completely take over.

Open Problems. We highlight here a set of future directions for SLAM where we believe machine learning and more specifically deep learning will be influential, or where the SLAM application will throw up challenges for deep learning.

Perceptual tool: It is clear that some perceptual problems that have been beyond the reach of off-the-shelf computer vision algorithms can now be addressed. For example, object recognition for the imagenet classes [213] can now, to an extent, be treated as a black box that works well from the perspective of the roboticist or SLAM researcher. Likewise semantic labeling of pixels in a variety of scene types reaches performance levels of around 80% accuracy or more [74]. We have already commented extensively on a move towards more semantically meaningful maps for SLAM systems, and these black-box tools will hasten that. But there is more at stake: deep networks show more promise for connecting raw sensor data to understanding, or connecting raw sensor data to actions, than anything that has preceded them.

Practical deployment: Successes in deep learning have mostly revolved around lengthy training times on supercomputers and inference on special-purpose GPU hardware for a one-off result. A challenge for SLAM researchers (or indeed anyone who wants to embed the impressive results in their system) is how to provide sufficient computing power in an embedded system. Do we simply wait for the technology to catch up, or do we investigate smaller, cheaper networks that can produce "good enough" results, and consider the impact of sensing over an extended period?

Online and life-long learning: An even greater and important challenge is that of online learning and adaptation, that will be essential to any future long-term SLAM system. SLAM systems typically operate in an open-world with continuous observation, where new objects and scenes can be encountered. But to date deep networks are usually trained on closed-world scenarios with, say, a fixed number of object classes. A significant challenge is to harness the power of deep networks in a one-shot or zero-shot scenario (i.e. one or even zero training examples of a new class) to enable life-long learning for a continuously moving, continuously observing SLAM system.

Similarly, existing networks tend to be trained on a vast corpus of labelled data, yet it cannot always be guaranteed that a suitable dataset exists or is practical to label for the supervised training. One area where some progress has recently been made is that of single-view depth estimation: Garg *et al.* [90] have recently shown how a deep network for single-view depth estimation can be trained simply by observing a large corpus of stereo pairs, without the need to observe or calculate depth explicitly. It remains to be seen if similar methods can be developed for tasks such as semantic scene labelling.

Bootstrapping: Prior information about a scene has increasingly been shown to provide a significant boost to SLAM systems. Examples in the literature to date include known objects [56, 217] or prior knowledge about the expected structure in the scene, like smoothness as in DTAM [184], Manhattan constraints as in [79], or even the expected relationships between objects [9]. It is clear that deep learning is capable of distilling such prior knowledge for specific tasks such as estimating scene labels or scene depths. How best to extract and use this information is a significant open problem. It is more pertinent in SLAM than in some other fields because in SLAM we have solid grasp of the mathematics of the scene geometry – the question then is how to fuse this well-understood geometry with the outputs of a deep network. One particular challenge that must be solved is to characterize the uncertainty of estimates derived from a deep network.

SLAM offers a challenging context for exploring potential connections between deep learning architectures and recursive state estimation in large-scale graphical models. For example, Krishan *et al.* [142] have recently proposed Deep Kalman Filters; perhaps it might one day be possible to create an end-to-end SLAM system using a deep architecture, without explicit feature modeling, data association, etc.

X. Conclusion

The problem of simultaneous localization and mapping has seen great progress over the last 30 years. Along the way, several important questions have been answered, while many new and interesting questions have been raised, with the development of new applications, new sensors, and new computational tools.

Revisiting the question "is SLAM necessary?", we believe the answer depends on the application, but quite often the answer is a resounding *yes*. SLAM and related techniques, such

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还没有。 Censi 等人的初步研究。 [45]表明，给定任务的性能还取决于可用于传感的功率。它还表明，最佳传感架构可能具有多个传感器，这些传感器可以根据所需的性能水平瞬时打开和关闭，或者通过不同的物理原理测量相同的现象以实现鲁棒性[68]。
B。深度学习

如果一篇论文声称考虑 SLAM 的未来方向而不提及深度学习，那将是失职的。它对计算机视觉的影响是革命性的，并且在撰写本文时，它已经在传统机器人技术（包括 SLAM）领域取得了重大进展。

研究人员已经证明，可以学习深度神经网络来直接从原始图像对[52]回归从移动机器人获取的两个图像之间的帧间姿态，从而有效地取代视觉里程计的标准几何结构。同样，可以使用回归森林[247]和深度卷积神经网络[129]来定位相机的6DoF，并根据输入图像的函数从单个视图估计场景（实际上是地图）的深度[27,70,158]。

在我们看来，这并不意味着传统的 SLAM 已经消亡，而且现在说这些方法是否只是简单地展示原则上可以做什么的好奇心，但它们不会取代传统的、易于理解的方法，或者它们是否会完全取代，还为时过早。

开放问题。我们在这里强调了 SLAM 的一系列未来方向，我们相信机器学习，更具体地说是深度学习将产生影响，或者 SLAM 应用程序将为深度学习带来挑战。

_感知工具：_很明显，一些现有的计算机视觉算法无法解决的感知问题现在可以得到解决。例如，imagenet 类 [213] 的对象识别现在在某种程度上可以被视为黑匣子，从机器人专家或 SLAM 研究人员的角度来看，它工作得很好。同样，各种场景类型中像素的语义标记达到了约 80% 准确度或更高的性能水平 [74]。我们已经对 SLAM 系统向语义更有意义的地图的发展进行了广泛的评论，而这些黑盒工具将加速这一进程。但还有更多的利害关系：深度网络比之前的任何网络都更有希望将原始传感器数据连接到理解，或将原始传感器数据连接到行动。

_实际部署：_深度学习的成功主要取决于超级计算机上的长时间训练以及专用 GPU 硬件上的推理以获得一次性结果。 SLAM 研究人员（或者实际上任何想要将令人印象深刻的结果嵌入到他们的系统中的人）面临的挑战是如何在嵌入式系统中提供足够的计算能力。我们是简单地等待技术迎头赶上，还是研究更小、更便宜、能够产生“足够好”结果的网络，并考虑长期传感的影响？

_在线和终身学习：_更大、更重要的挑战是在线学习和适应，这对于任何未来的长期 SLAM 系统都至关重要。 SLAM 系统通常在一个持续观察的开放世界中运行，在那里可以遇到新的物体和场景。但迄今为止，深度网络通常是在具有固定数量的对象类的封闭世界场景中进行训练的。 运行支撑框架（Harness） 面临的一个重大挑战是在单次或零次场景（即新类的一个甚至零个训练示例）中深度网络的力量，以实现持续移动、持续观察的 SLAM 系统的终身学习。

同样，现有网络倾向于在大量标记数据上进行训练，但不能总是保证存在合适的数据集或可用于标记监督训练。最近取得一些进展的一个领域是单视图深度估计：Garg 等人[90]最近展示了如何通过观察大量立体对语料库来简单地训练用于单视图深度估计的深度网络，而不需要显式地观察或计算深度。是否可以为语义场景标记等任务开发类似的方法还有待观察。

Bootstrapping： 越来越多的证据表明，有关场景的先验信息可以为 SLAM 系统提供显着的提升。迄今为止文献中的示例包括已知对象 [56, 217] 或有关场景中预期结构的先验知识，例如 DTAM [184] 中的平滑度、

[79] 中的曼哈顿约束，甚至对象之间的预期关系 [9]。显然，深度学习能够为特定任务（例如估计场景标签或场景深度）提炼此类先验知识。如何最好地提取和使用这些信息是一个重要的开放问题。它在 SLAM 中比在其他一些领域更相关，因为在 SLAM 中我们对场景几何的数学有扎实的掌握--接下来的问题是如何将这种易于理解的几何与深度网络的输出融合起来。必须解决的一个特殊挑战是表征从深层网络得出的估计的不确定性。

SLAM 为探索深度学习架构和大规模图形模型中的递归状态估计之间的潜在联系提供了一个具有挑战性的环境。例如，Krishan 等人[142]最近提出了深度卡尔曼滤波器；也许有一天可以使用深层架构创建一个端到端的 SLAM 系统，而不需要显式的特征建模、数据关联等。

十、结论

同步定位和地图绘制问题在过去 30 年中取得了巨大进展。一路走来，随着新应用、新传感器和新计算工具的发展，一些重要问题得到了解答，同时也提出了许多新的有趣的问题。

重新审视“SLAM 是否必要？”这个问题，我们相信答案取决于应用，但通常答案是响亮的_是_。 SLAM及相关技术，例如

as visual-inertial odometry, are being increasingly deployed in a variety of real-world settings, from self-driving cars to mobile devices. SLAM techniques will be increasingly relied upon to provide reliable metric positioning in situations where infrastructure-based solutions such as GPS are unavailable or do not provide sufficient accuracy. One can envision cloud-based location-as-a-service capabilities coming online, and maps becoming commoditized, due to the value of positioning information for mobile devices and agents.

In some applications, such as self-driving cars, precision localization is often performed by matching current sensor data to a high definition map of the environment that is created in advance [154]. If the a priori map is accurate, then online SLAM is not required. Operations in highly dynamic environments, however, will require dynamic online map updates to deal with construction or major changes to road infrastructure. The distributed updating and maintenance of visual maps created by large fleets of autonomous vehicles is a compelling area for future work.

One can identify tasks for which different flavors of SLAM formulations are more suitable than others. For instance, a topological map can be used to analyze reachability of a given place, but it is not suitable for motion planning and low-level control; a locally-consistent metric map is well-suited for obstacle avoidance and local interactions with the environment, but it may sacrifice accuracy; a globally-consistent metric map allows the robot to perform global path planning, but it may be computationally demanding to compute and maintain.

One may even devise examples in which SLAM is unnecessary altogether and can be replaced by other techniques, e.g., visual servoing for local control and stabilization, or "teach and repeat" to perform repetitive navigation tasks. A more general way to choose the most appropriate SLAM system is to think about SLAM as a mechanism to compute a sufficient statistic that summarizes all past observations of the robot, and in this sense, which information to retain in this compressed representation is deeply task-dependent.

As to the familiar question "is SLAM solved?", in this position paper we argue that, as we enter the *robust-perception age*, the question cannot be answered without specifying a robot/environment/performance combination. For many applications and environments, numerous major challenges and important questions remain open. To achieve truly robust perception and navigation for long-lived autonomous robots, more research in SLAM is needed. As an academic endeavor with important real-world implications, SLAM is *not* solved.

The unsolved questions involve four main aspects: *robust performance*, *high-level understanding*, *resource awareness*, and *task-driven inference*. From the perspective of robustness, the design of fail-safe, self-tuning SLAM systems is a formidable challenge with many aspects being largely unexplored. For long-term autonomy, techniques to construct and maintain large-scale time-varying maps, as well as policies that define when to remember, update, or forget information, still need a large amount of fundamental research; similar problems arise, at a different scale, in severely resource-constrained robotic systems.

Another fundamental question regards the design of metric and semantic representations for the environment. Despite the fact that the interaction with the environment is paramount for most applications of robotics, modern SLAM systems are not able to provide a tightly-coupled high-level understanding of the geometry and the semantic of the surrounding world; the design of such representations must be task-driven and currently a tractable framework to link task to optimal representations is lacking. Developing such a framework will bring together the robotics and computer vision communities.

Besides discussing many accomplishments and future challenges for the SLAM community, we also examined opportunities connected to the use of novel sensors, new tools (e.g., convex relaxations and duality theory, or deep learning), and the role of active sensing. SLAM still constitutes an indispensable backbone for most robotics applications and despite the amazing progress over the past decades, existing SLAM systems are far from providing insightful, actionable, and compact models of the environment, comparable to the ones effortlessly created and used by humans.

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REFERENCES

[1] P. A. Absil, R. Mahony, and R. Sepulchre. *Optimization Algorithms on Matrix Manifolds*. Princeton University Press, 2007.

[2] S. Agarwal, N. Snavely, I. Simon, S. M. Seitz, and R. Szeliski. Bundle adjustment in the large. In *European Conference on Computer Vision (ECCV)*, pages 29–42. Springer, 2010.

[3] S. Anderson, T. D. Barfoot, C. H. Tong, and S. Särkkä. Batch Nonlinear Continuous-time Trajectory Estimation As Exactly Sparse Gaussian Process Regression. *Autonomous Robots (AR)*, 39(3):221–238, Oct. 2015.

[4] S. Anderson, F. Dellaert, and T. D. Barfoot. A hierarchical wavelet decomposition for continuous-time SLAM. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 373–380. IEEE, 2014.

[5] R. Aragues, J. Cortes, and C. Sagues. Distributed Consensus on Robot Networks for Dynamically Merging Feature-Based Maps. *IEEE Transactions on Robotics (TRO)*, 28(4):840–854, 2012.

[6] J. Aulinas, Y. Petillot, J. Salvi, and X. Lladó. The SLAM Problem: A Survey. In *Proceedings of the International Conference of the Catalan Association for Artificial Intelligence*, pages 363–371. IOS Press, 2008.

[7] T. Bailey and H. F. Durrant-Whyte. Simultaneous Localisation and Mapping (SLAM): Part II. *Robotics and Autonomous Systems (RAS)*, 13(3):108–117, 2006.

[8] R. Bajcsy. Active Perception. *Proceedings of the IEEE*, 76(8):966–1005, 1988.

[9] S. Y. Bao, M. Bagra, Y. W. Chao, and S. Savarese. Semantic structure from motion with points, regions, and objects. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2703–2710. IEEE, 2012.

[10] A. G. Barto and R. S. Sutton. Goal seeking components for adaptive intelligence: An initial assessment. Technical Report Technical Report

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作为视觉惯性里程计，正越来越多地部署在各种现实环境中，从自动驾驶汽车到移动设备。在基于基础设施的解决方案（例如 GPS）不可用或不能提供足够精度的情况下，SLAM 技术将越来越依赖于提供可靠的度量定位。由于移动设备和智能体的定位信息的价值，人们可以预见基于云的位置即服务功能即将上线，地图将变得商品化。

在某些应用中，例如自动驾驶汽车，通常通过将当前传感器数据与预先创建的高清环境地图进行匹配来执行精确定位[154]。如果先验地图是准确的，那么就不需要在线SLAM。然而，在高度动态的环境中运营将需要动态在线地图更新，以应对道路基础设施的建设或重大变化。由大型自动驾驶车辆创建的视觉地图的分布式更新和维护是未来工作的一个引人注目的领域。

人们可以识别出不同风格的 SLAM 公式比其他公式更适合的任务。例如，拓扑图可用于分析给定地点的可达性，但它不适用于运动规划和低级控制；局部一致的度量地图非常适合避障和与环境的局部交互，但它可能会牺牲准确性；全局一致的度量地图允许机器人执行全局路径规划，但计算和维护可能需要大量计算。

人们甚至可以设计一些例子，其中 SLAM 完全没有必要，可以用其他技术代替，例如用于本地控制和稳定的视觉伺服，或“教导和重复”来执行重复的导航任务。选择最合适的 SLAM 系统的更通用方法是将 SLAM 视为一种计算足够统计量的机制，该统计量总结了机器人过去的所有观察结果，从这个意义上说，在这种压缩表示中保留哪些信息是深度依赖于任务的。

至于熟悉的问题“SLAM 解决了吗？”，在这篇立场文件中，我们认为，当我们进入_鲁棒感知时代_时，如果不指定机器人/环境/性能组合，这个问题就无法回答。对于许多应用程序和环境来说，许多重大挑战和重要问题仍然悬而未决。为了实现长寿命自主机器人真正强大的感知和导航，需要对 SLAM 进行更多研究。作为一项具有重要现实意义的学术努力，SLAM 尚未得到解决。

未解决的问题涉及四个主要方面：鲁棒性能、高层次理解、_资源感知_和_任务驱动推理_。从鲁棒性的角度来看，故障安全、自调整 SLAM 系统的设计是一项艰巨的挑战，许多方面在很大程度上尚未得到探索。对于长期自治，构建和维护大规模时变地图的技术，以及定义何时记住、更新或忘记信息的策略，仍然需要大量的基础研究

;在资源严重受限的机器人系统中，也会出现不同规模的类似问题。

另一个基本问题是关于度量的设计

以及环境的语义表示。尽管与环境的交互对于大多数机器人应用来说至关重要，但现代 SLAM 系统无法提供对周围世界的几何和语义的紧密耦合的高级理解；这种表示的设计必须是任务驱动的，目前缺乏将任务与最佳表示联系起来的易于处理的框架。开发这样的框架将把机器人和计算机视觉社区聚集在一起。

除了讨论 SLAM 社区的许多成就和未来挑战之外，我们还研究了与使用新型传感器、新工具（例如凸松弛和对偶理论或深度学习）以及主动传感的作用相关的机会。 SLAM 仍然是大多数机器人应用不可或缺的支柱，尽管在过去几十年中取得了惊人的进步，但现有的 SLAM 系统还远远不能提供有洞察力、可操作且紧凑的环境模型，与人类轻松创建和使用的模型相比。

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参考文献

- [1] P. A. Absil、R. Mahony 和 R. Sepulchre。 矩阵流形上的优化算法。普林斯顿大学出版社，2007。

- [2] S. Agarwal、N. Snavely、I. Simon、S. M. Seitz 和 R. Szeliski。捆绑调整幅度大。在《欧洲计算机视觉会议 (ECCV)》中，第 29-42 页。施普林格，2010。

- [3] S. Anderson、T. D. Barfoot、C. H. Tong 和 S. Sárkka。批量非线性连续时间轨迹估计作为精确稀疏高斯过程回归。 自主机器人 (AR)，39(3):221-238，2015 年 10 月。

- [4] S. Anderson、F. Dellaert 和 T. D. Barfoot。连续时间 SLAM 的分层小波分解。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 373-380 页。 IEEE，2014。

- [5] R. Aragues、J. Cortes 和 C. Sagues。用于动态合并基于特征的地图的机器人网络分布式共识。IEEE 机器人学报 (TRO)，28(4):840-854，2012 年。

- [6] J. Aulinas、Y. Petillot、J. Salvi 和 X. Llad´o。 SLAM 问题：一项调查。见_加泰罗尼亚人工智能协会国际会议记录_，第 363-371 页。 IOS 出版社，2008 年。

- [7] T. Bailey 和 H. F. Durrant-Whyte。同步定位和建图 (SLAM)：第二部分

。 机器人和自主系统 (RAS)，13(3):108-117，2006 年。

- [8] R. Bajcsy。主动感知。 IEEE 会议录，76(8):966-1005，1988。

- [9] S. Y. Bao、M. Bagra、Y. W. Chao 和 S. Savarese。来自点、区域和对象运动的语义结构。在《IEEE 计算机视觉和模式识别会议 (CVPR) 会议记录》中，第 2703-2710 页。 IEEE，2012。

- [10] A. G. 巴托和 R. S. 萨顿。适应性智能的目标寻求组件：初步评估。技术报告 技术报告

AFWAL-TR-81-1070, Air Force Wright Aeronautical Laboratories. Wright-Patterson Air Force Base, Jan. 1981.

[11] C. Bibby and I. Reid. Simultaneous Localisation and Mapping in Dynamic Environments (SLAMIDE) with Reversible Data Association. In *Proceedings of Robotics: Science and Systems Conference (RSS)*, pages 105–112, Atlanta, GA, USA, June 2007.

[12] C. Bibby and I. D. Reid. A hybrid SLAM representation for dynamic marine environments. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 1050–4729. IEEE, 2010.

[13] T. O. Binford. Visual perception by computer. In *IEEE Conference on Systems and Controls*. IEEE, 1971.

[14] T. E. Bishop and P. Favaro. The light field camera: Extended depth of field, aliasing, and superresolution. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 34(5):972–986, 2012.

[15] J. L. Blanco, J. A. Fernández-Madrigal, and J. Gonzalez. A Novel Measure of Uncertainty for Mobile Robot SLAM with RaoBlackwellized Particle Filters. *The International Journal of Robotics Research (IJRR)*, 27(1):73–89, 2008.

[16] J. Bloomenthal. *Introduction to Implicit Surfaces*. Morgan Kaufmann, 1997.

[17] A. Bodis-Szomoru, H. Riemenschneider, and L. Van-Gool. Efficient edge-aware surface mesh reconstruction for urban scenes. *Journal of Computer Vision and Image Understanding*, 66:91–106, 2015.

[18] M. Bosse, P. Newman, J. J. Leonard, and S. Teller. Simultaneous Localization and Map Building in Large-Scale Cyclic Environments Using the Atlas Framework. *The International Journal of Robotics Research (IJRR)*, 23(12):1113–1139, 2004.

[19] M. Bosse and R. Zlot. Continuous 3D Scan-Matching with a Spinning 2D Laser. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 4312–4319. IEEE, 2009.

[20] M. Bosse and R. Zlot. Keypoint design and evaluation for place recognition in 2D LIDAR maps. *Robotics and Autonomous Systems (RAS)*, 57(12):1211–1224, 2009.

[21] M. Bosse, R. Zlot, and P. Flick. Zebedee: Design of a Spring-Mounted 3D Range Sensor with Application to Mobile Mapping. *IEEE Transactions on Robotics (TRO)*, 28(5):1104–1119, 2012.

[22] F. Bourgault, A. A. Makarenko, S. B. Williams, B. Grocholsky, and H. F. Durrant-Whyte. Information Based Adaptive Robotic Exploration. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 540–545. IEEE, 2002.

[23] C. Brand, M. J. Schuster, H. Hirschmuller, and M. Suppa. Stereo-Vision Based Obstacle Mapping for Indoor/Outdoor SLAM. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 2153–0858. IEEE, 2014.

[24] W. Burgard, M. Moors, C. Stachniss, and F. E. Schneider. Coordinated Multi-Robot Exploration. *IEEE Transactions on Robotics (TRO)*, 21(3):376–386, 2005.

[25] C. Cadena, L. Carlone, H. Carrillo, Y. Latif, J. Neira, I. D. Reid, and J. J. Leonard. Robotics: Science and Systems (RSS), Workshop "The problem of mobile sensors: Setting future goals and indicators of progress for SLAM". http://ylatif.github.io/movingsensors/, Google Plus Community: https://plus.google.com/communities/102832228492942322585, June 2015.

[26] C. Cadena, A. Dick, and I. D. Reid. A Fast, Modular Scene Understanding System using Context-Aware Object Detection. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 4859–4866. IEEE, 2015.

[27] C. Cadena, A. Dick, and I. D. Reid. Multi-modal Auto-Encoders as Joint Estimators for Robotics Scene Understanding. In *Proceedings of Robotics: Science and Systems Conference (RSS)*, pages 377–386, 2016.

[28] N. Carlevaris-Bianco and R. M. Eustice. Generic factor-based node marginalization and edge sparsification for pose-graph SLAM. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 5748–5755. IEEE, 2013.

[29] L. Carlone. Convergence Analysis of Pose Graph Optimization via Gauss-Newton methods. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 965–972. IEEE, 2013.

[30] L. Carlone, R. Aragues, J. A. Castellanos, and B. Bona. A fast and accurate approximation for planar pose graph optimization. *The International Journal of Robotics Research (IJRR)*, 33(7):965–987, 2014.

[31] L. Carlone, G. Calafiore, C. Tommolillo, and F. Dellaert. Planar Pose Graph Optimization: Duality, Optimal Solutions, and Verification. *IEEE Transactions on Robotics (TRO)*, 32(3):545–565, 2016.

[32] L. Carlone and A. Censi. From Angular Manifolds to the Integer Lattice: Guaranteed Orientation Estimation With Application to Pose Graph Optimization. *IEEE Transactions on Robotics (TRO)*, 30(2):475–492, 2014.

[33] L. Carlone, A. Censi, and F. Dellaert. Selecting good measurements via ℓ_1 relaxation: a convex approach for robust estimation over graphs. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 2667 –2674. IEEE, 2014.

[34] L. Carlone and F. Dellaert. Duality-based Verification Techniques for 2D SLAM. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 4589–4596. IEEE, 2015.

[35] L. Carlone, J. Du, M. Kaouk, B. Bona, and M. Indri. Active SLAM and Exploration with Particle Filters Using Kullback-Leibler Divergence. *Journal of Intelligent & Robotic Systems*, 75(2):291–311, 2014.

[36] L. Carlone, D. Rosen, G. Calafiore, J. J. Leonard, and F. Dellaert. Lagrangian Duality in 3D SLAM: Verification Techniques and Optimal Solutions. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 125–132. IEEE, 2015.

[37] L. Carlone, R. Tron, K. Daniilidis, and F. Dellaert. Initialization Techniques for 3D SLAM: a Survey on Rotation Estimation and its Use in Pose Graph Optimization. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 4597–4604. IEEE, 2015.

[38] J. C. Carr, R. K. Beatson, J. B. Cherrie, T. J. Mitchell, W. R. Fright, B. C. McCallum, and T. R. Evans. Reconstruction and Representation of 3D Objects with Radial Basis Functions. In *SIGGRAPH*, pages 67–76. ACM, 2001.

[39] H. Carrillo, O. Birbach, H. Taubig, B. Bauml, U. Frese, and J. A. Castellanos. On Task-oriented Criteria for Configurations Selection in Robot Calibration. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 3653–3659. IEEE, 2013.

[40] H. Carrillo, P. Dames, K. Kumar, and J. A. Castellanos. Autonomous Robotic Exploration Using Occupancy Grid Maps and Graph SLAM Based on Shannon and Rényi Entropy. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 487–494. IEEE, 2015.

[41] H. Carrillo, Y. Latif, J. Neira, and J. A. Castellanos. Fast Minimum Uncertainty Search on a Graph Map Representation. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 2504–2511. IEEE, 2012.

[42] H. Carrillo, Y. Latif, M. L. Rodríguez, J. Neira, and J. A. Castellanos. On the Monotonicity of Optimality Criteria during Exploration in Active SLAM. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 1476–1483. IEEE, 2015.

[43] H. Carrillo, I. Reid, and J. A. Castellanos. On the Comparison of Uncertainty Criteria for Active SLAM. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 2080–2087. IEEE, 2012.

[44] R. O. Castle, D. J. Gawley, G. Klein, and D. W. Murray. Towards simultaneous recognition, localization and mapping for hand-held and wearable cameras. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 4102–4107. IEEE, 2007.

[45] A. Censi, E. Mueller, E. Frazzoli, and S. Soatto. A Power-Performance Approach to Comparing Sensor Families, with application to comparing neuromorphic to traditional vision sensors. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 3319–3326. IEEE, 2015.

[46] A. Chiuso, G. Picci, and S. Soatto. Wide-sense Estimation on the Special Orthogonal Group. *Communications in Information and Systems*, 8:185–200, 2008.

[47] S. Choudhary, L. Carlone, C. Nieto, J. Rogers, H. I. Christensen, and F. Dellaert. Distributed Trajectory Estimation with Privacy and Communication Constraints: a Two-Stage Distributed Gauss-Seidel Approach. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 5261–5268. IEEE, 2015.

[48] W. Churchill and P. Newman. Experience-based navigation for long-term localisation. *The International Journal of Robotics Research (IJRR)*, 32(14):1645–1661, 2013.

[49] T. Cieslewski, L. Simon, M. Dymczyk, S. Magnenat, and R. Siegwart. Map API - Scalable Decentralized Map Building for Robots. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 6241–6247. IEEE, 2015.

[50] J. Civera, D. Gálvez-López, L. Riazuelo, J. D. Tardós, and J. M. M. Montiel. Towards Semantic SLAM Using a Monocular Camera. In *Proceedings of the IEEE/RSJ International Conference on Intelligent*

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- AFWAL-TR-81-1070，空军莱特航空实验室。莱特-帕特森空军基地，1981 年 1 月。

- [11] C.毕比和I.里德。具有可逆数据关联的动态环境中的同步定位和绘图 (SLAMIDE)。收录于《机器人学报：科学与系统会议 (RSS)》，第 105-112 页，美国佐治亚州亚特兰大，2007 年 6 月。

- [12] C.毕比和 I.D.里德。动态海洋环境的混合 SLAM 表示。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 1050-4729 页。 IEEE，2010。

- [13] T.O.宾福德。计算机视觉感知。在 IEEE 系统与控制会议 中。 IEEE，1971。

- [14] T. E. Bishop 和 P. Favaro。光场相机：扩展景深、混叠和超分辨率。 IEEE 模式分析和机器智能汇刊，34(5):972-986，2012。

- [15] J. L. Blanco、J. A. Fern´andez-Madrigal 和 J. Gonzalez。使用 RaoBlackwellized 粒子滤波器测量移动机器人 SLAM 不确定性的新方法。 国际机器人研究杂志 (IJRR)，27(1):73-89，2008 年。

- [16] J.布卢门撒尔。 隐式曲面简介 。摩根·考夫曼，1997。

- [17] A. Bodis-Szomoru、H. Riemenschneider 和 L. Van-Gool。针对城市场景的高效边缘感知表面网格重建。 计算机视觉与图像理解杂志，66：91-106，2015。

- [18] M. Bosse、P. Newman、J. J. Leonard 和 S. Teller。使用 Atlas 框架在大规模循环环境中同时定位和地图构建。 国际机器人研究杂志 (IJRR)，23(12):1113-1139，2004 年。

- [19] M. Bosse 和 R. Zlot。使用旋转 2D 激光进行连续 3D 扫描匹配。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 4312-4319 页。 IEEE，2009。

- [20] M. Bosse 和 R. Zlot。二维激光雷达地图中地点识别的关键点设计和评估。 机器人和自主系统 (RAS)，57(12):1211-1224，2009。

- [21] M. Bosse、R. Zlot 和 P. Flick。 Zebedee：适用于移动测绘的 SpringMounted 3D 距离传感器设计。 IEEE 机器人学报 (TRO)，28(5):1104-1119，2012。

- [22] F. Bourgault、A. A. Makarenko、S. B. Williams、B. Grocholsky 和 H. F. Durrant-Whyte。基于信息的自适应机器人探索。在《IEEE/RSJ 国际智能机器人和系统会议 (IROS) 会议记录》中，第 540-545 页。 IEEE，2002。

- [23] C. Brand、M. J. Schuster、H. Hirschmuller 和 M. Suppa。基于立体视觉的室内/室外 SLAM 障碍物测绘。在《IEEE/RSJ 国际智能机器人和系统会议 (IROS) 会议记录》中，第 2153-0858 页。 IEEE，2014。

- [24] W. Burgard、M. Moors、C. Stachniss 和 F. E. Schneider。协调多机器人探索。 IEEE 机器人学报 (TRO)，21(3):376-386，2005 年。

- [25] C. Cadena、L. Carlone、H. Carrillo、Y. Latif、J. Neira、I. D. Reid 和 J. J. Leonard。机器人学：科学与系统 (RSS)，研讨会“移动传感器的问题：设定 SLAM 的未来目标和进展指标”。 http://ylatif.github.io/ movingsensors/，Google Plus 社区：https://plus.google.com/communities/102832228492942322585，2015 年 6 月。

- [26] C. Cadena、A. Dick 和 I. D. Reid。使用上下文感知对象检测的快速模块化场景理解系统。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 4859-4866 页。 IEEE，2015。

- [27] C. Cadena、A. Dick 和 I. D. Reid。多模态自动编码器作为机器人场景理解的联合估计器。收录于《机器人学报：科学与系统会议 (RSS)》，第 377-386 页，2016 年。

- [28] N. Carlevaris-Bianco 和 R. M. Eustice。位姿图 SLAM 的基于通用因子的节点边缘化和边缘稀疏化。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 5748-5755 页。 IEEE，2013。

- [29] L.卡隆。通过高斯-牛顿方法进行位姿图优化的收敛性分析。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 965-972 页。 IEEE，2013。

- [30] L. Carlone、R. Aragues、J. A. Castellanos 和 B. Bona。平面位姿图优化的快速且准确的近似。 国际机器人研究杂志 (IJRR)，33(7):965-987，2014 年。

- [31] L. Carlone、G. Calafiore、C. Tommolillo 和 F. Dellaert。平面位姿图优化：对偶性、最优解和验证。 IEEE 机器人学报 (TRO)，32(3):545-565，2016。

- [32] L. Carlone 和 A. Censi。从角度流形到整数晶格：保证方向估计及其应用于位姿图优化。 IEEE 机器人学报 (TRO)，30(2):475-492，2014 年。

- [33] L. Carlone、A. Censi 和 F. Dellaert。通过 $\ell 1$ 松弛选择良好的测量：一种用于对图形进行稳健估计的凸方法。在《IEEE/RSJ 国际智能机器人和系统会议 (IROS) 会议记录》中，第 2667 -2674 页。 IEEE，2014。

- [34] L. Carlone 和 F. Dellaert。基于对偶的 2D SLAM 验证技术。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 4589-4596 页。 IEEE，2015。

- [35] L. Carlone、J. Du、M. Kaouk、B. Bona 和 M. Indri。使用 Kullback-Leibler 散度的主动 SLAM 和粒子滤波器探索。 智能与机器人系统杂志，75(2):291-311，2014。

- [36] L. Carlone、D. Rosen、G. Calafiore、J. J. Leonard 和 F. Dellaert。 3D SLAM 中的拉格朗日对偶：验证技术和最佳解决方案。在《IEEE/RSJ 国际智能机器人和系统会议 (IROS) 会议记录》中，第 125-132 页。 IEEE，2015。

- [37] L. Carlone、R. Tron、K. Daniilidis 和 F. Dellaert。 3D SLAM 的初始化技术：旋转估计及其在位姿图优化中的应用的调查。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 4597-4604 页。 IEEE，2015。

- [38] J. C. Carr、R. K. Beatson、J. B. Cherrie、T. J. Mitchell、W. R. Fright、B. C. McCallum 和 T. R. Evans。使用径向基函数重建和表示 3D 对象。在 SIGGRAPH 中，第 67-76 页。美国CM，2001。

- [39] H. Carrillo、O. Birbach、H. Taubig、B. Bauml、U. Frese 和 J. A. Castellanos。机器人标定中面向任务的配置选择标准研究

。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 3653-3659 页。 IEEE，2013。

- [40] H. Carrillo、P. Dames、K. Kumar 和 J. A. Castellanos。基于 Shannon 和 R´enyi 熵的使用占用网格地图和图 SLAM 的自主机器人探索。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 487-494 页。 IEEE，2015。

- [41] H. Carrillo、Y. Latif、J. Neira 和 J. A. Castellanos。图形地图表示上的快速最小不确定性搜索。在《IEEE/RSJ 国际智能机器人和系统会议 (IROS) 会议记录》中，第 2504-2511 页。 IEEE，2012。

- [42] H. Carrillo、Y. Latif、M. L. Rodr´ıguez、J. Neira 和 J. A. Castellanos。关于主动 SLAM 探索过程中最优性准则的单调性。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 1476-1483 页。 IEEE，2015。

- [43] H. Carrillo、I. Reid 和 J. A. Castellanos。主动 SLAM 不确定性准则的比较。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 2080-2087 页。 IEEE，2012。

- [44] R. O. Castle、D. J. Gawley、G. Klein 和 D. W. Murray。实现手持式和可穿戴式相机的同步识别、定位和绘图。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 4102-4107 页。 IEEE，2007。

- [45] A. Censi、E. Mueller、E. Frazzoli 和 S. Soatto。比较传感器系列的功耗-性能方法，用于比较神经形态传感器与传统视觉传感器。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 3319-3326 页。 IEEE，2015。

- [46] A. Chiuso、G. Picci 和 S. Soatto。特殊正交群的广义估计。 信息和系统通信，8:185-200，2008 年。

- [47] S. Choudhary、L. Carlone、C. Nieto、J. Rogers、H. I. Christensen 和 F. Dellaert。具有隐私和通信约束的分布式轨迹估计：两阶段分布式高斯-赛德尔方法。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 5261-5268 页。 IEEE，2015。

- [48] W. 丘吉尔和 P. 纽曼。基于经验的导航，可实现长期本地化。 国际机器人研究杂志 (IJRR)，32(14):1645-1661，2013 年。

- [49] T. Cieslewski、L. Simon、M. Dymczyk、S. Magnat 和 R. Siegwart。地图 API - 为机器人构建可扩展的去中心化地图。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 6241-6247 页。 IEEE，2015。

- [50] J. Civera、D. G´alvez-L´opez、L. Riazuelo、J. D. Tard´os 和 J. M. M. Montiel。使用单目相机实现语义 SLAM。在_IEEE/RSJ 国际智能会议论文集_

Robots and Systems (IROS), pages 1277–1284. IEEE, 2011.

[51] A. Cohen, C. Zach, S. Sinha, and M. Pollefeys. Discovering and Exploiting 3D Symmetries in Structure from Motion. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2012.

[52] G. Costante, M. Mancini, P. Valigi, and T. A. Ciarfuglia. Exploring Representation Learning With CNNs for Frame-to-Frame Ego-Motion Estimation. *IEEE Robotics and Automation Letters*, 1(1):18–25, Jan 2016.

[53] M. Cummins and P. Newman. FAB-MAP: Probabilistic Localization and Mapping in the Space of Appearance. *The International Journal of Robotics Research (IJRR)*, 27(6):647–665, 2008.

[54] A. Cunningham, V. Indelman, and F. Dellaert. DDF-SAM 2.0: Consistent distributed smoothing and mapping. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 5220–5227. IEEE, 2013.

[55] B. Curless and M. Levoy. A volumetric method for building complex models from range images. In *SIGGRAPH*, pages 303–312. ACM, 1996.

[56] A. Dame, V. A. Prisacariu, C. Y. Ren, and I. D. Reid. Dense Reconstruction using 3D Object Shape Priors. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1288–1295. IEEE, 2013.

[57] D. G. Dansereau, D. L. Bongiorno, O. Pizarro, and S. B. Williams. Light Field Image Denoising Using a Linear 4D Frequency-Hyperfan all-in-focus Filter. In *Proceedings of the SPIE Conference on Computational Imaging*, volume 8657, pages 86570P1–86570P14, 2013.

[58] D. G. Dansereau, S. B. Williams, and P. Corke. Simple Change Detection from Mobile Light Field Cameras. *Computer Vision and Image Understanding*, 2016.

[59] A. Davison, I. Reid, N. Molton, and O. Stasse. MonoSLAM: Real-Time Single Camera SLAM. *IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI)*, 29(6):1052–1067, 2007.

[60] F. Dayoub, G. Cielniak, and T. Duckett. Long-term experiments with an adaptive spherical view representation for navigation in changing environments. *Robotics and Autonomous Systems (RAS)*, 59(5):285–295, 2011.

[61] F. Dellaert. Factor Graphs and GTSAM: A Hands-on Introduction. Technical Report GT-RIM-CP&R-2012-002, Georgia Institute of Technology, Sept. 2012.

[62] F. Dellaert, J. Carlson, V. Ila, K. Ni, and C. E. Thorpe. Subgraph-preconditioned Conjugate Gradient for Large Scale SLAM. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 2566–2571. IEEE, 2010.

[63] F. Dellaert and M. Kaess. Square Root SAM: Simultaneous Localization and Mapping via Square Root Information Smoothing. *The International Journal of Robotics Research (IJRR)*, 25(12):1181–1203, 2006.

[64] G. Dissanayake, S. Huang, Z. Wang, and R. Ranasinghe. A review of recent developments in Simultaneous Localization and Mapping. In *International Conference on Industrial and Information Systems*, pages 477–482. IEEE, 2011.

[65] F. Dong, S.-H. Ieng, X. Savatier, R. Etienne-Cummings, and R. Benosman. Plenoptic cameras in real-time robotics. *The International Journal of Robotics Research (IJRR)*, 32(2), 2013.

[66] J. Dong, E. Nelson, V. Indelman, N. Michael, and F. Dellaert. Distributed real-time cooperative localization and mapping using an uncertainty-aware expectation maximization approach. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 5807–5814. IEEE, 2015.

[67] R. Dubé, H. Sommer, A. Gawel, M. Bosse, and R. Siegwart. Non-uniform sampling strategies for continuous correction based trajectory estimation. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 4792–4798. IEEE, 2016.

[68] H. F. Durrant-Whyte. An Autonomous Guided Vehicle for Cargo Handling Applications. *The International Journal of Robotics Research (IJRR)*, 15(5):407–440, 1996.

[69] H. F. Durrant-Whyte and T. Bailey. Simultaneous Localisation and Mapping (SLAM): Part I. *IEEE Robotics and Automation Magazine*, 13(2):99–110, 2006.

[70] D. Eigen and R. Fergus. Predicting depth, surface normals and semantic labels with a common multi-scale convolutional architecture. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2015.

[71] A. Elfes. Occupancy Grids: A Probabilistic Framework for Robot Perception and Navigation. *Journal on Robotics and Automation*, RA-3(3):249–265, 1987.

[72] J. Engel, J. Schöps, and D. Cremers. LSD-SLAM: Large-scale direct monocular SLAM. In *European Conference on Computer Vision (ECCV)*, pages 834–849. Springer, 2014.

[73] R. M. Eustice, H. Singh, J. J. Leonard, and M. R. Walter. Visually Mapping the RMS Titanic: Conservative Covariance Estimates for SLAM Information Filters. *The International Journal of Robotics Research (IJRR)*, 25(12):1223–1242, 2006.

[74] M. Everingham, L. Van-Gool, C. K. I. Williams, J. Winn, and A. Zisserman. The PASCAL Visual Object Classes (VOC) Challenge. *International Journal of Computer Vision*, 88(2):303–338, 2010.

[75] N. Fairfield, G. Kantor, and D. Wettergreen. Real-Time SLAM with Octree Evidence Grids for Exploration in Underwater Tunnels. *Journal of Field Robotics (JFR)*, 24(1-2):03–21, 2007.

[76] N. Fairfield and D. Wettergreen. Active SLAM and Loop Prediction with the Segmented Map Using Simplified Models. In *Proceedings of International Conference on Field and Service Robotics (FSR)*, pages 173–182. Springer, 2010.

[77] H. J. S. Feder. *Simultaneous Stochastic Mapping and Localization*. PhD thesis, Massachusetts Institute of Technology, 1999.

[78] B. Ferris, D. Fox, and N. D. Lawrence. WiFi-SLAM using gaussian process latent variable models. In *International Joint Conference on Artificial Intelligence*, 2007.

[79] A. Flint, D. Murray, and I. D. Reid. Manhattan Scene Understanding Using Monocular, Stereo, and 3D Features. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 2228–2235. IEEE, 2011.

[80] J. Foley, A. van Dam, S. Feiner, and J. Hughes. *Computer Graphics: Principles and Practice*. Addison-Wesley, 1992.

[81] J. Folkesson and H. Christensen. Graphical SLAM for outdoor applications. *Journal of Field Robotics (JFR)*, 23(1):51–70, 2006.

[82] C. Forster, L. Carlone, F. Dellaert, and D. Scaramuzza. On-Manifold Preintegration for Real-Time Visual–Inertial Odometry. *IEEE Transactions on Robotics (TRO)*, PP(99):1–21, 2016.

[83] C. Forster, Z. Zhang, M. Gassner, M. Werlberger, and D. Scaramuzza. SVO: Semi-Direct Visual Odometry for Monocular and Multi-Camera Systems. *IEEE Transactions on Robotics (TRO)*, PP(99):1–17, 2016.

[84] D. Fox, J. Ko, K. Konolige, B. Limketkai, D. Schulz, and B. Stewart. Distributed Multirobot Exploration and Mapping. *Proceedings of the IEEE*, 94(7):1325–1339, July 2006.

[85] F. Fraundorfer and D. Scaramuzza. Visual Odometry. Part II: Matching, Robustness, Optimization, and Applications. *IEEE Robotics and Automation Magazine*, 19(2):78–90, 2012.

[86] J. Fredriksson and C. Olsson. Simultaneous Multiple Rotation Averaging using Lagrangian Duality. In *Asian Conference on Computer Vision (ACCV)*, pages 245–258. Springer, 2012.

[87] U. Frese. Interview: Is SLAM Solved? *KI - Künstliche Intelligenz*, 24(3):255–257, 2010.

[88] P. Furgale, T. D. Barfoot, and G. Sibley. Continuous-Time Batch Estimation using Temporal Basis Functions. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 2088–2095. IEEE, 2012.

[89] G. Gallego, J. Lund, E. Mueggler, H. Rebecq, T. Delbruck, and D. Scaramuzza. Event-based, 6-DOF Camera Tracking for High-Speed Applications. *CoRR*, abs/1607.03468:1–8, 2016.

[90] R. Garg, V. Kumar, G. Carneiro, and I. Reid. Unsupervised CNN for Single View Depth Estimation: Geometry to the Rescue. In *European Conference on Computer Vision (ECCV)*, pages 740–756, 2016.

[91] R. Garg, A. Roussos, and L. Agapito. Dense Variational Reconstruction of Non-Rigid Surfaces from Monocular Video. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1272–1279, 2013.

[92] J. J. Gibson. *The ecological approach to visual perception: classic edition*. Psychology Press, 2014.

[93] D. Golovin and A. Krause. Adaptive Submodularity: Theory and Applications in Active Learning and Stochastic Optimization. *Journal of Artificial Intelligence Research*, 42(1):427–486, 2011.

[94] Google. Project tango. https://www.google.com/atap/projecttango/, 2016.

[95] V. M. Govindu. Combining Two-view Constraints For Motion Estimation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 218–225. IEEE, 2001.

[96] O. Grasa, E. Bernal, S. Casado, I. Gil, and J. M. M. Montiel. Visual SLAM for Handheld Monocular Endoscope. *IEEE Transactions on Medical Imaging*, 33(1):135–146, 2014.

[97] G. Grisetti, R. Kümmerle, C. Stachniss, and W. Burgard. A Tutorial on Graph-based SLAM. *IEEE Intelligent Transportation Systems Magazine*, 2(4):31–43, 2010.

原文第 21 页 | 中文翻译初稿

20

- 机器人和系统 (IROS)，第 1277-1284 页。 IEEE，2011。

- [51] A. Cohen、C. Zach、S. Sinha 和 M. Pollefeys。从运动中发现和利用结构中的 3D 对称性。收录于 _IEEE 计算机视觉和模式识别会议 (CVPR) 会议记录_，2012 年。

- [52] G. Costante、M. Mancini、P. Valigi 和 T. A. Ciarfuglia。探索使用 CNN 进行表示学习以进行帧到帧自我运动估计。 IEEE 机器人与自动化快报，1(1):18-25，2016 年 1 月。

- [53] M.康明斯和P.纽曼。 FAB-MAP：外观空间中的概率定位和绘图。 国际机器人研究杂志 (IJRR)，27(6):647-665，2008。

- [54] A. Cunningham、V. Indelman 和 F. Dellaert。 DDF-SAM 2.0：一致的分布式平滑和映射。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 5220-5227 页。 IEEE，2013。

- [55] B. Curless 和 M. Levoy。一种从范围图像构建复杂模型的体积方法。在 SIGGRAPH 中，第 303-312 页。美国计算机学会，1996。

- [56] A. Dame、V. A. Prisacariu、C. Y. Ren 和 I. D. Reid。使用 3D 对象形状先验进行密集重建。在《IEEE 计算机视觉和模式识别会议 (CVPR) 会议记录》中，第 1288-1295 页。 IEEE，2013。

- [57] D. G. Dansereau、D. L. Bongiorno、O. Pizarro 和 S. B. Williams。使用线性 4D 频率 Hyperfan 全焦点滤波器进行光场图像去噪。收录于《SPIE 计算成像会议论文集》，第 8657 卷，第 86570P1-86570P14 页，2013 年。

- [58] D. G. Dansereau、S. B. Williams 和 P. Corke。移动光场相机的简单变化检测。 计算机视觉与图像理解，2016。

- [59] A.戴维森、I.里德、N.莫尔顿和O.斯塔斯。 MonoSLAM：实时单相机 SLAM。 IEEE 模式分析和机器智能汇刊 (PAMI)，29(6):1052-1067，2007 年。

- [60] F. Dayoub、G. Cielniak 和 T. Duckett。使用自适应球形视图表示进行长期实验，以在不断变化的环境中进行导航。 机器人和自主系统 (RAS)，59(5):285-295，2011。

- [61] F.德拉尔特。因子图和 GTSAM：实践介绍。技术报告 GT-RIM-CP&R-2012-002，佐治亚理工学院，2012 年 9 月。

- [62] F. Dellaert、J. Carlson、V. Ila、K. Ni 和 C. E. Thorpe。用于大规模 SLAM 的子图预处理共轭梯度。在《IEEE/RSJ 国际智能机器人和系统会议 (IROS) 会议记录》中，第 2566-2571 页。 IEEE，2010。

- [63] F. Dellaert 和 M. Kaess。平方根 SAM：通过平方根信息平滑进行同步定位和绘图。 国际机器人研究杂志 (IJRR)，25(12):1181-1203，2006 年。

- [64] G. Dissanayake、S. Huang、Z. Wang 和 R. Ranasinghe。对同步定位和测绘最新发展的回顾。见《工业和信息系统国际会议》，第 477-482 页。 IEEE，2011。

- [65] F. Dong, S.-H. Ieng、X.萨瓦蒂尔、R.艾蒂安-卡明斯和 R.贝诺斯曼。实时机器人技术中的全光相机。国际机器人研究杂志 (IJRR)，32(2)，2013。

- [66] J. Dong、E. Nelson、V. Indelman、N. Michael 和 F. Dellaert

。使用不确定性感知期望最大化方法进行分布式实时协作定位和映射。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 5807-5814 页。 IEEE，2015。

- [67] R. Dub´e、H. Sommer、A. Gawel、M. Bosse 和 R. Siegwart。基于连续校正的轨迹估计的非均匀采样策略。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 4792-4798 页。 IEEE，2016。

- [68] H. F. 杜兰特-怀特。用于货物装卸应用的自动导引车。 国际机器人研究杂志 (IJRR)，15(5):407-440，1996。

- [69] H. F. 杜兰特-怀特和 T. 贝利。同步定位和建图 (SLAM)：第 I 部分。IEEE 机器人与自动化杂志，13(2):99-110，2006。

- [70] D. Eigen 和 R. Fergus。使用通用的多尺度卷积架构预测深度、表面法线和语义标签。收录于《IEEE 国际计算机视觉会议 (ICCV) 会议记录》，2015 年。

- [71] A.埃尔夫斯。占用网格：机器人感知和导航的概率框架。 机器人与自动化杂志，RA3(3)：249-265，1987。

- [72] J. Engel、J. Schöps 和 D. Cremers。 LSD-SLAM：大规模直接单目SLAM。在《欧洲计算机视觉会议 (ECCV)》中，第 834-849 页。施普林格，2014。

- [73] R. M. 尤斯蒂斯、H. 辛格、J. J. 伦纳德和 M. R. 沃尔特。可视化绘制 RMS 泰坦尼克号：SLAM 信息过滤器的保守协方差估计。 国际机器人研究杂志 (IJRR)，25(12):1223-1242，2006 年。

- [74] M. Everingham、L. Van-Gool、C. K. I. Williams、J. Winn 和 A. Zisserman。 PASCAL 视觉对象类 (VOC) 挑战。 国际计算机视觉杂志，88(2):303-338，2010。

- [75] N. Fairfield、G. Kantor 和 D. Wettergreen。具有八叉树证据网格的实时 SLAM 用于水下隧道勘探。 现场机器人学杂志 (JFR)，24(1-2):03-21，2007 年。

- [76] N. Fairfield 和 D. Wettergreen。使用简化模型的分段地图进行主动 SLAM 和循环预测。摘自《现场和服务机器人国际会议记录 (FSR)》，第 173-182 页。施普林格，2010。

- [77] H. J. S. 费德。 同时随机映射和定位。博士论文，麻省理工学院，1999 年。

- [78] B. Ferris、D. Fox 和 N. D. Lawrence。 WiFi-SLAM 使用高斯过程潜变量模型。在_国际人工智能联合会议_，2007年。

- [79] A. 弗林特、D. 默里和 I. D. 里德。使用单目、立体和 3D 特征理解曼哈顿场景。在《IEEE 国际计算机视觉会议 (ICCV) 会议记录》中，第 2228-2235 页。IEEE，2011。

- [80] J. Foley、A. van Dam、S. Feiner 和 J. Hughes。 计算机图形学：原理与实践。艾迪生韦斯利，1992。

- [81] J.福克斯森和H.克里斯滕森。适用于户外应用的图形 SLAM。 现场机器人杂志 (JFR)，23(1):51-70，2006 年。

- [82] C. Forster、L. Carlone、F. Dellaert 和 D. Scaramuzza。用于实时视觉惯性里程计的歧管预积分。IEEE 机器人学报 (TRO)，PP(99)：1-21，2016 年。

- [83] C

。福斯特、Z. 张、M. 加斯纳、M. Werlberger 和 D. Scaramuzza。 SVO：用于单目和多摄像头系统的半直接视觉里程计。 IEEE 机器人学报 (TRO)，PP(99)：1-17，2016 年。

- [84] D. Fox、J. Ko、K. Konolige、B. Limketkai、D. Schulz 和 B. Stewart。分布式多机器人探索和绘图。 IEEE 会议记录，94(7):1325-1339，2006 年 7 月。

- [85] F. Fraundorfer 和 D. Scaramuzza。视觉里程计。第二部分：匹配、稳健性、优化和应用。 IEEE 机器人与自动化杂志，19(2):78-90，2012。

- [86] J. Fredriksson 和 C. Olsson。使用拉格朗日对偶进行同时多次旋转平均。载于《亚洲计算机视觉会议 (ACCV)》，第 245-258 页。施普林格，2012。

- [87] U.弗雷斯。专访：SLAM解决了吗？ KI - 科学情报，24(3):255-257，2010。

- [88] P. Furgale、T. D. Barfoot 和 G. Sibley。使用时间基函数的连续时间批量估计。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 2088-2095 页。 IEEE，2012。

- [89] G. Gallego、J. Lund、E. Mueggler、H. Rebecq、T. Delbruck 和 D. Scaramuzza。适用于高速应用的基于事件的 6 自由度相机跟踪。 CoRR，abs/1607.03468:1-8，2016。

- [90] R. Garg、V. Kumar、G. Carneiro 和 I. Reid。用于单视图深度估计的无监督 CNN：几何救援。摘自《欧洲计算机视觉会议 (ECCV)》，第 740-756 页，2016 年。

- [91] R. Garg、A. Roussos 和 L. Agapito。从单目视频非刚性表面的密集变分重建。参见《IEEE 计算机视觉和模式识别会议 (CVPR) 会议记录》，第 1272-1279 页，2013 年。

- [92] J. J. 吉布森。 视觉感知的生态方法：经典版。心理学出版社，2014。

- [93] D. Golovin 和 A. Krause。自适应子模块性：主动学习和随机优化的理论与应用。 人工智能研究杂志，42(1):427-486，2011。

- [94] 谷歌。探戈项目。 https://www.google.com/atap/projecttango/，2016 年。

- [95]V.M.戈文杜。结合两个视图约束进行运动估计。在《IEEE 计算机视觉和模式识别会议 (CVPR) 会议记录》中，第 218-225 页。 IEEE，2001。

- [96] O. Grasa、E. Bernal、S. Casado、I. Gil 和 J. M. M. Montiel。用于手持式单目内窥镜的视觉 SLAM。 IEEE 医学影像汇刊，33(1):135-146，2014 年。

- [97] G. Grisetti、R. Köummerle、C. Stachniss 和 W. Burgard。基于图的 SLAM 教程。 IEEE 智能交通系统杂志，2(4):31-43，2010。

[98] G. Grisetti, R. Kümmerle, C. Stachniss, U. Frese, and C. Hertzberg. Hierarchical Optimization on Manifolds for Online 2D and 3D Mapping. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 273–278. IEEE, 2010.

[99] G. Grisetti, C. Stachniss, and W. Burgard. Nonlinear Constraint Network Optimization for Efficient Map Learning. *IEEE Transactions on Intelligent Transportation Systems*, 10(3):428 –439, 2009.

[100] G. Grisetti, C. Stachniss, S. Grzonka, and W. Burgard. A Tree Parameterization for Efficiently Computing Maximum Likelihood Maps using Gradient Descent. In *Proceedings of Robotics: Science and Systems Conference (RSS)*, pages 65–72, 2007.

[101] J. S. Gutmann and K. Konolige. Incremental Mapping of Large Cyclic Environments. In *IEEE International Symposium on Computational Intelligence in Robotics and Automation (CIRA)*, pages 318–325. IEEE, 1999.

[102] C. Häne, C. Zach, A. Cohen, R. Angst, and M. Pollefeys. Joint 3D Scene Reconstruction and Class Segmentation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 97–104. IEEE, 2013.

[103] R. Hartley, J. Trumpf, Y. Dai, and H. Li. Rotation Averaging. *International Journal of Computer Vision*, 103(3):267–305, 2013.

[104] P. Henry, M. Krainin, E. Herbst, X. Ren, and D. Fox. RGB-D mapping: Using depth cameras for dense 3d modeling of indoor environments. In *Proceedings of the International Symposium on Experimental Robotics (ISER)*, 2010.

[105] J. A. Hesch, D. G. Kottas, S. L. Bowman, and S. I. Roumeliotis. Camera-IMU-based localization: Observability analysis and consistency improvement. *The International Journal of Robotics Research (IJRR)*, 33(1):182–201, 2014.

[106] K. L. Ho and P. Newman. Loop closure detection in SLAM by combining visual and spatial appearance. *Robotics and Autonomous Systems (RAS)*, 54(9):740–749, 2006.

[107] G. Huang, M. Kaess, and J. J. Leonard. Consistent sparsification for graph optimization. In *Proceedings of the European Conference on Mobile Robots (ECMR)*, pages 150–157. IEEE, 2013.

[108] G. P. Huang, A. I. Mourikis, and S. I. Roumeliotis. An observability-constrained sliding window filter for SLAM. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 65–72. IEEE, 2011.

[109] S. Huang and G. Dissanayake. A Critique of Current Developments in Simultaneous Localization and Mapping. *International Journal of Advanced Robotic Systems*, 13(5):1–13, 2016.

[110] S. Huang, Y. Lai, U. Frese, and G. Dissanayake. How far is SLAM from a linear least squares problem? In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 3011–3016. IEEE, 2010.

[111] S. Huang, H. Wang, U. Frese, and G. Dissanayake. On the Number of Local Minima to the Point Feature Based SLAM Problem. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 2074–2079. IEEE, 2012.

[112] P. J. Huber. *Robust Statistics*. Wiley, 1981.

[113] IEEE RAS Map Data Representation Working Group. IEEE Standard for Robot Map Data Representation for Navigation, Sponsor: IEEE Robotics and Automation Society. http://standards.ieee.org/findstds/standard/1873-2015.html/, June 2016. IEEE.

[114] IEEE Spectrum. Dyson's robot vacuum has 360-degree camera, tank treads, cyclone suction. http://spectrum.ieee.org/automaton/robotics/home-robots/dyson-the-360-eye-robot-vacuum, 2014.

[115] V. Ila, J. M. Porta, and J. Andrade-Cetto. Information-Based Compact Pose SLAM. *IEEE Transactions on Robotics (TRO)*, 26(1):78–93, 2010.

[116] V. Indelman, L. Carlone, and F. Dellaert. Planning in the Continuous Domain: a Generalized Belief Space Approach for Autonomous Navigation in Unknown Environments. *The International Journal of Robotics Research (IJRR)*, 34(7):849–882, 2015.

[117] V. Indelman, E. Nelson, J. Dong, N. Michael, and F. Dellaert. Incremental Distributed Inference from Arbitrary Poses and Unknown Data Association: Using Collaborating Robots to Establish a Common Reference. *IEEE Control Systems Magazine (CSM)*, 36(2):41–74, 2016.

[118] M. Irani and P. Anandan. All About Direct Methods. In *Proceedings of the International Workshop on Vision Algorithms: Theory and Practice*, pages 267–277. Springer, 1999.

[119] J. Jachnik, R. A. Newcombe, and A. J. Davison. Real-time surface light-field capture for augmentation of planar specular surfaces. In *IEEE and ACM International Symposium on Mixed and Augmented Reality (ISMAR)*, pages 91–97. IEEE, 2012.

[120] H. Johannsson, M. Kaess, M. Fallon, and J. J. Leonard. Temporally scalable visual SLAM using a reduced pose graph. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 54–61. IEEE, 2013.

[121] J. Johnson, R. Krishna, M. Stark, J. Li, M. Bernstein, and L. Fei-Fei. Image Retrieval using Scene Graphs. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3668–3678. IEEE, 2015.

[122] V. A. M. Jorge, R. Maffei, G. S. Franco, J. Daltrozo, M. Giambastiani, M. Kolberg, and E. Prestes. Ouroboros: Using potential field in unexplored regions to close loops. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 2125–2131. IEEE, 2015.

[123] L. P. Kaelbling, M. L. Littman, and A. R. Cassandra. Planning and acting in partially observable stochastic domains. *Artificial Intelligence*, 101(12):99–134, 1998.

[124] M. Kaess. Simultaneous Localization and Mapping with Infinite Planes. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 4605–4611. IEEE, 2015.

[125] M. Kaess, H. Johannsson, R. Roberts, V. Ila, J. J. Leonard, and F. Dellaert. iSAM2: Incremental Smoothing and Mapping Using the Bayes Tree. *The International Journal of Robotics Research (IJRR)*, 31:217–236, 2012.

[126] M. Kaess, A. Ranganathan, and F. Dellaert. iSAM: Incremental Smoothing and Mapping. *IEEE Transactions on Robotics (TRO)*, 24(6):1365–1378, 2008.

[127] M. Keidar and G. A. Kaminka. Efficient frontier detection for robot exploration. *The International Journal of Robotics Research (IJRR)*, 33(2):215–236, 2014.

[128] A. Kelly. *Mobile Robotics: Mathematics, Models, and Methods*. Cambridge University Press, 2013.

[129] A. Kendall, M. Grimes, and R. Cipolla. PoseNet: A Convolutional Network for Real-Time 6-DOF Camera Relocalization. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 2938–2946. IEEE, 2015.

[130] K. Khosoussi, S. Huang, and G. Dissanayake. Exploiting the Separable Structure of SLAM. In *Proceedings of Robotics: Science and Systems Conference (RSS)*, pages 208–218, 2015.

[131] A. Kim and R. M. Eustice. Real-time visual SLAM for autonomous underwater hull inspection using visual saliency. *IEEE Transactions on Robotics (TRO)*, 29(3):719–733, 2013.

[132] H. Kim, S. Leutenegger, and A. J. Davison. Real-Time 3D Reconstruction and 6-DoF Tracking with an Event Camera. In *European Conference on Computer Vision (ECCV)*, pages 349–364, 2016.

[133] J. H. Kim, C. Cadena, and I. D. Reid. Direct semi-dense SLAM for rolling shutter cameras. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 1308–1315. IEEE, May 2016.

[134] V. G. Kim, S. Chaudhuri, L. Guibas, and T. Funkhouser. Shape2Pose: Human-centric Shape Analysis. *ACM Transactions on Graphics (TOG)*, 33(4):120:1–120:12, 2014.

[135] G. Klein and D. Murray. Parallel tracking and mapping for small AR workspaces. In *IEEE and ACM International Symposium on Mixed and Augmented Reality (ISMAR)*, pages 225–234. IEEE, 2007.

[136] M. Klingensmith, I. Dryanovski, S. Srinivasa, and J. Xiao. Chisel: Real Time Large Scale 3D Reconstruction Onboard a Mobile Device using Spatially Hashed Signed Distance Fields. In *Proceedings of Robotics: Science and Systems Conference (RSS)*, pages 367–376, 2015.

[137] J. Knuth and P. Barooah. Collaborative localization with heterogeneous inter-robot measurements by Riemannian optimization. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 1534–1539. IEEE, 2013.

[138] J. Knuth and P. Barooah. Error Growth in Position Estimation from Noisy Relative Pose Measurements. *Robotics and Autonomous Systems (RAS)*, 61(3):229–224, 2013.

[139] D. G. Kottas, J. A. Hesch, S. L. Bowman, and S. I. Roumeliotis. On the consistency of vision-aided Inertial Navigation. In *Proceedings of the International Symposium on Experimental Robotics (ISER)*, pages 303–317. Springer, 2012.

[140] T. Krajník, J. P. Fentanes, O. M. Mozos, T. Duckett, J. Ekekrantz, and M. Hanheide. Long-term topological localisation for service robots in dynamic environments using spectral maps. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 4537–4542. IEEE, 2014.

[141] H. Kretzschmar, C. Stachniss, and G. Grisetti. Efficient information-theoretic graph pruning for graph-based SLAM with laser range finders. In *Proceedings of the IEEE/RSJ International Conference on Intelligent*

原文第 22 页 | 中文翻译初稿

- [98] G. Grisetti、R. Köummerle、C. Stachniss、U. Frese 和 C. Hertzberg。用于在线 2D 和 3D 绘图的流形的分层优化。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 273-278 页。 IEEE，2010。

- [99] G. Grisetti、C. Stachniss 和 W. Burgard。用于高效地图学习的非线性约束网络优化。 IEEE 智能交通系统汇刊，10(3):428 -439，2009。

- [100] G. Grisetti、C. Stachniss、S. Grzonka 和 W. Burgard。使用梯度下降有效计算最大似然图的树参数化。收录于《机器人学报：科学与系统会议 (RSS)》，第 65-72 页，2007 年。

- [101] J. S. Gutmann 和 K. Konolige。大循环环境的增量映射。载于_IEEE 机器人与自动化计算智能国际研讨会 (CIRA)_，第 318-325 页。 IEEE，1999。

- [102] C. Háne、C. Zach、A. Cohen、R. Angst 和 M. Pollefeys。联合 3D 场景重建和类分割。在_IEEE 计算机视觉和模式识别会议记录 (CVPR)_ 中，第 97-104 页。 IEEE，2013。

- [103] R. Hartley、J. Trumpf、Y. Dai 和 H. Li。旋转平均。 国际计算机视觉杂志，103(3):267-305，2013。

- [104] P. Henry、M. Krainin、E. Herbst、X. Ren 和 D. Fox。 RGB-D 映射：使用深度相机对室内环境进行密集 3D 建模。载于_国际实验机器人研讨会 (ISER) 论文集_，2010 年。

- [105] J. A. Hesch、D. G. Kottas、S. L. Bowman 和 S. I. Roumeliotis。基于相机 IMU 的定位：可观测性分析和一致性改进。 国际机器人研究杂志 (IJRR)，33(1):182-201，2014 年。

- [106] K. L. Ho 和 P. Newman。通过结合视觉和空间外观来进行 SLAM 中的闭环检测。 机器人和自主系统 (RAS)，54(9):740-749，2006 年。

- [107] G. Huang、M. Kaess 和 J. J. Leonard。用于图优化的一致稀疏化。在《欧洲移动机器人会议 (ECMR) 会议记录》中，第 150-157 页。 IEEE，2013。

- [108] G. P. Huang、A. I. Mourikis 和 S. I. Roumeliotis。 SLAM 的可观测性约束滑动窗口滤波器。在《IEEE/RSJ 国际智能机器人和系统会议 (IROS) 会议记录》中，第 65-72 页。 IEEE，2011。

- [109] S. Huang 和 G. Dissanayake。对同步定位和测绘当前发展的批评。 国际先进机器人系统杂志，13(5):1-13，2016。

- [110] S. Huang、Y. Lai、U. Frese 和 G. Dissanayake。 SLAM 距离线性最小二乘问题有多远？在《IEEE/RSJ 国际智能机器人和系统会议 (IROS) 会议记录》中，第 3011-3016 页。 IEEE，2010。

- [111] S. Huang、H. Wang、U. Frese 和 G. Dissanayake。基于局部极小值到点特征的SLAM问题的研究。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 2074-2079 页。 IEEE，2012。

- [112] P. J. 胡贝尔。 稳健的统计数据。 Wiley，1981。[113] IEEE RAS 地图数据表示工作组

。用于导航的机器人地图数据表示的 IEEE 标准，赞助商：IEEE 机器人与自动化协会。 http://standards.ieee.org/findstds/standard/1873-2015.html/，2016 年 6 月。IEEE。

- [114] IEEE 频谱。戴森扫地机器人配备 360 度摄像头、坦克履带、旋风吸力。http://spectrum.ieee.org/automaton/robotics/home-robots/dyson-the-360-eye-robot-vacuum，2014 年。

- [115] V. Ila、J. M. Porta 和 J. Andrade-Cetto。基于信息的紧凑姿态 SLAM。 IEEE 机器人学报 (TRO)，26(1):78-93，2010。

- [116] V. Indelman、L. Carlone 和 F. Dellaert。连续域规划：未知环境中自主导航的广义置信空间方法。 国际机器人研究杂志 (IJRR)，34(7):849-882，2015。

- [117] V. Indelman、E. Nelson、J. Dong、N. Michael 和 F. Dellaert。从任意姿势和未知数据关联进行增量分布式推理：使用协作机器人建立共同参考。 IEEE 控制系统杂志 (CSM)，36(2):41-74，2016 年。

- [118] M.伊拉尼和 P.阿南丹。关于直接方法的一切。摘自《视觉算法国际研讨会论文集：理论与实践》，第 267-277 页。施普林格，1999。

- [119] J. Jachnik、R. A. Newcombe 和 A. J. Davison。实时表面光场捕获，用于增强平面镜面。在 _IEEE 和 ACM 国际混合与增强现实研讨会 (ISMAR)_ 中，第 91-97 页。 IEEE，2012。

- [120] H. Johannsson、M. Kaess、M. Fallon 和 J. J. Leonard。使用简化姿态图的时间可扩展视觉 SLAM。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 54-61 页。 IEEE，2013。

- [121] J. Johnson、R. Krishna、M. Stark、J. Li、M. Bernstein 和 L. FeiFei。使用场景图进行图像检索。在_IEEE 计算机视觉和模式识别会议记录 (CVPR)_ 中，第 3668-3678 页。 IEEE，2015。

- [122] V. A. M. Jorge、R. Maffei、G. S. Franco、J. Daltrozo、M. Giambastiani、M. Kolberg 和 E. Prestes。衔尾蛇：利用未探索区域的势场来闭合回路。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 2125-2131 页。 IEEE，2015。

- [123] L. P. Kaelbling、M. L. Littman 和 A. R. Cassandra。在部分可观察的随机域中进行规划和行动。人工智能，101(12)：99-134，1998。

- [124] M.凯斯。无限平面的同步定位和测绘。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 4605-4611 页。 IEEE，2015。

- [125] M. Kaess、H. Johannsson、R. Roberts、V. Ila、J. J. Leonard 和 F. Dellaert。 iSAM2：使用贝叶斯树的增量平滑和映射。 国际机器人研究杂志 (IJRR)，31:217-236，2012 年。

- [126] M. Kaess、A. Ranganathan 和 F. Dellaert。 iSAM：增量平滑和映射。 IEEE 机器人学报 (TRO)，24(6):1365-1378，2008。

- [127] M. Keidar 和 G. A. Kaminka。机器人探索的高效前沿检测。 国际机器人研究杂志 (IJRR)，33(2):215-236，2014 年。

- [128] A.凯利。 移动机器人：数学、模型和方法

。剑桥大学出版社，2013。

- [129] A. Kendall、M. Grimes 和 R. Cipolla。 PoseNet：用于实时 6-DOF 相机重定位的卷积网络。在《IEEE 国际计算机视觉会议 (ICCV) 会议记录》中，第 2938-2946 页。 IEEE，2015。

- [130] K. Khosoussi、S. Huang 和 G. Dissanayake。利用 SLAM 的可分离结构。收录于《机器人学报：科学与系统会议 (RSS)》，第 208-218 页，2015 年。

- [131] A. Kim 和 R. M. Eustice。使用视觉显着性进行自主水下船体检查的实时视觉 SLAM。 IEEE 机器人学报 (TRO)，29(3):719-733，2013 年。

- [132] H. Kim、S. Leutenegger 和 A. J. Davison。使用事件相机进行实时 3D 重建和 6-DoF 跟踪。摘自《欧洲计算机视觉会议 (ECCV)》，第 349-364 页，2016 年。

- [133] J. H. Kim、C. Cadena 和 I. D. Reid。用于卷帘快门相机的直接半密集 SLAM。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 1308-1315 页。 IEEE，2016 年 5 月。

- [134] V. G. Kim、S. Chaudhuri、L. Guibas 和 T. Funkhouser。 Shape2Pose：以人为中心的形状分析。 ACM 图形交易 (TOG)，33(4):120:1-120:12，2014 年。

- [135] G.克莱因和D.默里。小型 AR 工作空间的并行跟踪和映射。在_IEEE 和 ACM 国际混合与增强现实研讨会 (ISMAR)_ 中，第 225-234 页。 IEEE，2007。

- [136] M. Klingensmith、I. Dryanovski、S. Srinivasa 和 J. Xiao。 Chisel：使用空间散列有符号距离场在移动设备上进行实时大规模 3D 重建。收录于《机器人学报：科学与系统会议 (RSS)》，第 367-376 页，2015 年。

- [137] J. Knuth 和 P. Barooah。通过黎曼优化实现异构机器人间测量的协作定位。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 1534-1539 页。 IEEE，2013。

- [138] J. Knuth 和 P. Barooah。来自嘈杂的相对姿势测量的位置估计的误差增长。 机器人和自主系统 (RAS)，61(3):229-224，2013 年。

- [139] D. G. Kottas、J. A. Hesch、S. L. Bowman 和 S. I. Roumeliotis。视觉辅助惯性导航一致性研究[J]. 见_国际实验机器人研讨会 (ISER) 论文集_，第 303-317 页。施普林格，2012。

- [140] T. Krajn´ık、J. P. Fentanes、O. M. Mozos、T. Duckett、J. Ekekrantz 和 M. Hanheide。使用光谱图对动态环境中的服务机器人进行长期拓扑定位。在《IEEE/RSJ 国际智能机器人和系统会议 (IROS) 会议记录》中，第 4537-4542 页。 IEEE，2014。

- [141] H. Kretzschmar、C. Stachniss 和 G. Grisetti。使用激光测距仪对基于图的 SLAM 进行有效的信息理论图修剪。在_IEEE/RSJ 国际智能会议论文集_

Robots and Systems (IROS), pages 865–871. IEEE, 2011.

[142] R. G. Krishnan, U. Shalit, and D. Sontag. Deep Kalman Filters. In *NIPS 2016 Workshop: Advances in Approximate Bayesian Inference*, pages 1–7. NIPS, 2016.

[143] F. R. Kschischang, B. J. Frey, and H. A. Loeliger. Factor Graphs and the Sum-Product Algorithm. *IEEE Transactions on Information Theory*, 47(2):498–519, 2001.

[144] B. Kueng, E. Mueggler, G. Gallego, and D. Scaramuzza. Low-Latency Visual Odometry using Event-based Feature Tracks. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2016.

[145] Kuka Robotics. Kuka Navigation Solution. http://www.kuka-robotics.com/res/robotics/Products/PDF/EN/KUKA_Navigation_Solution_EN.pdf, 2016.

[146] R. Kümmerle, G. Grisetti, H. Strasdat, K. Konolige, and W. Burgard. g2o: A General Framework for Graph Optimization. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 3607–3613. IEEE, 2011.

[147] A. Kundu, Y. Li, F. Dellaert, F. Li, and J. M. Rehg. Joint Semantic Segmentation and 3D Reconstruction from Monocular Video. In *European Conference on Computer Vision (ECCV)*, pages 703–718. Springer, 2014.

[148] K. Lai, L. Bo, and D. Fox. Unsupervised Feature Learning for 3D Scene Labeling. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 3050–3057, 2014.

[149] K. Lai and D. Fox. Object Recognition in 3D Point Clouds Using Web Data and Domain Adaptation. *The International Journal of Robotics Research (IJRR)*, 29(8):1019–1037, 2010.

[150] Y. Latif, C. Cadena, , and J. Neira. Robust loop closing over time for pose graph slam. *The International Journal of Robotics Research (IJRR)*, 32(14):1611–1626, 2013.

[151] M. T. Lazaro, L. Paz, P. Pinies, J. A. Castellanos, and G. Grisetti. Multi-robot SLAM using condensed measurements. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 1069–1076. IEEE, 2013.

[152] C. Leung, S. Huang, and G. Dissanayake. Active SLAM using Model Predictive Control and Attractor Based Exploration. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 5026–5031. IEEE, 2006.

[153] C. Leung, S. Huang, N. Kwok, and G. Dissanayake. Planning Under Uncertainty using Model Predictive Control for Information Gathering. *Robotics and Autonomous Systems (RAS)*, 54(11):898–910, 2006.

[154] J. Levinson. *Automatic Laser Calibration, Mapping, and Localization for Autonomous Vehicles*. PhD thesis, Stanford University, 2011.

[155] C. Li, C. Brandli, R. Berner, H. Liu, M. Yang, S. C. Liu, and T. Delbruck. Design of an RGBW Color VGA Rolling and Global Shutter Dynamic and Active-Pixel Vision Sensor. In *IEEE International Symposium on Circuits and Systems*. IEEE, 2015.

[156] P. Lichtsteiner, C. Posch, and T. Delbruck. A 128×128 120 dB 15 μs latency asynchronous temporal contrast vision sensor. *IEEE Journal of Solid-State Circuits*, 43(2):566–576, 2008.

[157] J. J. Lim, H. Pirsiavash, and A. Torralba. Parsing IKEA objects: Fine pose estimation. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 2992–2999. IEEE, 2013.

[158] F. Liu, C. Shen, and G. Lin. Deep convolutional neural fields for depth estimation from a single image. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2015.

[159] M. Liu, S. Huang, G. Dissanayake, and H. Wang. A convex optimization based approach for pose SLAM problems. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 1898–1903. IEEE, 2012.

[160] S. Lowry, N. Sünderhauf, P. Newman, J. J. Leonard, D. Cox, P. Corke, and M. J. Milford. Visual Place Recognition: A Survey. *IEEE Transactions on Robotics (TRO)*, 32(1):1–19, 2016.

[161] F. Lu and E. Milios. Globally consistent range scan alignment for environment mapping. *Autonomous Robots (AR)*, 4(4):333–349, 1997.

[162] Y. Lu and D. Song. Visual Navigation Using Heterogeneous Landmarks and Unsupervised Geometric Constraints. *IEEE Transactions on Robotics (TRO)*, 31(3):736–749, 2015.

[163] S. Lynen, T. Sattler, M. Bosse, J. Hesch, M. Pollefeys, and R. Siegwart. Get out of my lab: Large-scale, real-time visual-inertial localization. In *Proceedings of Robotics: Science and Systems Conference (RSS)*, pages 338–347, 2015.

[164] D. J. C. MacKay. *Information Theory, Inference & Learning Algorithms*. Cambridge University Press, 2002.

[165] M. Magnabosco and T. P. Breckon. Cross-spectral visual simultaneous localization and mapping (slam) with sensor handover. *Robotics and Autonomous Systems (RAS)*, 61(2):195–208, 2013.

[166] M. Maimone, Y. Cheng, and L. Matthies. Two years of Visual Odometry on the Mars Exploration Rovers. *Journal of Field Robotics*, 24(3):169–186, 2007.

[167] L. Marques, U. Nunes, and A. T. de Almeida. Olfaction-based mobile robot navigation. *Thin solid films*, 418(1):51–58, 2002.

[168] D. Martinec and T. Pajdla. Robust Rotation and Translation Estimation in Multiview Reconstruction. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1–8. IEEE, 2007.

[169] R. Martinez-Cantin, N. de Freitas, E. Brochu, J. Castellanos, and A. Doucet. A Bayesian Exploration-exploitation Approach for Optimal Online Sensing and Planning with a Visually Guided Mobile Robot. *Autonomous Robots (AR)*, 27(2):93–103, 2009.

[170] M. Mazuran, W. Burgard, and G. D. Tipaldi. Nonlinear factor recovery for long-term SLAM. *The International Journal of Robotics Research (IJRR)*, 35(1-3):50–72, 2016.

[171] J. Mendel. *Lessons in Estimation Theory for Signal Processing, Communications, and Control*. Prentice Hall, 1995.

[172] P. Merrell and D. Manocha. Model Synthesis: A General Procedural Modeling Algorithm. *IEEE Transactions on Visualization and Computer Graphics*, 17(6):715–728, 2010.

[173] M. J. Milford and G. F. Wyeth. SeqSLAM: Visual route-based navigation for sunny summer days and stormy winter nights. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 1643–1649. IEEE, 2012.

[174] J. M. M. Montiel, J. Civera, and A. J. Davison. Unified Inverse Depth Parametrization for Monocular SLAM. In *Proceedings of Robotics: Science and Systems Conference (RSS)*, pages 81–88, Aug. 2006.

[175] A. I. Mourikis and S. I. Roumeliotis. A Multi-State Constraint Kalman Filter for Vision-aided Inertial Navigation. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 3565–3572. IEEE, 2007.

[176] O. M. Mozos, R. Triebel, P. Jensfelt, A. Rottmann, and W. Burgard. Supervised semantic labeling of places using information extracted from sensor data. *Robotics and Autonomous Systems (RAS)*, 55(5):391–402, 2007.

[177] E. Mueggler, G. Gallego, and D. Scaramuzza. Continuous-Time Trajectory Estimation for Event-based Vision Sensors. In *Proceedings of Robotics: Science and Systems Conference (RSS)*, 2015.

[178] E. Mueller, A. Censi, and E. Frazzoli. Low-latency heading feedback control with neuromorphic vision sensors using efficient approximated incremental inference. In *Proceedings of the IEEE Conference on Decision and Control (CDC)*, pages 992–999. IEEE, 2015.

[179] R. Mur-Artal, J. M. M. Montiel, and J. D. Tardós. ORB-SLAM: a Versatile and Accurate Monocular SLAM System. *IEEE Transactions on Robotics (TRO)*, 31(5):1147–1163, 2015.

[180] J. Neira and J. D. Tardós. Data Association in Stochastic Mapping Using the Joint Compatibility Test. *IEEE Transactions on Robotics and Automation (TRA)*, 17(6):890–897, 2001.

[181] E. D. Nerurkar, S. I. Roumeliotis, and A. Martinelli. Distributed maximum a posteriori estimation for multi-robot cooperative localization. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 1402–1409. IEEE, 2009.

[182] R. Newcombe, D. Fox, and S. M. Seitz. DynamicFusion: Reconstruction and tracking of non-rigid scenes in real-time. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 343–352. IEEE, 2015.

[183] R. A. Newcombe, A. J. Davison, S. Izadi, P. Kohli, O. Hilliges, J. Shotton, D. Molyneaux, S. Hodges, D. Kim, and A. Fitzgibbon. KinectFusion: Real-Time Dense Surface Mapping and Tracking. In *IEEE and ACM International Symposium on Mixed and Augmented Reality (ISMAR)*, pages 127–136, Basel, Switzerland, Oct. 2011.

[184] R. A. Newcombe, S. J. Lovegrove, and A. J. Davison. DTAM: Dense Tracking and Mapping in Real-Time. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 2320–2327. IEEE, 2011.

[185] P. Newman, J. J. Leonard, J. D. Tardos, and J. Neira. Explore and Return: Experimental Validation of Real-Time Concurrent Mapping and Localization. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 1802–1809. IEEE, 2002.

[186] R. Ng, M. Levoy, M. Bredif, G. Duval, M. Horowitz, and P. Hanrahan. Light Field Photography with a Hand-Held Plenoptic Camera. In *Stanford University Computer Science Tech Report CSTR 2005-02*, 2005.

[187] K. Ni and F. Dellaert. Multi-Level Submap Based SLAM Using Nested Dissection. In *Proceedings of the IEEE/RSJ International Conference*

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- 机器人和系统 (IROS)，第 865-871 页。 IEEE，2011。

- [142] R. G. Krishnan、U. Shalit 和 D. Sontag。深度卡尔曼滤波器。在 NIPS 2016 研讨会：近似贝叶斯推理的进展 中，第 1-7 页。国家信息处理计划，2016。

- [143] F. R. Kschischang、B. J. Frey 和 H. A. Loeliger。因子图和和积算法。 IEEE 信息论汇刊，47(2):498-519，2001。

- [144] B. Kueng、E. Mueggler、G. Gallego 和 D. Scaramuzza。使用基于事件的特征轨迹的低延迟视觉里程计。收录于《IEEE/RSJ 国际智能机器人与系统会议 (IROS) 会议记录》，2016 年。

- [145] 库卡机器人公司。库卡导航解决方案。 http://www.kuka-robotics。com/res/robotics/Products/PDF/EN/KUKA <u>导航解决方案 EN。</u> pdf，2016 年。

- [146] R. Kümmerle、G. Grisetti、H. Strasdat、K. Konolige 和 W. Burgard。 g2o：图优化的通用框架。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 3607-3613 页。 IEEE，2011。

- [147] A. Kundu、Y. Li、F. Dellaert、F. Li 和 J. M. Rehg。单目视频的联合语义分割和 3D 重建。在《欧洲计算机视觉会议 (ECCV)》中，第 703-718 页。施普林格，2014。

- [148] K. Lai、L. Bo 和 D. Fox。用于 3D 场景标记的无监督特征学习。参见《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》，第 3050-3057 页，2014 年。

- [149] K. Lai 和 D. Fox。使用 Web 数据和域适应的 3D 点云中的对象识别。 国际机器人研究杂志 (IJRR)，29(8):1019-1037，2010。

- [150] Y. Latif、C. Cadena、和 J. Neira。姿势图猛击随着时间的推移稳健的循环闭合。 国际机器人研究杂志 (IJRR)，32(14):1611-1626，2013 年。

- [151] M. T. Lazaro、L. Paz、P. Pinies、J. A. Castellanos 和 G. Grisetti。使用压缩测量的多机器人 SLAM。在《IEEE/RSJ 国际智能机器人和系统会议 (IROS) 会议记录》中，第 1069-1076 页。 IEEE，2013。

- [152] C. Leung、S. Huang 和 G. Dissanayake。使用模型预测控制和基于吸引子的探索的主动 SLAM。在《IEEE/RSJ 国际智能机器人和系统会议 (IROS) 会议记录》中，第 5026-5031 页。 IEEE，2006。

- [153] C. Leung、S. Huang、N. Kwok 和 G. Dissanayake。使用模型预测控制进行信息收集的不确定性下的规划。 机器人和自主系统 (RAS)，54(11):898-910，2006。

- [154] J.莱文森。 自动驾驶车辆的自动激光校准、测绘和定位。博士论文，斯坦福大学，2011 年。

- [155] C. Li、C. Brandli、R. Berner、H. Liu、M. Yang、S. C. Liu 和 T. Delbruck。 RGBW 彩色 VGA 滚动和全局快门动态和主动像素视觉传感器的设计。在_IEEE国际电路与系统研讨会_中。 IEEE，2015。

- [156] P. Lichtsteiner、C. Posch 和 T. Delbruck。 128 × 128 120 dB 15 μ s 延迟异步时间对比视觉传感器。 IEEE 固态电路杂志，43(2):566-576，2008。

- [157] J. J. Lim、H. Pirsiavash 和 A. Torralba。解析宜家物体：精细姿势估计

。在《IEEE 国际计算机视觉会议 (ICCV) 会议记录》中，第 2992-2999 页。 IEEE，2013。

- [158] F. Liu，C. Shen，G. Lin。用于从单个图像进行深度估计的深度卷积神经场。收录于《IEEE 计算机视觉和模式识别会议 (CVPR) 会议论文集》，2015 年。

- [159] M. Liu、S. Huang、G. Dissanayake 和 H. Wang。一种基于凸优化的方法来解决 SLAM 问题。在《IEEE/RSJ 国际智能机器人和系统会议 (IROS) 会议记录》中，第 1898-1903 页。 IEEE，2012。

- [160] S. Lowry、N. Süunderhauf、P. Newman、J. J. Leonard、D. Cox、P. Corke 和 M. J. Milford。视觉地点识别：一项调查。 IEEE 机器人学报 (TRO)，32(1):1-19，2016 年。

- [161] F. Lu 和 E. Milios。用于环境映射的全局一致的范围扫描对齐。 自主机器人 (AR)，4(4):333-349，1997。

- [162] Y. Lu 和 D. Song。使用异构地标和无监督几何约束的视觉导航。 IEEE 机器人学报 (TRO)，31(3):736-749，2015。

- [163] S. Lynen、T. Sattler、M. Bosse、J. Hesch、M. Pollefeys 和 R. Siegwart。走出我的实验室：大规模实时视觉惯性定位。收录于《机器人学报：科学与系统会议 (RSS)》，第 338-347 页，2015 年。

- [164]D.J.C.麦凯。 信息论、推理和学习算法。剑桥大学出版社，2002。

- [165] M. Magnabosco 和 T. P. Breckon。通过传感器切换进行跨光谱视觉同步定位和映射（SLAM）。机器人和

自治系统（RAS），61（2）：195-208，2013。

- [166] M. Maimone、Y. Cheng 和 L. Matthies。火星探索漫游车上的两年视觉里程计。 现场机器人杂志，24(3):169-186，2007。

- [167] L. Marques、U. Nunes 和 A. T. de Almeida。基于嗅觉的移动机器人导航。 固体薄膜，418(1):51-58, 2002。

- [168] D. Martinec 和 T. Pajdla。多视图重建中的鲁棒旋转和平移估计。在《IEEE 计算机视觉和模式识别会议 (CVPR) 会议记录》中，第 1-8 页。 IEEE，2007。

- [169] R. Martinez-Cantin、N. de Freitas、E. Brochu、J. Castellanos 和 A. Doucet。使用视觉引导移动机器人实现最佳在线感知和规划的贝叶斯探索开发方法。 自主机器人 (AR)，27(2):93-103，2009。

- [170] M. Mazuran、W. Burgard 和 G. D. Tipaldi。长期 SLAM 的非线性因子恢复。 国际机器人研究杂志 (IJRR)，35(1-3):50-72，2016 年。

- [171] J.孟德尔。 信号处理、通信和控制的估计理论课程。普伦蒂斯·霍尔，1995。

- [172] P. Merrell 和 D. Manocha。模型综合：通用过程建模算法。 IEEE 可视化和计算机图形学汇刊，17(6):715-728，2010。

- [173] M. J. 米尔福德和 G. F. 怀斯。 SeqSLAM：基于视觉路线的导航，适合阳光明媚的夏日和暴风雨的冬夜。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 1643-1649 页。 IEEE，2012。

- [174] J. M. M. Montiel、J. Civera 和 A. J. Davison。单目 SLAM 的统一逆深度参数化

。收录于《机器人学报：科学与系统会议 (RSS)》，第 81-88 页，2006 年 8 月。

- [175] A. I. Mourikis 和 S. I. Roumeliotis。用于视觉辅助惯性导航的多状态约束卡尔曼滤波器。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 3565-3572 页。 IEEE，2007。

- [176] O. M. Mozos、R. Triebel、P. Jensfelt、A. Rottmann 和 W. Burgard。使用从传感器数据中提取的信息对地点进行监督语义标记。 机器人和自主系统 (RAS)，55(5):391-402，2007 年。

- [177] E. Mueggler、G. Gallego 和 D. Scaramuzza。基于事件的视觉传感器的连续时间轨迹估计。在《机器人学报：科学与系统会议 (RSS)》中，2015 年。

- [178] E. Mueller、A. Censi 和 E. Frazzoli。使用高效近似增量推理的神经形态视觉传感器进行低延迟航向反馈控制。在《IEEE 决策与控制会议 (CDC) 会议记录》中，第 992-999 页。 IEEE，2015。

- [179] R. Mur-Artal、J. M. M. Montiel 和 J. D. Tard´os。 ORB-SLAM：多功能且精确的单目 SLAM 系统。 IEEE 机器人学报 (TRO)，31(5):1147-1163，2015。

- [180] J. Neira 和 J. D. Tard´os。使用联合兼容性测试的随机映射中的数据关联。 IEEE 机器人与自动化汇刊 (TRA)，17(6):890-897，2001 年。

- [181] E. D. Nerurkar、S. I. Roumeliotis 和 A. Martinelli。多机器人协作定位的分布式最大后验估计。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 1402-1409 页。 IEEE，2009。

- [182] R. Newcombe、D. Fox 和 S. M. Seitz。 DynamicFusion：实时重建和跟踪非刚性场景。在《IEEE 计算机视觉和模式识别会议 (CVPR) 会议记录》中，第 343-352 页。 IEEE，2015。

- [183] R. A. Newcombe、A. J. Davison、S. Izadi、P. Kohli、O. Hilliges、J. Shotton、D. Molyneaux、S. Hodges、D. Kim 和 A. Fitzgibbon。 KinectFusion：实时密集表面测绘和跟踪。载于_IEEE 和 ACM 国际混合与增强现实研讨会 (ISMAR)_，第 127-136 页，瑞士巴塞尔，2011 年 10 月。

- [184] R. A. Newcombe、S. J. Lovegrove 和 A. J. Davison。 DTAM：实时密集跟踪和测绘。在《IEEE 国际计算机视觉会议 (ICCV) 会议记录》中，第 2320-2327 页。IEEE，2011。

- [185] P. Newman、J. J. Leonard、J. D. Tardos 和 J. Neira。探索与回归：实时并发测绘和定位的实验验证。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 1802-1809 页。 IEEE，2002。

- [186] R. Ng、M. Levoy、M. Bredif、G. Duval、M. Horowitz 和 P. Hanrahan。使用手持式全光相机进行光场摄影。载于_斯坦福大学计算机科学技术报告 CSTR 2005-02_，2005 年。

- [187] K. Ni 和 F. Dellaert。使用嵌套解剖的基于多级子图的 SLAM。在_IEEE/RSJ 国际会议记录_

on Intelligent Robots and Systems (IROS), pages 2558–2565. IEEE, 2010.

[188] M. Nießner, M. Zollhöfer, S. Izadi, and M. Stamminger. Real-time 3D Reconstruction at Scale Using Voxel Hashing. *ACM Transactions on Graphics (TOG)*, 32(6):169:1–169:11, Nov. 2013.

[189] D. Nistér and H. Stewénius. Scalable Recognition with a Vocabulary Tree. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, volume 2, pages 2161–2168. IEEE, 2006.

[190] A. Nüchter. *3D Robotic Mapping: the Simultaneous Localization and Mapping Problem with Six Degrees of Freedom*, volume 52. Springer, 2009.

[191] E. Olson and P. Agarwal. Inference on networks of mixtures for robust robot mapping. *The International Journal of Robotics Research (IJRR)*, 32(7):826–840, 2013.

[192] E. Olson, J. J. Leonard, and S. Teller. Fast Iterative Alignment of Pose Graphs with Poor Initial Estimates. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 2262–2269. IEEE, 2006.

[193] Open Geospatial Consortium (OGC). CityGML 2.0.0. http://www.citygml.org/, Jun 2016.

[194] Open Geospatial Consortium (OGC). OGC IndoorGML standard. http://www.opengeospatial.org/standards/indoorgml, Jun 2016.

[195] S. Patil, G. Kahn, M. Laskey, J. Schulman, K. Goldberg, and P. Abbeel. Scaling up Gaussian Belief Space Planning Through Covariance-Free Trajectory Optimization and Automatic Differentiation. In *Proceedings of the International Workshop on the Algorithmic Foundations of Robotics (WAFR)*, pages 515–533. Springer, 2015.

[196] A. Patron-Perez, S. Lovegrove, and G. Sibley. A Spline-Based Trajectory Representation for Sensor Fusion and Rolling Shutter Cameras. *International Journal of Computer Vision*, 113(3):208–219, 2015.

[197] L. Paull, G. Huang, M. Seto, and J. J. Leonard. Communication-Constrained Multi-AUV Cooperative SLAM. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 509–516. IEEE, 2015.

[198] A. Pázman. *Foundations of Optimum Experimental Design*. Springer, 1986.

[199] J. R. Peters, D. Borra, B. Paden, and F. Bullo. Sensor Network Localization on the Group of Three-Dimensional Displacements. *SIAM Journal on Control and Optimization*, 53(6):3534–3561, 2015.

[200] C. J. Phillips, M. Lecce, C. Davis, and K. Daniilidis. Grasping surfaces of revolution: Simultaneous pose and shape recovery from two views. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 1352–1359. IEEE, 2015.

[201] S. Pillai and J. J. Leonard. Monocular SLAM Supported Object Recognition. In *Proceedings of Robotics: Science and Systems Conference (RSS)*, pages 310–319, 2015.

[202] G. Piovan, I. Shames, B. Fidan, F. Bullo, and B. Anderson. On frame and orientation localization for relative sensing networks. *Automatica*, 49(1):206–213, 2013.

[203] M. Pizzoli, C. Forster, and D. Scaramuzza. REMODE: Probabilistic, Monocular Dense Reconstruction in Real Time. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 2609–2616. IEEE, 2014.

[204] L. Polok, V. Ila, M. Solony, P. Smrz, and P. Zemcik. Incremental Block Cholesky Factorization for Nonlinear Least Squares in Robotics. In *Proceedings of Robotics: Science and Systems Conference (RSS)*, pages 328–336, 2013.

[205] C. Posch, D. Matolin, and R. Wohlgenannt. An asynchronous time-based image sensor. In *IEEE International Symposium on Circuits and Systems (ISCAS)*, pages 2130–2133. IEEE, 2008.

[206] A. Pronobis and P. Jensfelt. Large-scale semantic mapping and reasoning with heterogeneous modalities. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 3515–3522. IEEE, 2012.

[207] H. Rebecq, T. Horstschaefer, G. Gallego, and D. Scaramuzza. EVO: A Geometric Approach to Event-based 6-DOF Parallel Tracking and Mapping in Real-Time. *IEEE Robotics and Automation Letters*, 2016.

[208] A. Rényi. On Measures Of Entropy And Information. In *Proceedings of the 4th Berkeley Symposium on Mathematics, Statistics and Probability*, pages 547–561, 1960.

[209] A. G. Requicha. Representations for Rigid Solids: Theory, Methods, and Systems. *ACM Computing Surveys (CSUR)*, 12(4):437–464, 1980.

[210] L. Riazuelo, J. Civera, and J. M. M. Montiel. C2TAM: A cloud framework for cooperative tracking and mapping. *Robotics and Autonomous Systems (RAS)*, 62(4):401–413, 2014.

[211] D. M. Rosen, C. DuHadway, and J. J. Leonard. A convex relaxation for approximate global optimization in Simultaneous Localization and Mapping. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 5822–5829. IEEE, 2015.

[212] D. M. Rosen, M. Kaess, and J. J. Leonard. Robust Incremental Online Inference Over Sparse Factor Graphs: Beyond the Gaussian Case. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 1025–1032. IEEE, 2013.

[213] O. Russakovsky, J. Deng, H. Su, J. Krause, S. Satheesh, S. Ma, Z. Huang, A. Karpathy, A. Khosla, M. Bernstein, A. C. Berg, and L. Fei-Fei. ImageNet Large Scale Visual Recognition Challenge. *International Journal of Computer Vision*, 115(3):211–252, 2015.

[214] S. Agarwal et al. Ceres Solver. http://ceres-solver.org, June 2016. Google.

[215] D. Sabatta, D. Scaramuzza, and R. Siegwart. Improved Appearance-based Matching in Similar and Dynamic Environments Using a Vocabulary Tree. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 1008–1013. IEEE, 2010.

[216] S. Saeedi, M. Trentini, M. Seto, and H. Li. Multiple-Robot Simultaneous Localization and Mapping: A Review. *Journal of Field Robotics (JFR)*, 33(1):3–46, 2016.

[217] R. Salas-Moreno, R. Newcombe, H. Strasdat, P. Kelly, and A. Davison. SLAM++: Simultaneous Localisation and Mapping at the Level of Objects. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1352–1359. IEEE, 2013.

[218] D. Scaramuzza and F. Fraundorfer. Visual Odometry [Tutorial]. Part I: The First 30 Years and Fundamentals. *IEEE Robotics and Automation Magazine*, 18(4):80–92, 2011.

[219] S. Sengupta and P. Sturgess. Semantic octree: Unifying recognition, reconstruction and representation via an octree constrained higher order MRF. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 1874–1879. IEEE, 2015.

[220] J. Shah and M. Mäntylä. *Parametric and Feature-Based CAD/CAM: Concepts, Techniques, and Applications*. Wiley, 1995.

[221] V. Shapiro. Solid Modeling. In G. E. Farin, J. Hoschek, and M. S. Kim, editors, *Handbook of Computer Aided Geometric Design*, chapter 20, pages 473–518. Elsevier, 2002.

[222] C. Shen, J. F. O'Brien, and J. R. Shewchuk. Interpolating and Approximating Implicit Surfaces from Polygon Soup. In *SIGGRAPH*, pages 896–904. ACM, 2004.

[223] G. Sibley, C. Mei, I. D. Reid, and P. Newman. Adaptive relative bundle adjustment. In *Proceedings of Robotics: Science and Systems Conference (RSS)*, pages 177–184, 2009.

[224] A. Singer. Angular synchronization by eigenvectors and semidefinite programming. *Applied and Computational Harmonic Analysis*, 30(1):20–36, 2010.

[225] A. Singer and Y. Shkolnisky. Three-Dimensional Structure Determination from Common Lines in Cryo-EM by Eigenvectors and Semidefinite Programming. *SIAM Journal of Imaging Sciences*, 4(2):543–572, 2011.

[226] J. Sivic and A. Zisserman. Video Google: A Text Retrieval Approach to Object Matching in Videos. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 1470–1477. IEEE, 2003.

[227] B. M. Smith, L. Zhang, H. Jin, and A. Agarwala. Light field video stabilization. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 341–348. IEEE, 2009.

[228] S. Soatto. Steps Towards a Theory of Visual Information: Active Perception, Signal-to-Symbol Conversion and the Interplay Between Sensing and Control. *CoRR*, abs/1110.2053:1–214, 2011.

[229] S. Soatto and A. Chiuso. Visual Representations: Defining Properties and Deep Approximations. *CoRR*, abs/1411.7676:1–20, 2016.

[230] L. Song, B. Boots, S. M. Siddiqi, G. J. Gordon, and A. J. Smola. Hilbert Space Embeddings of Hidden Markov Models. In *International Conference on Machine Learning (ICML)*, pages 991–998, 2010.

[231] N. Srinivasan, L. Carlone, and F. Dellaert. Structural Symmetries from Motion for Scene Reconstruction and Understanding. In *Proceedings of the British Machine Vision Conference (BMVC)*, pages 1–13. BMVA Press, 2015.

[232] C. Stachniss. *Robotic Mapping and Exploration*. Springer, 2009.

[233] C. Stachniss, G. Grisetti, and W. Burgard. Information Gain-based Exploration Using Rao-Blackwellized Particle Filters. In *Proceedings of Robotics: Science and Systems Conference (RSS)*, pages 65–72, 2005.

[234] C. Stachniss, S. Thrun, and J. J. Leonard. Simultaneous Localization and Mapping. In B. Siciliano and O. Khatib, editors, *Springer Handbook of Robotics*, chapter 46, pages 1153–1176. Springer, 2nd edition, 2016.

[235] H. Strasdat, A. J. Davison, J. M. M. Montiel, and K. Konolige. Double

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关于智能机器人和系统 (IROS)，第 2558-2565 页。 IEEE，2010。

- [188] M. Nießner、M. Zollhofer、S. Izadi 和 M. Stamminger。使用体素哈希进行大规模实时 3D 重建。 ACM 图形交易 (TOG)，32(6):169:1-169:11，2013 年 11 月。

- [189] D. Nist´er 和 H. Stew´enius。使用词汇树进行可扩展识别。参见《IEEE 计算机视觉和模式识别会议记录 (CVPR)》，第 2 卷，第 2161-2168 页。 IEEE，2006。

- [190] A.诺希特。 3D 机器人测绘：六自由度的同时定位和测绘问题，第 52 卷。Springer，2009 年。

- [191] E. 奥尔森和 P. 阿加瓦尔。对混合物网络进行推理以实现稳健的机器人映射。 国际机器人研究杂志 (IJRR)，32(7):826-840，2013 年。

- [192] E. Olson、J. J. Leonard 和 S. Teller。初始估计较差的位姿图的快速迭代对齐。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 2262-2269 页。 IEEE，2006。

- [193] 开放地理空间联盟（OGC）。 CityGML 2.0.0。 http://www.citygml.org/，2016 年 6 月。

- [194] 开放地理空间联盟（OGC）。 OGC IndoorGML 标准。 http://www.opengeospatial.org/standards/indoorgml，2016 年 6 月。

- [195] S. Patil、G. Kahn、M. Laskey、J. Schulman、K. Goldberg 和 P. Abbeel。通过无协方差轨迹优化和自动微分扩大高斯置信空间规划。在《机器人算法基础国际研讨会 (WAFR) 会议记录》中，第 515-533 页。施普林格，2015。

- [196] A. Patron-Perez、S. Lovegrove 和 G. Sibley。用于传感器融合和卷帘快门相机的基于样条的轨迹表示。 国际计算机视觉杂志，113(3):208-219，2015。

- [197] L. Paull、G. Huang、M. Seto 和 J. J. Leonard。通信受限的多 AUV 协作 SLAM。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 509-516 页。 IEEE，2015。

- [198] A. P´azman。 最佳实验设计的基础 。施普林格，1986。

- [199] J. R. Peters、D. Borra、B. Paden 和 F. Bullo。三维位移群上的传感器网络定位。 SIAM 控制与优化杂志，53(6):3534-3561，2015。

- [200] C. J. Phillips、M. Lecce、C. Davis 和 K. Daniilidis。抓取旋转表面：从两个视图同时恢复姿势和形状。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 1352-1359 页。 IEEE，2015。

- [201] S. 皮莱和 J. J. 伦纳德。单目SLAM支持物体识别。收录于《机器人学报：科学与系统会议 (RSS)》，第 310-319 页，2015 年。

- [202] G. Piovan、I. Shames、B. Fidan、F. Bullo 和 B. Anderson。关于相对传感网络的框架和方向定位。 自动机，49（1）：206-213，2013。

- [203] M. Pizzoli、C. Forster 和 D. Scaramuzza。 REMODE：实时概率、单目密集重建。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 2609-2616 页。 IEEE，2014。

- [204] L. Polok、V. Ila、M. Solony、P. Smrz 和 P. Zemcik

。机器人中非线性最小二乘的增量块 Cholesky 分解。收录于《机器人学报：科学与系统会议 (RSS)》，第 328-336 页，2013 年。

- [205] C. Posch、D. Matolin 和 R. Wohlgenannt。基于时间的异步图像传感器。见_IEEE 国际电路与系统研讨会 (ISCAS)_，第 2130-2133 页。 IEEE，2008。

- [206] A. Pronobis 和 P. Jensfelt。具有异构模式的大规模语义映射和推理。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 3515-3522 页。 IEEE，2012。

- [207] H. Rebecq、T. Horstschaefer、G. Gallego 和 D. Scaramuzza。 EVO：基于事件的六自由度并行实时跟踪和映射的几何方法。 IEEE 机器人与自动化快报，2016 年。

- [208] A.R´enyi。关于熵和信息的度量。载于《第四届伯克利数学、统计和概率研讨会论文集》，第 547-561 页，1960 年。

- [209] A.G.雷基查。刚性固体的表示：理论、方法和系统。 ACM 计算调查 (CSUR)，12(4):437-464，1980。

- [210] L. Riazuelo、J. Civera 和 J. M. M. Montiel。 C2TAM：用于协作跟踪和绘图的云框架。 机器人和自主系统 (RAS)，62(4):401-413，2014 年。

- [211] D. M. Rosen、C. DuHadway 和 J. J. Leonard。同步定位中近似全局优化的凸松弛

映射。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 5822-5829 页。 IEEE，2015。

- [212] D. M. 罗森、M. 凯斯和 J. J. 伦纳德。稀疏因子图上的鲁棒增量在线推理：超越高斯情况。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 1025-1032 页。 IEEE，2013。

- [213] O. Russakovsky、J. Deng、H. Su、J. Krause、S. Satheesh、S. Ma、Z. Huang、A. Karpathy、A. Khosla、M. Bernstein、A. C. Berg 和 L. Fei-Fei。 ImageNet 大规模视觉识别挑战赛。 国际计算机视觉杂志，115(3):211-252，2015。

- [214] S. Agarwal 等人。谷神星求解器。 http://ceres-solver.org，2016 年 6 月。谷歌。

- [215] D. Sabatta、D. Scaramuzza 和 R. Siegwart。使用词汇树改进相似和动态环境中基于外观的匹配。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 1008-1013 页。 IEEE，2010。

- [216] S. Saeedi、M. Trentini、M. Seto 和 H. Li。多机器人同时定位和测绘：回顾。 现场机器人杂志 (JFR)，33(1):3-46，2016。

- [217] R. Salas-Moreno、R. Newcombe、H. Strasdat、P. Kelly 和 A. Davison。 SLAM++：对象级别的同步定位和建图。在《IEEE 计算机视觉和模式识别会议 (CVPR) 会议记录》中，第 1352-1359 页。 IEEE，2013。

- [218] D. Scaramuzza 和 F. Fraundorfer。视觉里程计[教程]。第一部分：前 30 年和基础知识。 IEEE 机器人与自动化杂志，18(4):80-92，2011。

- [219] S. Sengupta 和 P. Sturgess。语义八叉树：通过八叉树约束的高阶 MRF 统一识别、重建和表示

。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 1874-1879 页。 IEEE，2015。

- [220] J. Shah 和 M. Mântylâa。 基于参数和特征的 CAD/CAM：概念、技术和应用。威利，1995。

- [221] V.夏皮罗。实体建模。 G. E. Farin、J. Hoschek 和 M. S. Kim 编辑，《计算机辅助几何设计手册》，第 20 章，第 473-518 页。爱思唯尔，2002。

- [222] C. Shen、J. F. O'Brien 和 J. R. Shewchuk。对多边形汤中的隐式曲面进行插值和逼近。在 SIGGRAPH 中，第 896-904 页。美国CM，2004。

- [223] G. 西布利、C. 梅、I. D. 里德和 P. 纽曼。自适应相对束调整。收录于《机器人学报：科学与系统会议 (RSS)》，第 177-184 页，2009 年。

- [224] A. 辛格。通过特征向量和半定规划进行角度同步。 应用和计算谐波分析，30(1):20-36，2010。

- [225] A.辛格和Y.什科尔尼斯基。通过特征向量和半定规划从冷冻电镜中的公共线确定三维结构。 SIAM 成像科学杂志，4(2):543-572，2011。

- [226] J. Sivic 和 A. Zisserman。 Video Google：视频中对象匹配的文本检索方法。在《IEEE 国际计算机视觉会议 (ICCV) 会议记录》中，第 1470-1477 页。 IEEE，2003。

- [227] B. M. 史密斯、L. 张、H. Jin 和 A. Agarwala。光场视频稳定。在《IEEE 国际计算机视觉会议 (ICCV) 会议记录》中，第 341-348 页。 IEEE，2009。

- [228] S.索阿托。迈向视觉信息理论的步骤：主动感知、信号到符号的转换以及传感与控制之间的相互作用。 CoRR，abs/1110.2053:1-214，2011。

- [229] S. Soatto 和 A. Chiuso。视觉表示：定义属性和深度近似。 CoRR，abs/1411.7676:1-20，2016。

- [230] L. Song、B. Boots、S. M. Siddiqi、G. J. Gordon 和 A. J. Smola。隐马尔可夫模型的希尔伯特空间嵌入。参见《国际机器学习会议 (ICML)》，第 991-998 页，2010 年。

- [231] N. Srinivasan、L. Carlone 和 F. Dellaert。用于场景重建和理解的运动结构对称性。参见《英国机器视觉会议 (BMVC) 会议记录》，第 1-13 页。 BMVA 出版社，2015 年。

- [232] C.斯塔尼斯。 机器人测绘和探索。 Springer，2009。[233] C. Stachniss、G. Grisetti 和 W. Burgard。使用 Rao-Blackwellized 粒子滤波器进行基于信息增益的探索。收录于《机器人学报：科学与系统会议 (RSS)》，第 65-72 页，2005 年。

- [234] C. Stachniss、S. Thrun 和 J. J. Leonard。同步定位和绘图。 B. Siciliano 和 O. Khatib，编辑，《施普林格机器人手册》，第 46 章，第 1153-1176 页。施普林格，第二版，2016 年。

- [235] H. Strasdat、A. J. Davison、J. M. M. Montiel 和 K. Konolige。双倍的

Window Optimisation for Constant Time Visual SLAM. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 2352 – 2359. IEEE, 2011.

[236] H. Strasdat, J. M. M. Montiel, and A. J. Davison. Visual SLAM: Why filter? *Computer Vision and Image Understanding (CVIU)*, 30(2):65–77, 2012.

[237] C. Strub, F. Wörgötter, H. Ritterz, and Y. Sandamirskaya. Correcting Pose Estimates during Tactile Exploration of Object Shape: a Neuro-robotic Study. In *International Conference on Development and Learning and on Epigenetic Robotics*, pages 26–33, Oct 2014.

[238] N. Sünderhauf and P. Protzel. Towards a robust back-end for pose graph SLAM. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 1254–1261. IEEE, 2012.

[239] S. Thrun. Exploration in Active Learning. In M. A. Arbib, editor, *Handbook of Brain Science and Neural Networks*, pages 381–384. MIT Press, 1995.

[240] S. Thrun, W. Burgard, and D. Fox. *Probabilistic Robotics*. MIT Press, 2005.

[241] S. Thrun and M. Montemerlo. The GraphSLAM Algorithm With Applications to Large-Scale Mapping of Urban Structures. *The International Journal of Robotics Research (IJRR)*, 25(5-6):403–430, 2005.

[242] G. D. Tipaldi and K. O. Arras. Flirt-interest regions for 2D range data. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 3616–3622. IEEE, 2010.

[243] C. H. Tong, P. Furgale, and T. D. Barfoot. Gaussian process Gauss-Newton for non-parametric simultaneous localization and mapping. *The International Journal of Robotics Research (IJRR)*, 32(5):507–525, 2013.

[244] B. Triggs, P. McLauchlan, R. Hartley, and A. Fitzgibbon. Bundle Adjustment - A Modern Synthesis. In W. Triggs, A. Zisserman, and R. Szeliski, editors, *Vision Algorithms: Theory and Practice*, pages 298–375. Springer, 2000.

[245] R. Tron, B. Afsari, and R. Vidal. Intrinsic Consensus on SO(3) with Almost Global Convergence. In *Proceedings of the IEEE Conference on Decision and Control (CDC)*, pages 2052–2058. IEEE, 2012.

[246] R. Valencia, J. Vallvé, G. Dissanayake, and J. Andrade-Cetto. Active Pose SLAM. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 1885–1891. IEEE, 2012.

[247] J. Valentin, M. Niener, J. Shotton, A. Fitzgibbon, S. Izadi, and P. Torr. Exploiting Uncertainty in Regression Forests for Accurate Camera Relocalization. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 4400–4408. IEEE, 2015.

[248] I. Vallivaara, J. Haverinen, A. Kemppainen, and J. Röning. Simultaneous localization and mapping using ambient magnetic field. In *Multisensor Fusion and Integration for Intelligent Systems (MFI), IEEE Conference on*, pages 14–19. IEEE, 2010.

[249] J. Vallvé and J. Andrade-Cetto. Potential information fields for mobile robot exploration. *Robotics and Autonomous Systems (RAS)*, 69:68–79, 2015.

[250] J. van den Berg, S. Patil, and R. Alterovitz. Motion Planning Under Uncertainty Using Iterative Local Optimization in Belief Space. *The International Journal of Robotics Research (IJRR)*, 31(11):1263–1278, 2012.

[251] V. Vineet, O. Miksik, M. Lidegaard, M. Niessner, S. Golodetz, V. A. Prisacariu, O. Kahler, D. W. Murray, S. Izadi, P. Peerez, and P. H. S. Torr. Incremental dense semantic stereo fusion for large-scale semantic scene reconstruction. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 75–82. IEEE, 2015.

[252] N. Wahlström, T. B. Schön, and M. P. Deisenroth. Learning deep dynamical models from image pixels. In *IFAC Symposium on System Identification (SYSID)*, pages 1059–1064, 2015.

[253] C. C. Wang, C. Thorpe, S. Thrun, M. Hebert, and H. F. Durrant-Whyte. Simultaneous localization, mapping and moving object tracking. *The International Journal of Robotics Research (IJRR)*, 26(9):889–916, 2007.

[254] L. Wang and A. Singer. Exact and Stable Recovery of Rotations for Robust Synchronization. *Information and Inference: A Journal of the IMA*, 2(2):145–193, 2013.

[255] Z. Wang, S. Huang, and G. Dissanayake. *Simultaneous Localization and Mapping: Exactly Sparse Information Filters*. World Scientific, 2011.

[256] J. W. Weingarten, G. Gruener, and R. Siegwart. A State-of-the-Art 3D Sensor for Robot Navigation. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 2155–2160. IEEE, 2004.

[257] T. Whelan, S. Leutenegger, R. F. Salas-Moreno, B. Glocker, and A. J. Davison. ElasticFusion: Dense SLAM Without A Pose Graph. In *Proceedings of Robotics: Science and Systems Conference (RSS)*, 2015.

[258] T. Whelan, J. B. McDonald, M. Kaess, M. F. Fallon, H. Johannsson, and J. J. Leonard. Kintinuous: Spatially Extended Kinect Fusion. In *RSS Workshop on RGB-D: Advanced Reasoning with Depth Cameras*, 2012.

[259] S. Williams, V. Indelman, M. Kaess, R. Roberts, J. J. Leonard, and F. Dellaert. Concurrent filtering and smoothing: A parallel architecture for real-time navigation and full smoothing. *The International Journal of Robotics Research (IJRR)*, 33(12):1544–1568, 2014.

[260] R. W. Wolcott and R. M. Eustice. Visual localization within LIDAR maps for automated urban driving. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 176–183. IEEE, 2014.

[261] R. Wood. RoboBees Project. http://robobees.seas.harvard.edu/, June 2015.

[262] B. Yamauchi. A Frontier-based Approach for Autonomous Exploration. In *Proceedings of IEEE International Symposium on Computational Intelligence in Robotics and Automation (CIRA)*, pages 146–151. IEEE, 1997.

[263] K. T. Yu, J. Leonard, and A. Rodriguez. Shape and Pose Recovery from Planar Pushing. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 1208–1215, Sept 2015.

[264] C. Zach, T. Pock, and H. Bischof. A Globally Optimal Algorithm for Robust TV-L1 Range Image Integration. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 1–8. IEEE, 2007.

[265] J. Zhang, M. Kaess, and S. Singh. On Degeneracy of Optimization-based State Estimation Problems. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 809–816, 2016.

[266] S. Zhang, Y. Zhan, M. Dewan, J. Huang, D. Metaxas, and X. Zhou. Towards Robust and Effective Shape Modeling: Sparse Shape Composition. *Medical Image Analysis*, 16(1):265–277, 2012.

[267] L. Zhao, S. Huang, and G. Dissanayake. Linear SLAM: A Linear Solution to the Feature-Based and Pose Graph SLAM based on Submap Joining. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 24–30. IEEE, 2013.

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恒定时间视觉 SLAM 的窗口优化。在《IEEE 国际计算机视觉会议 (ICCV) 会议记录》中，第 2352 - 2359 页。IEEE，2011。

- [236] H. Strasdat、J. M. M. Montiel 和 A. J. Davison。视觉 SLAM：为什么要过滤？ 计算机视觉和图像理解 (CVIU)，30(2):65-77，2012 年。

- [237] C. Strub、F. Wororgotter、H. Ritterz 和 Y. Sandamirskaya。在物体形状的触觉探索过程中校正姿势估计：神经机器人研究。载于《发展与学习及表观遗传机器人国际会议》，第 26-33 页，2014 年 10 月。

- [238] N. Süunderhauf 和 P. Protzel。面向姿势图 SLAM 的强大后端。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 1254-1261 页。 IEEE，2012。

- [239] S.特龙。主动学习中的探索。 M. A. Arbib，编辑，《脑科学和神经网络手册》，第 381-384 页。麻省理工学院出版社，1995。

- [240] S. Thrun、W. Burgard 和 D. Fox。 概率机器人。麻省理工学院出版社，2005 年。

- [241] S. Thrun 和 M. Montemerlo。 GraphSLAM 算法在城市结构大规模测绘中的应用。 国际机器人研究杂志 (IJRR)，25(5-6):403-430，2005。

- [242] G. D. Tipaldi 和 K. O. Arras。 2D 范围数据的调情兴趣区域。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 3616-3622 页。 IEEE，2010。

- [243] C. H. Tong、P. Furgale 和 T. D. Barfoot。用于非参数同时定位和建图的高斯过程 GaussNewton。 国际机器人研究杂志 (IJRR)，32(5):507-525，2013 年。

- [244] B. Triggs、P. McLauchlan、R. Hartley 和 A. Fitzgibbon。捆绑调整 - 现代综合。见 W. Triggs、A. Zisserman 和 R. Szeliski，编辑，视觉算法：理论与实践，第 298-375 页。施普林格，2000。

- [245] R. Tron、B. Afsari 和 R. Vidal。 SO(3) 的内在共识几乎全球趋同。在《IEEE 决策与控制会议 (CDC) 会议记录》中，第 2052-2058 页。 IEEE，2012。

- [246] R. Valencia、J. Vallv´e、G. Dissanayake 和 J. Andrade-Cetto。主动姿势 SLAM。在《IEEE/RSJ 国际智能机器人和系统会议 (IROS) 会议记录》中，第 1885-1891 页。 IEEE，2012。

- [247] J. Valentin、M. Niener、J. Shotton、A. Fitzgibbon、S. Izadi 和 P. Torr。利用回归森林中的不确定性进行精确的相机重定位。在《IEEE 计算机视觉和模式识别会议 (CVPR) 会议记录》中，第 4400-4408 页。 IEEE，2015。

- [248] I. Vallivaara、J. Haverinen、A. Kemppainen 和 J. Réoning。使用环境磁场进行同步定位和绘图。参见《智能系统多传感器融合与集成 (MFI)》，IEEE 会议_，第 14-19 页。 IEEE，2010。

- [257] T. Whelan、S. Leutenegger、R. F. Salas-Moreno、B. Glocker 和 A. J. Davison。 ElasticFusion：没有姿势图的密集 SLAM。在《机器人学报：科学与系统会议 (RSS)》中，2015 年。

- [258] T. Whelan、J. B. McDonald、M. Kaess、M. F. Fallon、H. Johannsson 和 J. J. Leonard。Kintinious：空间扩展的 Kinect 融合

。在 RGB-D RSS 研讨会：深度相机的高级推理，2012 年。

- [259] S. Williams、V. Indelman、M. Kaess、R. Roberts、J. J. Leonard 和 F. Dellaert。并发过滤和平滑：用于实时导航和完全平滑的并行架构。 国际机器人研究杂志 (IJRR)，33(12):1544-1568，2014 年。

- [260] R. W. 沃尔科特和 R. M. 尤斯蒂斯。用于自动城市驾驶的激光雷达地图中的视觉定位。在《IEEE/RSJ 国际智能机器人和系统会议 (IROS) 会议记录》中，第 176-183 页。 IEEE，2014。

- [261] R.伍德。机器人蜜蜂项目。 http://robobees.seas.harvard.edu/，2015 年 6 月。

- [262] B.山内。基于前沿的自主探索方法。在《IEEE 机器人与自动化计算智能国际研讨会 (CIRA) 会议记录》中，第 146-151 页。 IEEE，1997。

- [263] K. T. Yu、J. Leonard 和 A. Rodriguez。平面推动的形状和姿势恢复。摘自《IEEE/RSJ 国际智能机器人和系统会议 (IROS) 会议记录》，第 1208-1215 页，2015 年 9 月。

- [264] C.扎克、T.波克和H.比肖夫。用于鲁棒 TV-L1 范围图像集成的全局最优算法。在《IEEE 国际计算机视觉会议 (ICCV) 会议记录》中，第 1-8 页。IEEE，2007 年。

- [265] J. 张、M. Kaess 和 S. Singh。基于优化的状态估计问题的简并性。参见《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》，第 809-816 页，2016 年。

- [266] S. 张、Y. Zhan、M. Dewan、J. Huang、D. Metaxas 和 X. Zhou。迈向稳健且有效的形状建模：稀疏形状组合。 医学图像分析，16(1)：265-277，2012。

- [267] L. 赵、S. 黄和 G. Dissanayake。线性 SLAM：基于子图连接的基于特征和位姿图 SLAM 的线性解决方案。在《IEEE/RSJ 国际智能机器人和系统会议 (IROS) 会议记录》中，第 24-30 页。 IEEE，2013。

- [249] J. Vallv´e 和 J. Andrade-Cetto。移动机器人探索的潜在信息领域。 机器人和自主系统 (RAS)，69:68-79，2015。

- [250] J. van den Berg、S. Patil 和 R. Alterovitz。使用置信空间中的迭代局部优化进行不确定性下的运动规划。 国际机器人研究杂志 (IJRR)，31(11):1263-1278，2012。

- [251] V. Vineet、O. Miksik、M. Lidegaard、M. Niessner、S. Golodetz、V. A. Prisacariu、O. Kahler、D. W. Murray、S. Izadi、P. Peerez 和 P. H. S. Torr。用于大规模语义场景重建的增量密集语义立体融合。在《IEEE 国际机器人与自动化会议 (ICRA) 会议记录》中，第 75-82 页。 IEEE，2015。

- [252] N. Wahlström、T. B. Schöon 和 M. P. Deisenroth。从图像像素学习深度动态模型。 IFAC 系统识别研讨会 (SYSID)，第 1059-1064 页，2015 年。

- [253] C.C. Wang、C. Thorpe、S. Thrun、M. Hebert 和 H. F. Durrant-Whyte。同时定位、绘图和移动物体跟踪。 国际机器人研究杂志 (IJRR)，26(9):889-916，2007 年。

- [254] L. Wang 和 A. Singer。精确稳定的旋转恢复以实现稳健同步

。 信息与推理：IMA 杂志，2(2):145-193，2013 年。

- [255] Z. Wang、S. Huang 和 G. Dissanayake。 同时定位和绘图：精确稀疏的信息过滤器。世界科学，2011。

- [256] J. W. Weingarten、G. Gruener 和 R. Siegwart。用于机器人导航的最先进的 3D 传感器。在《IEEE/RSJ 国际智能机器人和系统会议 (IROS) 会议记录》中，第 2155-2160 页。 IEEE，2004。